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Digital Audio Broadcasting (DAB);
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Foreword

This Technical Specification (TS) has been produced by Joint Technical Committee (JTC) Broadcast of the European Broadcasting Union (EBU), Comité Européen de Normalisation ELECtrotechnique (CENELEC) and the European Telecommunications Standards Institute (ETSI).

NOTE 1: The EBU/ETSI JTC Broadcast was established in 1990 to co-ordinate the drafting of standards in the specific field of broadcasting and related fields. Since 1995 the JTC Broadcast became a tripartite body by including in the Memorandum of Understanding also CENELEC, which is responsible for the standardization of radio and television receivers. The EBU is a professional association of broadcasting organizations whose work includes the co-ordination of its members' activities in the technical, legal, programme-making and programme-exchange domains. The EBU has active members in about 60 countries in the European broadcasting area; its headquarters is in Geneva.

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The Eureka Project 147 was established in 1987, with funding from the European Commission, to develop a system for the broadcasting of audio and data to fixed, portable or mobile receivers. Their work resulted in the publication of European Standard, ETSI EN 300 401 [1], for DAB® (see note 2) which now has worldwide acceptance.

NOTE 2: DAB is a registered trademark owned by one of the Eureka Project 147 partners.

The DAB family of standards is supported by World DAB, an organization with members drawn from broadcasting organizations and telecommunication providers together with companies from the professional and consumer electronics industry.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document defines rules of implementation for certain service information features. These rules have been developed to provide a reliable and consistent experience for digital radio listeners; they provide implementation details for how the Fast Information Channel (FIC) signalling is used and how receivers will interpret and behave in response to receiving the FIC signalling.

This revision adds a preference capability to service linking and a recommendation to use EWS instead of alarm announcements.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found in the ETSI docbox.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI EN 300 401 (V2.1.1): "Radio Broadcasting Systems; Digital Audio Broadcasting (DAB) to mobile, portable and fixed receivers".

[2] ETSI TS 101 756: "Digital Audio Broadcasting (DAB); Registered Tables".

[3] Unicode standard.

[4] Unicode bidirectional algorithm, UAX#9.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] ISO 10646: "Information technology -- Universal Coded Character Set (UCS)".

[i.2] ETSI TS 104 089: "Digital Audio Broadcasting (DAB); Emergency Warning System (EWS); Definition and rules of behaviour".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

alarm announcement: announcement of type 'alarm' which has elevated priority

announcement: short audio message containing information categorized by an announcement type

Change Event Indication (CEI): set of FIG fields with particular values to indicate a change of database content for certain service information features

database entry: part of the service information addressed by a database key

database key: set of FIG fields that sub-divide a database for certain service information features

implicit linking: linking of DAB programme service and FM-RDS service with identical identifiers requiring no FIG 0/6 signalling

international link: linking of DAB and FM-RDS services from countries with different ECCs, or linking including DRM and AMSS services.

key service: DAB service carried in the tuned ensemble and placed as the first Id in the list of all services in a linkage set

linkage set: description of a network configuration consisting of lists of identifiers which carry the same (hard link) or related (soft link) content

national link: linking of DAB and FM-RDS services from countries with same ECCs

part-time service element: service element that cycles between an on-air and an off-air status

pre-tuning memory: information stored in a receiver from previous tuning actions providing details of ensembles, tuning frequencies and services

regular announcement: announcement of any type except 'alarm' with normal priority

same service: two DAB services with the same service identifier (SId) and extended country code (ECC)

service element: smallest addressable part of a service; a service component, either primary or secondary

NOTE: In a service that consists of only the primary service component, the term service element refers to the entire service.

service following: process for maintaining the same audio or data content that the user has selected in spite of the varying reception conditions that occur, for example, when travelling by car or train

service list: feature of a radio receiver where a list of service elements is used for service selection

service list entry: one item in a service list that represents a single service element

SI label: text label carried in any extension of FIG type 1 and FIG type 2

user controls: all elements of a user interface of a radio receiver that are used to display service information and provide for user control

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

AAC	Advanced Audio Coding
AMSS	Amplitude Modulation Signalling System
ASu	Announcement Support
ASw	Announcement Switching
C/N	Current/Next
CA	Conditional Access
CAId	Conditional Access Identifier
CEI	Change Event Indication
DAB	Digital Audio Broadcasting
DMB	Digital Multimedia Broadcasting
DRM	Digital Radio Mondiale
ECC	Extended Country Code
EId	Ensemble Identifier
EWS	Emergency Warning System
F	Frequency
FI	Frequency Information
FIB	Fast Information Block
FIC	Fast Information Channel
FIG	Fast Information Group
FM	Frequency Modulation
Id	Identifier
IdLP	Identifier List Preference level
IdLQ	Identifier List Qualifier
ILS	International Linkage Set
ISO	International Standards Organisation
LA	Linkage Actuator
LRI	Left Right Isolate
LSN	Linkage Set Number
LTO	Local Time Offset
LTR	Left To Right
MCI	Multiplex Configuration Information
MFN	Multi-Frequency Network
MJD	Modified Julian Date
MPEG	Moving Pictures Expert Group
MSC	Main Service Channel
OE	Other Ensemble
P/D	Programme/Data service flag
PAD	Programme Associated Data
PDI	Pop Directional Isolate
PI	Programme Identification code (RDS)
R&M	Range and Modulation
RDS	Radio Data System
Rfa	Reserved for future addition
RLI	Right Left Isolate
RTL	Right To Left
S/H	Soft/Hard
SC	Service Component
SCI	Service Component Information
SFN	Single Frequency Network
SI	Service Information
SId	Service Identifier
UCS	Universal Character Set
UTC	Co-ordinated Universal Time
UTF	Unicode Transformation Format

4 Overview

Service information (SI) in DAB is carried in the Fast Information Channel (FIC) as a series of Fast Information Groups (FIGs) carried in Fast Information Blocks (FIBs). Different FIGs are used for different service information, and several different FIGs may be needed to implement a particular service information feature, such as service linking or announcements. The dynamic label is also considered to be a service information feature. The present document provides rules of implementation for service information ***features*** and so groups the usage of the required FIGs together. Some FIG types are used by a number of different features and the rules are designed so that the FIG is always coded and decoded consistently.

The present document provides normative rules of behaviour for complex service information features:

- service following, using FIG 0/6, FIG 0/9, FIG 0/21 and FIG 0/24;
- service lists, using FIG 0/20;
- announcements, using FIG 0/18, FIG 0/19, FIG 0/25 and FIG 0/26 (and also FIG 0/21 and FIG 0/24);
- text labels, using FIG 1/x, FIG 2/x and dynamic labels.

The nominal repetition rate for SI FIGs is once per second. This rate applies to FIG 0/5, FIG 0/9, FIG 0/10, FIG 0/17, FIG 0/18, FIG 0/20, FIG 0/25, FIG 1/x and FIG 2/x. However, many of the FIGs used for the complex service information features described in the present document require several repetition rates, depending on the usage. Table 1 provides an overview of the repetition rates for these complex SI FIGs.

Table 1: Repetition rates for complex SI FIGs

FIG type/ext	Description	Usage	Nominal repetition rate
FIG 0/6	Service linking information	Short form, activation state	once per 10 seconds
		Short form, CEI	once per second for 5 seconds
		Long form, database definition	all information at least every 2 minutes
FIG 0/19	Announcement switching	Start of announcement	10 times per second for 5 seconds
		Continuation of announcement	once per second
		End of announcement	10 times per second for 2 seconds
FIG 0/21	Frequency information (FI)	Short form, CEI	once per second for 5 seconds
		Long form, database definition	all information at least every 2 minutes
FIG 0/24	OE services	Short form, CEI	once per second for 5 seconds
		Long form, database definition	all information at least every 2 minutes
FIG 0/26	OE Announcement switching	Start of announcement	10 times per second for 5 seconds
		Continuation of announcement	once per second
		End of announcement	10 times per second for 2 seconds

The FIC also carries Multiplex Configuration Information (MCI). This information describes the current multiplex in terms of services, service components and sub-channels. When the multiplex is reconfigured, the MCI also describes the next multiplex configuration. To ensure that receivers know the configuration of the multiplex quickly, all configuration information is *normally* signalled every 96 ms, except for FIG 0/8 for data service components and FIG 0/13 which are signalled at least every 1 s. During a reconfiguration, all configuration information for both the current and next configurations is *normally* signalled in 192 ms, except for FIG 0/8 for data service components and FIG 0/13 which are signalled for both the current and next configurations at least every 1 second (see ETSI EN 300 401 [1], clause 6.1).

The signalling in the FIC of both MCI and SI can achieve the nominal repetition rates for ensembles with up to 20 DAB+ audio services, each carrying a SlideShow in PAD, and with additional SI for date and time, language, programme type, service following and announcements. An example of the FIC coding for such an ensemble is given in annex F. However, due to the fixed capacity of the FIC, when a greater number of service components is present, the signalling rates will be lower. It is recommended that of the 12 FIBs available in each 96 ms period, a maximum of 10 are allocated to delivering the MCI and labels, with the remaining FIBs allocated to SI.

The configuration of the ensemble and the SI features supported determine the actual repetition rates signalled for each FIG, and these shall not fall below one third of the nominal rates defined in ETSI EN 300 401 [1] and table 1 of the present document.

A reasonable rule of thumb is to expect the repetition rates to decline to be about half of the nominal rates as the number of service components rises to around 40, and to further decline to be about one third of the nominal rates when about 60 service components are present in an ensemble. The number of services in the ensemble is provided by FIG 0/7, which can be used to verify when the complete MCI has been received. The total number of service components can then be determined from FIG 0/2 to give an indication of the likely repetition rates.

5 Service following

5.1 Introduction

5.1.0 General

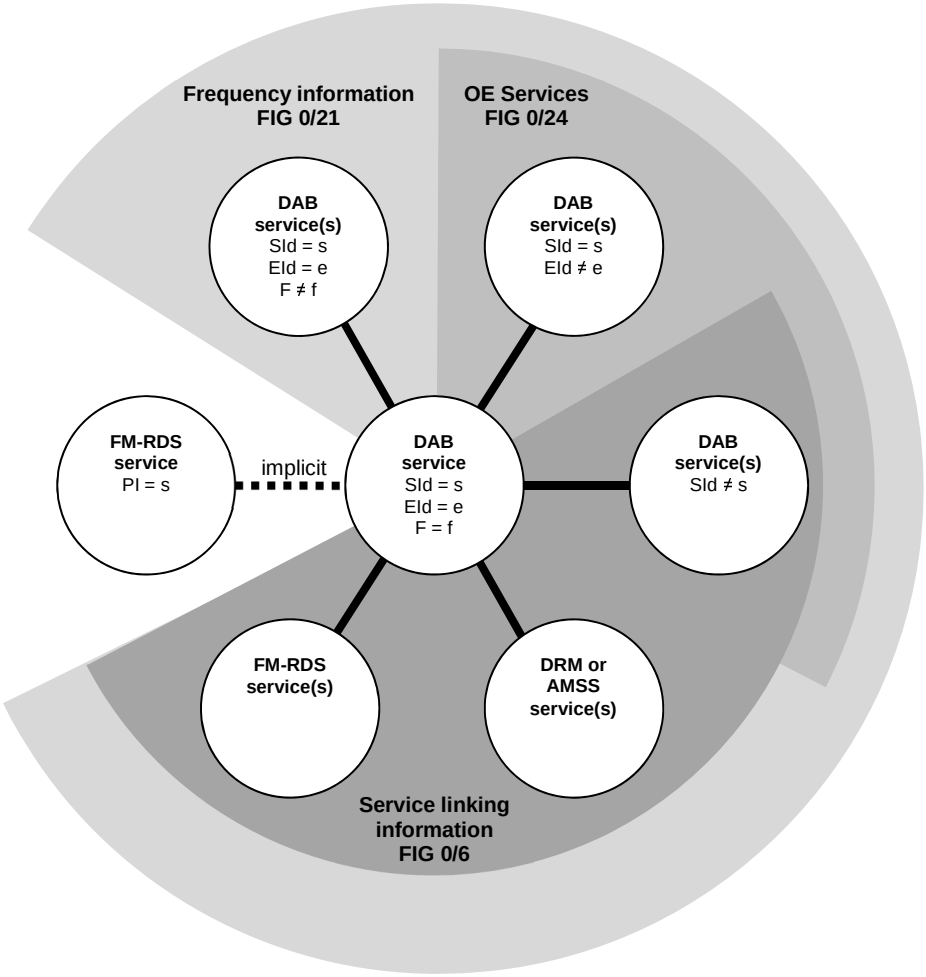
Service following is the term applied to maintaining the same audio or data content that the user has selected in spite of the varying reception conditions that occur, for example, when travelling by car or train. Many broadcast network topologies are possible, and the tuned service may be carried on an ensemble with multiple tuning frequencies, on more than one ensemble, may carry common programming with other DAB services, and for audio services, also be carried on FM-RDS or another bearer.

The best service following experience for the listener is achieved when the broadcaster minimizes the timing differences between different bearers, taking into consideration the different coding and decoding delays of the different systems. Reliable service following also requires that all the identifiers used are properly allocated in such a way as to make them unique within their respective scope.

DAB provides signalling that enables service providers to inform receivers about the broadcast networks and service configurations that allow service following to take place (much of which is static, since it refers to transmitter network configurations that change only infrequently), and also provides dynamic information to control which of that information is used at any given time to take account of changes during the day.

Three types of information may be involved with service following - Service linking information, Other Ensembles Services information and Frequency information. Service following provides information to allow precisely the same content to be followed, and when linking information is needed this is called hard linking. Service providers are also able to indicate suitable alternative content when service following is not possible by signalling related services using soft linking. Broadcasters also operate services with local or regional variations at some times of the day and may provide related services on different ensembles with overlapping coverage areas. For this case, broadcasters are able to indicate that some services are less preferred than others when making a service following decision.

Figure 1 shows the principal FIGs used for service following.



NOTE: For clarity, the figure does not show the role of FIG 0/9 in providing programme service and ensemble ECCs.

Figure 1: Service following and principal supporting FIGs

The DAB service at the centre of figure 1 represents the tuned service and it may have one or more of the alternate sources represented by the surrounding circles. To ensure that the DAB identifiers used in service following are globally unique, the respective ECCs for programme services and ensembles are delivered using FIG 0/9.

5.1.1 Service linking information

Service linking information, carried using FIG 0/6, allows the service provider to establish one or more sets of identifiers that carry identical, in the case of a hard link, or related, in the case of a soft link, content. The set of identifiers is called a linkage set. There may be several linkage sets that are valid at different times of day. Each linkage set is identified by the Linkage Set Number together with a set of flags, and by use of the Linkage Actuator, linkage sets can be activated and deactivated.

The receiver uses these linkage sets during service selection and service following to determine a set of candidate services, potentially on different bearers, that are equivalent or related to the selected service. The receiver selects an appropriate service from these candidates based on criteria such as service availability and quality.

Receivers may cache linkage sets for services of interest (e.g. the currently selected service and other recently or frequently selected services) across power cycles, radio frequencies and locations so that the information is available immediately when the receiver is turned on and when it switches from one radio frequency to another.

5.1.2 OE Services

In order to assist the receiver, the service provider may signal a list of geographically adjacent alternative ensembles using FIG 0/24 on which the current and other services can be found.

The receiver may use this information to identify ensembles carrying a service when it is selected by the user and during service following. This may reduce the number of radio frequencies that a receiver scans before it locates an ensemble carrying the service. This helps minimize the time delay between the user selecting a service and the start of audio output, particularly when selecting a service on another ensemble in a mobile environment. It also helps minimize the duration of any interruption to the audio output when service following occurs, especially in single tuner receivers.

It is therefore recommended that service providers signal OE Services to enable the best possible user experience during service selection and following. This information is especially useful when provided together with Frequency Information for other ensembles in the same or an adjacent coverage area.

5.1.3 Frequency information

In order to assist the receiver, the service provider may signal a list of alternative frequencies using FIG 0/21 on which the tuned ensemble and other ensembles may be found. FIG 0/21 may also signal frequency information for other bearers such as FM-RDS, DRM and AMSS.

The receiver may use this information to identify frequencies carrying an ensemble during service selection and service following. It may also use it to identify frequencies carrying linked FM-RDS, DRM or AMSS services during service following. This may reduce the number of frequencies that the receiver scans before it locates a service when it is selected by the user and during service following. This helps minimize the time delay between the user selecting a service and audio output and, especially in single-tuner receivers, the duration of any audio interruption during service following.

It is therefore recommended that service providers signal frequency information to enable the best possible user experience during service selection and following. This information is especially useful when provided together with OE Services for other ensembles in the same or an adjacent coverage area.

Where the same ensemble is available on different radio frequencies with overlapping coverage areas, it is also advantageous for the transmissions to be synchronized and for the continuity flag to be signalled in the frequency information. This enables the receiver to maintain continuous audio output during service following without any re-synchronization delay in the de-interleaver and audio decoder.

5.1.4 Country information, LTO and International Table

Service following requires receivers to compare identifiers. To ensure that DAB services are uniquely identified in all situations, the ECC shall be known. FIG 0/9 allows the ECC of each programme (audio) service to be provided (data services include the ECC in the SId). If all programme services have the same ECC, then the **Ensemble ECC** applies to all services, and no extension field is required. However, if any service carried in the ensemble has a different ECC to the **Ensemble ECC**, (for example, if it is carried in an ensemble comprising services from different countries) then the extension field shall be used to provide the **ECC** and list of SIds with that service specific ECC (up to 3 SIds). If more than 3 services have a particular service specific ECC, then other sub-fields with the same **ECC** shall be signalled. All programme services shall be provided with the correct ECC.

NOTE: The maximum size of the extension field is 25 bytes, which means that a maximum of six additional ECCs can be signalled with one service each, or a maximum of nine services can be assigned a single additional ECC, or other combinations.

Receivers shall assign the correct ECC to all programme services in the ensemble by decoding the FIG 0/9.

5.2 Service linking

5.2.0 General

The FIG 0/6 consists of a header field and one or more **service linking** fields.

The **service linking** fields can take the short form or the long form and this is determined by the state of the **Id list flag**.

The database key for FIG 0/6 consists of the **OE** and **P/D flags** from the header field and the **S/H, ILS, and LSN** fields from the **service linking** field.

Linkage sets shall only be signalled if they contain at least one service carried in the tuned ensemble (i.e. there is no provision of other ensemble signalling of linkage sets).

5.2.1 Short form - Activation state and change event indication

The short form has the **Id list flag** set to 0. It is used for signalling the activation state of all the linkage sets signalled in the ensemble and to indicate that changes to the database content of a linkage set will be made. The **C/N flag** in the FIG 0/6 header is used to determine which type of information is being signalled.

When the activation state of the linkage sets is being signalled the **C/N flag** is set to 1.

When the content of the database associated with a particular linkage set will be changed - due to a network configuration change - **the C/N flag** is set to 0.

5.2.2 Long form - Database definition

The long form has the **Id list flag** set to 1. It is used to build up the database of identifiers that are linked together to form linkage sets. The **C/N flag** is used in the database definition mode and the flag indicates the start of the database definition when set to 0 or a continuation of the database definition when set to 1. There shall be only one **service linking** field for the start of database for each database key and there may be one or more **service linking** fields for the continuation of database for each database key.

NOTE: It is not possible to mix start of database definition and continuation of database definition service linking fields in the same FIG since the **C/N flag** applies to the entire FIG.

5.2.3 Linkage sets

A linkage set is a collection of identifiers (DAB SIds, RDS PI codes, etc.) that correspond to alternative sources of the same content (hard link) or related content (soft link) - effectively a linkage set represents a particular transmission network configuration and so is essentially static. However, a particular service may have different alternative sources at different times of the day. Each of these different situations can be represented by a different linkage set, each corresponding to a particular network configuration. Each linkage set may be either activated or deactivated in order to reflect which is the current network configuration for the services carried in the ensemble.

All services carrying identical content at a given time are grouped together in a hard linkage set.

All services carrying related content at a given time are grouped together in a soft linkage set.

NOTE 1: All services in a linkage set are possible alternatives to each other when the linkage set is active.

All linkage sets shall have one DAB SId, and may also have other DAB SIds, RDS PI codes, DRM SIds and AMSS SIds up to a maximum of 128 Ids in total.

Linkage sets are identified using the **LSN** field in combination with four flags: the **P/D flag**, the **OE flag**, the **Soft/Hard (S/H) flag** and the **International Linkage Set (ILS) flag**.

NOTE 2: ETSI EN 300 401 [1] defines the OE flag as *Rfu*; therefore, it always takes the value 0.

The **LSN**s shall be coordinated between all broadcasters in a particular country such that they are unique in combination with the flags.

When defining a linkage set for programme services (**P/D flag** = 0), if all the Ids in the linkage set are for DAB and/or FM-RDS services with the Ensemble ECC, then the **ILS flag** shall be set to 0; otherwise (i.e. if any DAB and/or FM-RDS service has a different ECC, or if a DRM and/or AMSS service is included) the **ILS flag** shall be set to 1.

When defining a linkage set for data services (**P/D flag** = 1) then the **ILS flag** has no particular meaning but differentiates between two different linkage sets.

NOTE 3: The name of the **ILS flag** is historic; for programme services (**P/D flag** = 0) it is a switch between using 16-bit and 24-bit **Id** fields in the **Id List** field - when the **ILS flag** = 0, 16-bit fields are used and the ECC for each SId or PI code is determined from the FIG 0/9 **Ensemble ECC**; when the **ILS flag** = 1, 24-bit fields are used and the ECC is provided for each SId or PI code in the FIG 0/6 information.

To permit broadcasters to indicate that some services in a linkage set are preferred to others, the **Rfa** flag that follows the **IdLQ** flag (see ETSI EN 300 401 [1] figure 53) is repurposed as the **IdLP** flag with the following definition:

> **IdLP (Identifier List Preference level):** this 1-bit field shall indicate the preference level of the identifiers in the **Id list**, as follows:
>
> 0: the identifiers in the **Id list** have a normal preference level;
>
> 1: the identifiers in the **Id list** have a low preference level.

Broadcasters may indicate that some services in the linkage set are preferred to others when making a service following decision. The less preferred services in the linkage set are indicated by setting the **IdLP** flag to 1 for the Id list that carries those identifiers. The more preferred services are indicated by setting the **IdLP** flag to 0 in the respective Id list. If all identifiers in the linkage set have equal preference, the **IdLP** flag is set to 0 for all the Id lists.

Linkage sets are activated and deactivated according to the state of the **LA flag**. When a linkage set is activated, receivers may switch to any of the alternate sources of the content; when it is deactivated, they shall not. This feature allows service providers to signal linkage sets in advance of their use and control the receiver linkage behaviour by changing the state of the **LA flag** for each linkage set.

The linkage set definitions reflect different broadcast network configurations. If for a particular service, the alternative sources remain the same throughout the day, week and year, then only one linkage set is needed and it will be activated at all times. For other services there may be different alternatives at different times of day that reflect the programming needs of the service provider. Each broadcast network configuration is represented by a different linkage set, and each of these are defined and allocated an identifier - the **LSN** in combination with the four flags. As the day progresses and each different network configuration becomes valid, a different linkage set is activated to inform receivers of the valid alternatives. In this way the database information built up by the receivers remains static, but the validity of a particular broadcast network configuration is controlled dynamically by the indication of which linkage set is active.

Of all the linkage sets defined containing a particular service, only one hard linkage set (either National or International) and one soft linkage set (either National or International) shall be activated at a given time; all other linkage sets defined containing that particular service shall be deactivated.

If the only other source of a DAB service is an FM-RDS service and the SId and PI code are identical, then ***implicit linking*** may be used. It requires no service linking information to be sent, but provides the equivalent case to an activated hard linkage set in which the only entries are a DAB SId and an RDS PI code with the same value. This implicit linking exists for all DAB services unless it is overridden by an activated hard linkage set which includes the DAB SId and a **service linking** field with **IdLQ** = 01 [RDS PI codes]. Such a linkage set may contain zero PI codes in which case all service following to FM-RDS is prevented (the so called "dead link"). When implicit linking is overridden, the PI codes of all equivalent RDS services (including the PI code with the same value as the SId if it is equivalent) shall be present in the hard linkage set.

NOTE 4: In general, RDS services do not provide the ECC in addition to the PI code. Receivers linking between DAB and FM-RDS services will usually have to assume that the RDS service uses the same ECC as the DAB service.

For each ensemble that transmits a linkage set, the transmission order of the identifiers in the linkage set shall be determined according to the following sequence; if some services have a different preference level within a particular numbered sub-group, the identifiers of each preference level shall be grouped together:

1) The SId(s) of the DAB service(s) carried in the ensemble.
2) The SId(s) of the DAB service(s) carried in geographically adjacent ensembles.
3) The SIds of all other DAB services.
4) The PI code(s) of the RDS service(s) available in the coverage area of the ensemble.
5) The PI code(s) of the RDS service(s) available in the coverage area of geographically adjacent ensembles.
6) The PI codes of all other RDS services.
7) The SId(s) of the DRM or AMSS service(s) available in the coverage area of the ensemble.
8) The SId(s) of the DRM or AMSS service(s) available in the coverage area of geographically adjacent ensembles.

9) The SIds of all other DRM or AMSS services.

Once the transmission order is determined for each ensemble, it shall not be changed unless the members of the linkage set change.

5.2.4 Signalling a link

5.2.4.0 General

There are different phases to signalling a link. Each linkage set is signalled as a FIG 0/6 database entry. It shall be defined using the long form, and activated and deactivated using the short form. A linkage set may be changed by signalling the CEI (short form) corresponding to the database entry and then redefined using the long form. Database changes should be kept to a minimum. The **LA flag** shall always reflect the activation state of the linkage set, whether signalled in the long or short form. Listeners may tune to an ensemble at any time and therefore the database entry definition shall be repeated continually.

Every DAB ensemble that carries **any** of the services with an SId in the linkage set shall signal the linkage set continually.

5.2.4.1 Defining a linkage set

The transmission order list of the linkage set for the tuned ensemble is determined in the planning stage (see clause 5.2.3).

The identifier list shall be divided into FIG 0/6s carrying **service linking** fields according to the following cases:

Step A: a start of database FIG 0/6 (**C/N** = 0) contains a **service linking** field with:

- *if* the transmission order list contains one DAB SId only (a dead link): the **IdLQ** is set to 01 [RDS PI codes] and the first and only **Id** in the list is the DAB SId (key service); the **IdLP** flag is set to 0.
- *else if* the first two identifiers in the transmission order list have different preference levels: the **IdLQ** is set to 00 [DAB SIds] and the first and only **Id** in the list is the DAB SId (key service); the **IdLP** flag is set according to the preference level of the key service.
- *else if* the transmission order list contains one DAB SId and one or more RDS PI codes (and optionally one or more DRM/AMSS SIds): the **IdLQ** is set to 01 [RDS PI codes] and the first **Id** in the list is the DAB SId (key service) and all other **Id**s are RDS PI codes of the same preference level as the key service; the **IdLP** flag is set according to the preference level of the **Id**s in the **Id list**.
- *else if* the transmission order list contains one DAB SId and one or more DRM or AMSS SIds: the **IdLQ** is set to 11 [DRM or AMSS SIds], the first **Id** in the list is the DAB SId (key service) and all other **Id**s are DRM or AMSS SIds of the same preference level as the key service; the **IdLP** flag is set according to the preference level of the **Id**s in the **Id list**.
- *else* the **IdLQ** is set to 00 [DAB SIds] and the **Id list** consists of as many DAB SIds of the same preference level as will fit into the FIG following the transmission order list of the ensemble (it is recommended to include as many **Id**s as will fit, up to the full capacity of an FIB, for this FIG 0/6); the **IdLP** flag is set according to the preference level of the **Id**s in the **Id list**.

Step B: if the transmission order list contains untransmitted **Id**s then a continuation of database FIG 0/6 (**C/N** = 1) contains a **service linking** field with:

- *if* there are DAB SIds to signal: the **IdLQ** is set to 00 [DAB SIds] and the **Id list** contains as many DAB SIds of the same preference level as will fit into the FIG following the transmission order list of the ensemble; the **IdLP** flag is set according to the preference level of the **Id**s in the **Id list**.
- *else if* there are RDS PI codes to signal: the **IdLQ** is set to 01 [RDS PI codes] and the **Id list** contains as many RDS PI codes of the same preference level as will fit into the FIG following the transmission order list of the ensemble; the **IdLP** flag is set according to the preference level of the **Id**s in the **Id list**.

- *else* there are DRM or AMSS SIds to signal: the **IdLQ** is set to 11 [DRM or AMSS SIds] and the **Id list** contains as many DRM or AMSS SIds of the same preference level as will fit into the FIG following the transmission order list of the ensemble; the **IdLP** flag is set according to the preference level of the **Id**s in the **Id list**.

Repeat step B until all Ids in the transmission order list have been signalled, beginning new FIG 0/6s as required.

If capacity remains in any FIB, additional **service linking** fields for other linkage sets may be appended.

The complete database entry for each database key should be transmitted within 10 seconds and repeated at least every 2 minutes.

NOTE: Although the transmission order is consistent, reception conditions may mean that the continuation of database FIGs may not always be received in the transmitted sequence (missing FIGs).

5.2.4.2 Changing the linkage set

If the network configuration represented by a particular linkage set changes, then the transmission order of the linkage set for each ensemble shall be redetermined (see clause 5.2.3).

The CEI is signalled at least once per second for a period of five seconds. The CEI is defined as the short form **service linking** field with the **C/N flag** set to 0 and the database key - the **OE flag**, **P/D flag**, **S/H flag**, **ILS flag**, and **LSN** - defining the database entry corresponding to the linkage set being changed.

The linkage set is then defined according to clause 5.2.4.1.

5.2.4.3 Signalling the activation state of linkage sets

The activation state of all the linkage sets in the database shall be signalled at least every 10 seconds. This shall be accomplished by either repeating the definition of a linkage set at least every 10 seconds, or by signalling the activation state of the linkage set by using the short form of the **service linking** field with the **C/N flag** set to 1.

When network configuration changes occur, additional signalling is provided to alert receivers to the changes. When the state of the **LA flag** of a linkage set changes, a short form FIG 0/6 is sent. If one linkage set is activated and simultaneously another linkage set is deactivated, the short form FIG 0/6 is sent with two **service linking** fields; the first carries the details of the linkage set being deactivated, the second contains the linkage set being activated. These FIGs are sent once per second for five seconds to enable rapid linking and delinking.

5.2.5 Receiver behaviour for a link

5.2.5.0 General

When a receiver initially tunes to a particular ensemble, it shall begin to assemble the service linkage database entry associated with each database key- **OE flag**, **P/D flag**, **S/H flag**, **ILS flag**, **LSN**. Some linkage sets may be activated, others deactivated, as indicated by the **LA flag** for each database key. FIG 0/6 in the long form carries the lists of identifiers.

During this phase, it is necessary to wait for the FIG 0/6 in the long form with the **C/N flag** set to 0 indicating start of database in order to determine the SId(s) in the tuned ensemble to which the linkage set relates, although linkage sets may be built up whenever they are received. The receiver will never know that the database is complete, and a receiver should act upon the information in the database whenever it needs to. There is a limit of 128 Ids (SIds, PI codes, etc.) in each linkage set (**OE, P/D, S/H, ILS, LSN**).

Receivers can only receive the CEI associated with a particular database key if they are tuned to the ensemble at the time of the change. Therefore, receivers cannot assume that the database entry associated with a particular database key is unchanged from when the receiver was last tuned to the same ensemble and shall reacquire the information.

5.2.5.1 Reaction to CEI

When the CEI is received, that is a FIG 0/6 in short form with the **C/N flag** set to 0, receivers are informed that the part of the linkage database corresponding to the database key - **OE, P/D, S/H, ILS, LSN** - is about to be changed or deleted. The database entry corresponding to the database key is deleted (if present) and the receiver makes whatever preparation is needed to begin to build a new database entry for the indicated database key.

NOTE: Since network configurations change infrequently, the reception of CEI is likely to be unusual.

5.2.5.2 Reaction to definition

Receivers shall begin to assemble the information when they receive a FIG 0/6 in the long form with the **C/N flag** set to 0, indicating the start of the database entry corresponding to the database key - **OE, P/D, S/H, ILS, LSN**. When **P/D** = 0 and **ILS** = 0, the **Ensemble ECC** provided in FIG 0/9 shall apply to all services in the **Id List**. The receiver shall watch for FIG 0/6 in the long form with the **C/N flag** set to 1, indicating a continuation of the database entry, and aggregate the identifiers according to the bearer information indicated by the **IdLQ**.

NOTE: Care should be taken to recall that when a FIG 0/6 is received where the **P/D flag** is 0, the **OE flag** is 0 and the **C/N flag** is 0 that regardless of the **IdLQ** field, the first identifier is a DAB SId, but that when the **C/N flag** is 1 that all identifiers correspond to the **IdLQ** field.

The transmission order of the identifiers for a linkage set is consistent for a given ensemble, but can vary between ensembles. However, due to reception conditions, some FIGs may be lost and so some **service linking** fields may appear out of sequence.

5.2.5.3 Reaction to activation and deactivation

Reception of the short form of FIG 0/6 with the **C/N flag** set to 1 provides information on the activation and deactivation of defined linkage sets. The activation state of all linkage sets is transmitted every 10 seconds. When changes to the activation state take place, the activation state of the affected linkage sets is sent more frequently.

The flags **OE, P/D, S/H, ILS** and the **LSN** define which database entry is being activated or deactivated and the **LA flag** indicates the activation state - **LA** = 0 means deactivated and **LA** = 1 means activated. The receiver determines the activation state of a linkage set by the last received **LA flag**. It is the responsibility of the service provider to ensure the **LA flag** always correctly indicates the activation state.

5.3 OE Services

5.3.0 General

The FIG 0/24 consists of a header field and one or more **OE Services** fields.

The **OE Services** fields can take the short form or the long form and this is determined by the state of the **Number of EIds** field.

The database key for FIG 0/24 consists of the **OE** and **P/D flags** from the header field and the **SId** field from the **OE Services** field.

FIG 0/24 provides supporting information for service following and OE alarm announcements.

5.3.1 Short form - Change event indication

The short form has the **Number of EIds** field equal to 0. It is used for signalling changes to the database entry of the service indicated by the database key. The **C/N flag** is set to 0.

5.3.2 Long form - Database definition

The long form has the **Number of EIds** field not equal to 0. It is used to build up the database of EIds that also carry the service indicated in the **SId** field. The **C/N flag** is used in the database definition mode and the flag indicates the start of the database definition when set to 0 or a continuation of the database definition when set to 1. Every database entry requires an **OE Services** field with a start of database indicator; depending on the number of **EId**s there may be one or more **OE Services** fields with a continuation of database indicator. The order of transmission of the **EId**s shall be kept constant so that receivers may build confidence in the information they store.

5.3.3 OE Services fields

The **P/D flag** determines the length of the **SId** field that corresponds with a service in the tuned ensemble when the **OE flag** = 0, or in another ensemble when the **OE flag** = 1. It is most probable that the information will change very infrequently with a network configuration change when a service is added to or withdrawn from an ensemble.

NOTE: The ECC is not signalled by FIG 0/24, neither for the SId nor for the EId(s). It cannot be assumed that the ECC is identical for all identifiers.

The database key for FIG 0/24 consists of the **OE** and **P/D flags** from the header field and the **SId** field.

5.3.4 Signalling

5.3.4.0 General

There are different phases to signalling. The database shall be defined (using the long form) and may be changed by signalling the CEI (short form) corresponding to the database entry and then redefined using the long form. Database changes should be kept to a minimum. Listeners may tune to an ensemble at any time and therefore the database definition shall be repeated continually.

5.3.4.1 Defining the database

The database is defined using the long form **OE Services** field. For each database entry there is a start of database FIG 0/24 (**C/N** = 0) containing an **OE Services** field, and, if all **EId**s do not fit into the FIG, one or more continuation of database FIG 0/24 (**C/N** = 1) containing an **OE service** field corresponding to the same database key. The complete database entry for each key should be transmitted within 10 seconds.

Once defined, the **EId**s for each database entry shall be transmitted in the same order on each repetition, and shall be transmitted at least every two minutes to maintain the database and allow newly tuned receivers to acquire the information.

5.3.4.2 Changing the database

When a database entry needs to be changed, the CEI is signalled at least once per second for a period of five seconds. CEI is defined as the short form **OE Services** field with the **C/N flag** set to 0 and the database key - the **OE flag**, **P/D flag**, and **SId** - defining which part of the database is being changed.

The database entry is then defined according to clause 5.3.4.1.

5.3.5 Receiver behaviour

5.3.5.0 General

When a receiver initially tunes to a particular ensemble, it shall begin to assemble the OE Services database entry associated with each database key- **OE flag**, **P/D flag**, **SId**. The receiver will never know that the database is complete, and a receiver should act upon the information in the database whenever it needs to.

Receivers can only receive the CEI associated with a particular database key if they are tuned to the ensemble at the time of the change. Therefore, receivers cannot assume that the database entry associated with a particular database key is unchanged from when the receiver was last tuned to the same ensemble and shall reacquire the information.

5.3.5.1 Reaction to CEI

When the CEI is received, that is a FIG 0/24 in short form with the **C/N flag** set to 0, receivers are informed that the part of the OE Services database corresponding to the database key - **OE, P/D, SId** - is about to be changed or deleted. The database entry corresponding to the database key is deleted (if present) and the receiver makes whatever preparation is needed to begin to build a new database entry in the database for the indicated database key.

NOTE: Since network configurations change infrequently, the reception of CEI is likely to be unusual.

5.3.5.2 Reaction to definition

Receivers shall begin to assemble the information when they receive a FIG 0/24 in the long form with the **C/N flag** set to 0, indicating the start of the database entry corresponding to the database key - **OE, P/D, SId**. The receiver shall watch for FIG 0/24 in the long form with the **C/N flag** set to 1, indicating a continuation of the database entry, and aggregate the **EId**s.

5.4 Frequency information

5.4.0 General

The FIG 0/21 consists of a header field and one or more **Frequency Information** fields.

The **Frequency Information** fields can take the short form or the long form and this is determined by the state of the **Length of Freq list** field.

The database key for FIG 0/21 consists of the **OE** and **P/D flags** from the header field and the **Rfa**, **Id** field and **R&M** field from the **Frequency information** field. The **P/D flag** is always 0 for FIG 0/21.

FIG 0/21 provides supporting information for service following and OE announcements.

5.4.1 Short form - Change event indication

The short form has the **Length of Freq list** field equal to 0. It is used for signalling changes to the database content of the frequencies indicated by the **Rfa**, **Id** field and **R&M** field. The **C/N flag** is set to 0.

5.4.2 Long form - Database definition

The long form has the **Length of Freq list** field not equal to 0. It is used to build up the database of frequency information about DAB ensembles, FM-RDS or other services. The **C/N flag** is used in the database definition mode and the flag indicates the start of the database definition when set to 0 or a continuation of the database definition when set to 1. Every database definition requires a **Frequency information** field with a start of database indicator; depending on the number of frequencies there may be one or more **Frequency information** fields with a continuation of database indicator. The order of transmission of the frequencies shall be kept constant so that receivers may build confidence in the information they store.

5.4.3 Frequency information fields

For DAB ensembles, the **OE flag** is used to indicate whether the information applies to the tuned ensemble (**OE flag** = 0) or to another ensemble (**OE flag** = 1). For other bearers (FM with RDS, AM with AMSS, DRM), the **OE flag** is used to indicate whether the information applies to a service (primary service component) in the tuned ensemble that is carried on another bearer (**OE flag** = 0) or to any other service (**OE flag** = 1). It is most probable that the information will change very infrequently with a network configuration change when a service is added to or withdrawn from an ensemble, or introduced or withdrawn on another bearer.

The **continuity flag** shall be set correctly in order to assist receivers with making switching decisions. For DAB frequencies, the **continuity flag** indicates whether continuous (i.e. uninterrupted) audio output is possible or not when switching frequencies; for other bearers it indicates whether an appropriate time delay has been added to allow seamless, (or near seamless) audio output or not when switching frequencies. For DAB, the **continuity flag** should only be set when a DAB ensemble is transmitted on several frequencies with co-timed and synchronized signals.

Synchronized means that the frame start (null symbol) is sent at precisely the same moment in time, the interleaving is identical and the guard interval is respected on all frequencies. See clause A.1 for a use case of an MFN which uses the **continuity flag**.

The database key for FIG 0/21 consists of the **OE** and **P/D flags** from the header field and the **Rfa**, **Id** field and **R&M** field.

5.4.4 Signalling

5.4.4.0 General

There are different phases to signalling. The database shall be defined (using the long form) and maybe changed by signalling the CEI (short form) corresponding to the database entry and then redefined using the long form. Database changes should be kept to a minimum. Listeners may tune to an ensemble at any time and therefore the database definition shall be repeated continually.

5.4.4.1 Defining the database

The database is defined using the long form **Frequency information** field. For each database entry there is a start of database FIG 0/21 (**C/N** = 0) containing a **Frequency information** field, and, if all frequencies do not fit into the FIG, one or more continuation of database FIG 0/21 (**C/N** = 1) containing a **Frequency information** field corresponding to the same database key. The complete database entry for each key should be transmitted within 10 seconds.

Once defined, the frequencies for each database entry shall be transmitted in the same order on each repetition, and shall be transmitted at least every two minutes to maintain the database and allow newly tuned receivers to acquire the information.

5.4.4.2 Changing the database

When a database entry needs to be changed, the CEI is signalled at least once per second for a period of five seconds. CEI is defined as the short form **Frequency information** field with the **C/N flag** set to 0 and the database key - the **OE flag**, **P/D flag**, and **SId** - defining which part of the database is being changed.

The database entry is then defined according to clause 5.4.4.1.

5.4.5 Receiver behaviour

5.4.5.0 General

When a receiver initially tunes to a particular ensemble, it shall begin to assemble the Frequency information database entry associated with each database key- **OE flag**, **P/D flag**, **Rfa**, **Id** field and **R&M** field. The receiver will never know that the database is complete, and a receiver should act upon the information in the database whenever it needs to.

Receivers can only receive the CEI associated with a particular database key if they are tuned to the ensemble at the time of the change. Therefore, receivers cannot assume that the database entry associated with a particular database key is unchanged from when the receiver was last tuned to the same ensemble and shall reacquire the information.

5.4.5.1 Reaction to CEI

When the CEI is received, that is a FIG 0/21 in short form with the **C/N flag** set to 0, receivers are informed that the part of the Frequency information database corresponding to the database key - **OE, P/D, Rfa, Id** field and **R&M** field - is about to be changed or deleted. The database entry corresponding to the database key is deleted (if present) and the receiver makes whatever preparation is needed to begin to build a new database entry for the indicated database key.

NOTE: Since network configurations change infrequently, the reception of CEI is likely to be unusual.

5.4.5.2 Reaction to definition

Receivers shall begin to assemble the information when they receive a FIG 0/21 in the long form with the **C/N flag** set to 0, indicating the start of the database entry corresponding to the database key - **OE, P/D, Rfa, Id** field and **R&M**

field. The receiver shall watch for FIG 0/21 in the long form with the **C/N flag** set to 1, indicating a continuation of the database entry, and aggregate the information.

5.5 Transmission network examples

5.5.1 DAB/DAB service following

There is no such thing as a "standard" DAB network. Depending on very many variables, networks will be built that provide the radio services that are needed. Some examples are listed below. In all cases, it is essential that proper allocation of identifiers is made such that there are no conflicts of information within the country.

A DAB service may be carried in an ensemble which is transmitted on different frequencies. In this case, service providers may assist receivers by providing the Frequency information for the ensemble using FIG 0/21. Clause A.1 illustrates this use case.

Another DAB service may be carried on different ensembles. In this case, service providers may assist receivers by providing the EIds of the other ensembles using FIG 0/24 and may provide the Frequency Information for each ensemble using FIG 0/21. Clause A.2 illustrates this use case. Clause A.12 illustrates a similar service following case crossing an international border (see also clause A.11).

Another DAB service may have programme splits according to geography available on different ensembles. At some times of the day, all the ensembles carry the same audio, but at other times, the service splits into regional variants. In this case service linking information is provided with FIG 0/6 to define the network configurations and activate and deactivate different combinations. The ensembles on which each service variation is carried is provided with FIG 0/24 and the frequencies of the ensembles with FIG 0/21. Clause A.3 illustrates this case. A similar situation may occur where both national and regional or local variants are carried on different ensembles with overlapping coverage areas. In this case the broadcaster can indicate that some services are less preferred than others as alternatives when service following. Clause A.13 illustrates this case.

Another scenario is that a service is provided in both DAB and DAB+ in the same ensemble and in DAB+ only on another ensemble. The service provider can assist receivers by providing service linking information that connects the SIds, ensemble information and frequency information as above. Clause A.4 illustrates this use case.

A service provider may have several related radio services which sometimes carry the same content but sometimes carry different content. The service provider would like to help his regular listener in one location to find a similar service when in another location, and for this he provides service linking information with soft links using FIG 0/6. He may provide additional information to help receivers tune to the related services. When the listener presses a preset to select his usual service, the receiver may use the soft linking information to find a suitable alternative when out of area. Clause A.5 illustrates this case.

5.5.2 DAB/FM-RDS service following

In order to allow receivers to continue to provide a service carried on DAB when moving beyond the digital coverage area, service providers with the same content available from FM-RDS can provide service following information. In general, FIG 0/6 is used to provide the linkage between DAB services and the corresponding FM-RDS services; this is illustrated in clause A.6. In the special case that identical content is carried on DAB and FM-RDS and the SId and PI code are identical, then a service provider can choose not to signal any FM-RDS service linking information, relying instead on implicit linking. In this case no service linking information is signalled and the receiver may switch between the DAB and FM-RDS sources as determined by algorithms in the receiver. Implicit linking cannot cope with regionalization using PI code switching - an example is illustrated in clause A.7, nor the case where the content on DAB and FM-RDS is not identical at certain times of day - this is illustrated in clause A.8. For these cases, FIG 0/6 is used to define a hard linkage set that overrides implicit linking when activated.

Because the coverage of DAB and FM transmissions may be different, and service providers may localize or regionalize their content on one or both bearers, the relationship between services carrying identical audio content may vary during the day. Each possible combination of DAB SId and RDS PI codes is expressed by a linkage set and as the day progresses and the different combinations become valid, the LA of each linkage set is either activated or deactivated. In this way, the database entries remain constant, but the active alternative services are dynamically controlled. There are very many possible combinations.

There may be situations where allocation of SIds and PI codes has not been coordinated and service providers with no FM-RDS services may find receivers that implement implicit linkage switch listeners from DAB to FM-RDS but the two services have different content. To prevent this condition, service providers may signal an FM-RDS link without any PI codes - this is an example of a "dead link" and is illustrated in clause A.9, which shows linked DAB services without links to the corresponding FM-RDS services.

5.5.3 DAB/DRM service following

In order to allow receivers to continue to provide a service carried on DAB when moving beyond the DAB coverage area, service providers with the same content available from DRM can provide service following information. The services may be data services, as illustrated in clause A.10, or audio services as illustrated in clause A.11. In both cases, care is needed to provide the service identifiers correctly.

5.6 Receiver reference model

5.6.0 General

Clause 5.6 describes a conceptual model of a receiver that implements service following according to these Rules of Implementation.

The purpose of the model is not to specify a particular receiver implementation, but rather to define the expected behaviour of a conformant receiver. It is expected that real receivers may differ from the reference model in their detailed design and implementation, but that they will exhibit the same user-visible behaviour as the model.

A receiver designed for mobile reception may implement a service following procedure to maintain listening during a journey despite the changing broadcast signal environment. Conceptually the receiver consists of a control circuit that monitors the reception quality and selects another service using the information base to recover from a degradation or loss of signal. The service following procedure shall be designed to provide the best possible user experience requiring least user intervention along a given travel route.

A receiver designed for either mobile reception or domestic reception may implement mechanisms in order to provide the listener with the best service selection response, even when located away from the "home" location, by using stored information. The selection procedure shall be designed to provide the best possible user experience requiring least user intervention when selecting a service that is no longer available.

5.6.1 Reception monitoring

A mobile receiver shall provide a means to monitor the received signal. This may include one or more aspects of signal reception, such as signal strength, carrier-to-noise ratio, bit-error rates, etc. For each quality indicator a threshold level allows detection of a loss of signal. The decision circuit may include hysteresis for the ON/OFF decision making. Care should be taken to ensure that the effective sensitivity of the receiver is not reduced by the service following process.

5.6.2 Information base

To identify a suitable alternative service or frequency resource, a receiver can make use of various information sources. These may include service following information received through the FIC, learning memory and pre-stored data to build the information base.

The information base consists of information elements combining frequency information, extended country code, service identifiers and ensemble identifiers to unambiguously identify a broadcast signal. Since LSNs are allocated on a national basis, it is necessary to store the extended country code and the country code from the ensemble identifier along with the LSNs in order to eliminate ambiguity in the information base in areas where ensembles from more than one country are receivable.

The present document discusses solely information provided via the FIC forming the service following information. This may take some time to acquire and thus receivers benefit from storing in memory relevant service following information - this is especially true for the selection procedure. It is assumed that the service following information represents the dominant part of the information base.

5.6.3 Service following procedure

5.6.3.1 Overview

The service following procedure deploys a multi-stage algorithm to keep track of a selected service and signal and where this fails, to identify and select an alternative signal and/or service. The service following procedure is an automated routine controlling tuning and decoding units to maintain and (re-)establish a minimal service reception.

The implementation of the service following procedure will vary according to the hardware resources available. Additional tuners and signal processing chains permit alternatives to be monitored in parallel with monitoring the signal quality of the tuned service, and in this case, the service following procedure may run continuously. If the selected service is indicated via an active hard link to have a low preference level, checks can be made to determine if a service with a normal preference level is available in sufficient quality to switch to. For single tuner implementations, the service following procedure begins with a loss of signal.

The service following procedure starts to test each element in the information base for signal and service availability. The service following procedure continues until service reception is re-established or a user intervention occurs. The procedure should prioritize a DAB service over other bearers such that DAB is always selected when available in sufficient quality.

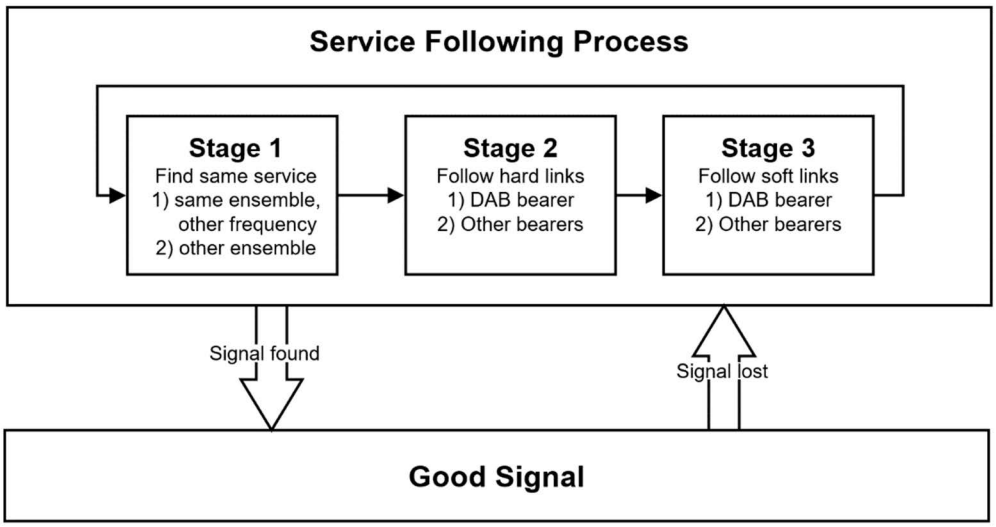


Figure 2: Service following procedure flow diagram

The full service following procedure includes three stages, and each stage is executed if the previous stage fails.

Stage 1: tries to recover the selected service 'as selected', testing only information elements that identify 'exactly the same' service at another frequency. The receiver should only retune to decode ensembles carrying the same service (i.e. same ECC-SId) as the user selected service. This should prioritize frequencies that transmit the same ensemble as the user selected.

Stage 2: searches for hard linked services as given in the information base. The receiver tries to identify services that are hard-linked with the selected service on either the DAB bearer or other bearers. Ideally, suitable candidate services will already have been identified and monitored to determine their suitability. This will allow a change of service, but will present the same audio content.

Stage 3: searches for soft linked services as given in the information base. Soft linked services are related to the user selected service, but do not carry the same audio content; consequently the decision to select may require user intervention. As for Stage 2, DAB alternatives should be prioritized and other bearers should be evaluated secondly.

The service following procedure has a cyclic design to account for the receiver moving randomly within the signal coverage, where no assumptions on signal availability can be made. For single-tuner solutions, the procedure always starts at stage 1, when the user selected service is no longer available. When parallel monitoring of alternatives is possible, and the selected service is indicated via an active hard link to have a low preference level, the receiver jumps to stage 2 to determine if there is an alternative service with a normal preference level available of suitable quality and switches to it if that is the case; otherwise the procedure starts at stage 1 when the user selected service has reached a certain quality level.

Subsequently, the receiver will try to find a signal that contains the user selected service or, if this fails, a suitable alternative according to the service following information. The procedure ends if a suitable service can be presented to the user or the user selects another service. If in the course of this procedure the receiver has tuned away from a DAB service, it is expected to return to a DAB service independently of the signal quality on other bearers.

The DAB service identifier (SId) has national scope and is globally unique only when associated with the Extended Country Code (ECC). Therefore, when identifying services, the combination of ECC and SId shall be used. Only services that match in both the ECC and the SId shall be considered correctly identified, either as the same service or as a linked service. The same principle applies to the ensemble identifier, it has national scope and is only globally unique when associated with the ensemble ECC. Therefore when matching ensembles, both the EId and ECC shall match.

Typically, all ensembles and services received at a given location will have the same ECC, so that only the SId is differing between any two services. However, when services from different countries are received, either when the receiver is crossing borders or when an ensemble carries services from different countries, more than one ECC may be relevant and the signalled ECCs shall be taken into account.

For programme service linkage sets with the ILS flag = 1, both the extended country code and the DAB or RDS service identifier are carried in the FIG 0/6 Id list. The receiver shall ensure that the ECC is stored with the SId.

The Target Id in the service following process is the combination of the ECC and SId of the user selected service. In a linkage situation, a Candidate Id is the combination of the Ensemble ECC and the SId or PI code provided by the linkage set for National links (ILS flag = 0), or the ECC + SId or ECC + PI code or DRM or AMSS Id provided by the linkage set for international linkage sets (ILS = 1).

When linking to FM-RDS, although the Candidate Id will include the ECC, it is generally not possible to verify the ECC because most RDS services do not signal it. Therefore receivers shall make suitable provisions to ensure correct service matching in implicit or signalled RDS linkage situations.

5.6.3.2 Stage 1: Find same service

In this stage, the receiver attempts to locate a DAB ensemble containing a service with ECC-SId equal to the Target Id.

It may use FI (FIG 0/21) to find the same ensemble on another frequency; OE Services information (FIG 0/24) and FI to find another ensemble carrying the same service; and cached MCI and SI from previously-tuned ensembles to determine a list of candidate ensembles. From that list the receiver attempts to identify an ensemble that carries the Target Id.

The receiver may use any available quality information and any criteria it chooses to filter or determine the priority order of candidates. For example, different receiver implementations may make different trade-offs between maximizing the initial reception quality and minimizing the initial selection time. Similarly, different receiver implementations may have different resources available (e.g. number of tuners, processing power, storage) that influence such trade-offs.

Note that information from previously-tuned ensembles may be incomplete or out-of-date and that previously-tuned ensembles may or may not be receivable at the current location. Different receiver implementations may use different strategies for determining how much information to store and when to refresh it. For example, a single-tuner receiver performing a service following decision may make a first quick pass using cached information and then perform another slower pass during which all information is refreshed, whereas a dual-tuner receiver with sufficient storage may cache all information and operate continuously in the background. Similar considerations apply to the following stages, with the difference that the Candidate Id represents a different service that has a linkage relationship to the selected service.

5.6.3.3 Stage 2: Follow hard links

At this stage, the receiver has failed to select the Target Id on any DAB ensemble, or is seeking an alternate service because the selected service has a low preference level. The receiver attempts to find a service that is hard-linked to the Target Id and is therefore carrying identical audio content to the Target Id.

Stage 2.1

The receiver uses linkage information received for the selected service to assemble a list of Candidate Ids from the activated linkage sets. From that list the receiver attempts to identify ensembles and frequencies that carry a linked DAB service using its information base. DAB services with a normal preference level will be evaluated before DAB services with a low preference level, but all DAB alternatives will be evaluated before alternatives on FM-RDS or other bearers.

The receiver may use any available quality information and any criteria it chooses to filter or determine the priority order of Candidate Ids. Finally it will select an available service according to that order.

Stage 2.2

The receiver uses linkage information received for the selected service to assemble a list of Candidate Ids from the activated linkage sets on FM-RDS or other bearers that are hard-linked to the Target SId. If no hard linkage set with **IdLQ** = FM-RDS is activated then the PI code matching the SId part of the Target Id is a candidate (implicit linking).

Then the receiver uses its information base to determine a list of candidate frequencies for each candidate linked service.

From that list, the receiver attempts to select each candidate service on each candidate frequency in turn.

The receiver may use any available quality information and any criteria it chooses to filter or determine the priority order of candidates. Finally it will select an available service according to that order.

5.6.3.4 Stage 3: Follow soft links

At this stage, the receiver has failed to select the Target Id on any DAB ensemble and has also failed to select an equivalent hard-linked service on DAB or other bearers. This means a service with the same audio content is not available.

In this situation, a related service may represent the best alternative according to the user's expectations or preferences. A related service may be a regional variant of the selected service, it may be a language variant, or another service from the same broadcaster.

Consideration should be given when designing algorithms for following soft links as to whether the listener will understand why the audio has changed and whether the receiver will be able to stay with the related service for a reasonable period of time.

Single tuner receivers are unable to evaluate alternative services until signal loss has occurred and it is recommended that such receivers do not include stage 3 in service following procedures (although stage 3 should be included in service selection procedures).

Stages 3.1 and 3.2 are identical to stages 2.1 and 2.2 respectively, with the exception that stage 3 uses soft-linked services.

5.6.3.5 Handling of bearer system transitions

The service following procedure is designed to present the best available alternative to the user selected service. In the course of this procedure the alternative service may represent a significant departure from the initial user selection. The procedure shall be designed in a way that the receiver respects the initially user selected service at all times.

In the case of a bearer transition, the receiver has moved away from the DAB system and is receiving a service on another bearer system, e.g. FM-RDS. If service following rules on the other bearer system are inconsistent with the rules applicable to DAB, it is left to the discretion of the implementer how conflicts or ambiguities are best handled in the interest of the user.

For example, on FM-RDS a service may be regionalized. The user selected DAB service may only be linked to a particular regional variant of that service. Whilst tuned to FM-RDS, a receiver may select another regional variant as part of the RDS service following procedure. As regionalization does not exist on DAB, returning to DAB on the other regional variant would represent a change of service and could conflict with the user expectation.

A similar situation may occur when linkage information is temporally dynamic. After having followed a link, the receiver should ensure the validity of the linkage information.

In general, a user will expect that the audio content does not suddenly change to a different programme. If the tuned service has changed via a series of service following steps and no longer satisfies the user, he has the option to select another service.

5.6.4 Selection procedure

If the user stores a service as a preset on the receiver (either mobile or domestic), and the service is not found when the user subsequently selects the preset, the receiver shall attempt all service discovery and linking methods to locate that service, or an alternative service advertising the preset service as a link (either hard or soft). Similarly, if the user selects a service from the service list which is not currently available, the receiver shall also attempt to locate the service, or an alternative service advertising the chosen service as a link (either hard or soft).

Since relevant linkage information may not be available on the currently tuned ensemble, the storage of service following information in memory is advantageous in providing the receiver with a set of candidate alternatives to evaluate. In general, the current state of the Linkage Actuator will not be known because the information was gathered at a previous time.

The same three stage procedure shall be used as in service following (see clause 5.6.3), although all linkage sets shall be searched, both activated and deactivated, and all receiver types should implement stage 3 to follow soft links. The **IdLP** flag shall be used to determine the order of services to be evaluated: services with a normal preference level shall be evaluated before services with a low preference level.

EXAMPLE: A Service is stored on a preset with SId = 0xC586, EId = 0xC181, ECC = 0xE0, F = 227 360 kHz.

The user moves to a different location, and presses the preset button.

The receiver attempts to locate ensemble EId = 0xC181, ECC = 0xE0 at F = 227 360 kHz and fails.

Stage 1: The receiver attempts to discover EId = 0xC181on alternative frequencies and then other ensembles carrying SId = 0xC586, ECC = 0xE0 from information stored in the information base. If that fails, the receiver examines hard linkage sets (either acquired or stored) to locate SId = 0xC586, ECC = 0xE0.

Stage 2: The receiver finds a hard linkage set with services ECC = 0xE0, SId = 0xC386, 0xC486, 0xC586. All services have a normal preference level. As 0xC586 exists in this list, any of the other services in this list (0xC386, 0xC486 with ECC = 0xE0) are possible alternatives that will provide the same audio if this linkage set is active at the moment of selection. If that fails, the receiver should examine soft linkage sets (either acquired or stored) to locate SId = 0xC586, ECC = 0xE0.

Stage 3: The receiver finds a soft linkage set with services ECC = 0xE0, SId = 0xC386, 0xC486, 0xC586, 0xC686, 0xC786. All services have a normal preference level. As 0xC586 exists in this list, any of these services are possible alternatives, although only providing related content.

The receiver is able to tune to SId = 0xC686, EId = 0xC194, ECC = 0xE0 at F = 218 640 kHz and therefore provides this service as an acceptable alternative. (This behaviour emulates recalling an FM preset when REG PI code following is enabled.)

5.6.5 Linkage sets

The receiver maintains a number of linkage sets as part of an information base used for service following decisions.

Each linkage set contains the identifiers for a set of one or more services, possibly on different bearers, that are equivalent (hard linked) or related (soft linked).

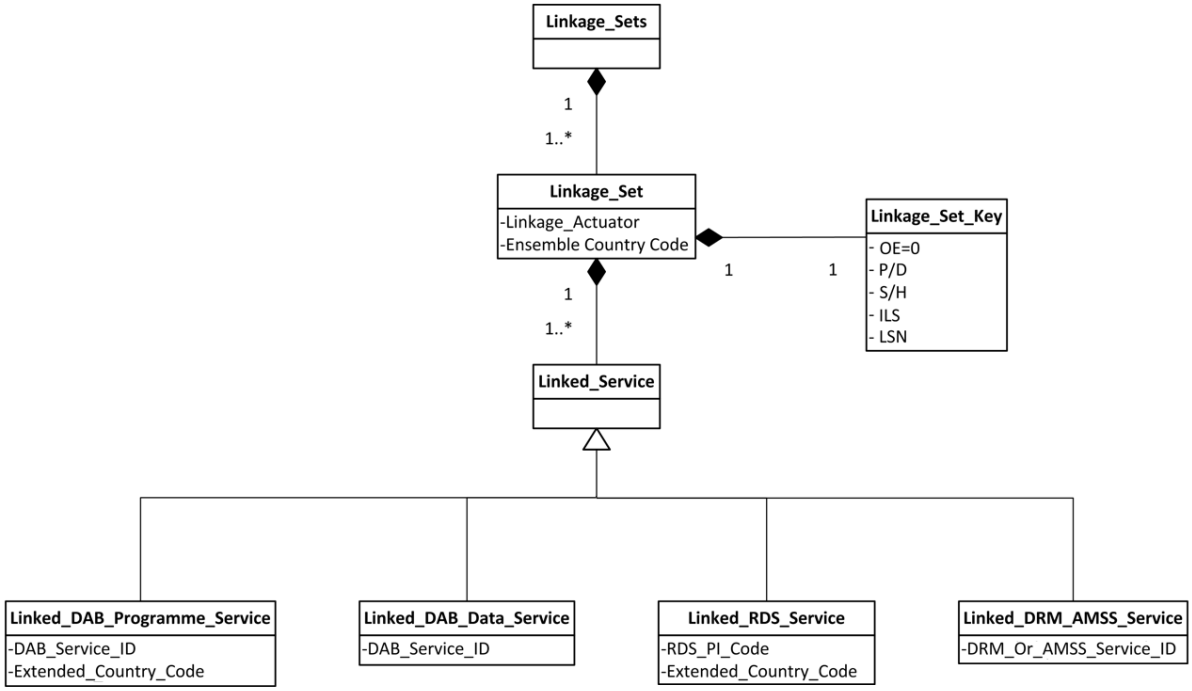


Figure 3: Structure of linkage sets

Each linkage set is uniquely identified by a database key comprising the **P/D flag**, **OE flag**, **S/H flag**, **ILS flag**, and **LSN**.

Each linkage set contains one or more identifiers for services which carry the same (in the case of hard links) or related (in the case of soft links) content.

Each id list entry in a Linkage Set, except for the first one at the start of the database, may identify either a DAB Programme service, a DAB Data service, an FM-RDS service, a DRM service or an AMSS service.

When a Linkage Set has ILS flag = 0 and P/D flag = 0, all the DAB and/or FM-RDS services listed have the Ensemble ECC, which is provided in FIG 0/9. When a Linkage Set has ILS flag = 1, all services are listed with full identification - DAB and FM-RDS services are signalled with the service ECC.

Of all the linkage sets defined containing a particular service, only one hard linkage set (either National or International) and one soft linkage set (either National or International) may be activated at a given time; all other linkage sets defined containing that particular service shall be deactivated.

5.6.6 Linkage set management

The receiver obtains the linkage sets from information supplied by service providers and signalled in FIG 0/6.

Figure 4 defines the process by which a receiver may obtain and manage all of the linkage sets transmitted on the tuned ensemble.



Figure 4: FIG 0/6 process tree

Receiver implementations may differ in the number of linkage sets they can store at any one time. As a minimum, a basic receiver shall maintain the currently active linkage set for the currently selected audio programme service. A more advanced receiver with additional storage capacity may maintain additional linkage sets - for example those associated with services stored on presets or otherwise determined to be of interest to the user.

A receiver implementation need only store the identifiers for bearers that it is capable of receiving and intends to use for service following. For example, a receiver that does not support service following from DAB to DRM need not store any DRM service identifiers in the linkage sets.

6 Service lists

6.1 Introduction

The efficient maintenance of the list of service elements that a listener may choose is an important aspect of the satisfaction that a user gains from using a digital radio. It is important that the options available to the listener are up to date, following the changing broadcast schedule, the roll-out of the transmission networks and the current location of the receiver. However, maintenance of the service list is quite a complex task and proper signalling by broadcasters and proper decoding and behaviour by receivers is required to make this happen well.

Clause 6 describes in detail the correct signalling and behaviours to enable efficient maintenance of service lists in receivers.

Managing the list of service elements comprises several requirements: automatically adding new service elements when they appear; removing those no longer available; following changes to the Service Identifier, changes to labels or the ensemble on which a service is carried; dealing correctly with part-time or "pop-up" service elements.

6.2 Managing ensemble changes

6.2.1 Describing an ensemble

A DAB ensemble carries a number of services, each of which has one (and only one) primary service component and may have one or more secondary service components. The current arrangement of the services carried in the ensemble is described by FIG 0/2 with the **C/N flag** set to 0 (Current). FIG 0/8 provides additional information to correctly identify secondary service components.

When changes are to be made to these services at a reconfiguration, the next arrangement of the services is described by FIG 0/2 and FIG 0/8 with the **C/N flag** set to 1 (Next) in the period of six seconds before the change actually occurs (both current and next are signalled during this period). The precise timing of the reconfiguration of the ensemble is signalled using FIG 0/0.

The service labels and service component labels are the mechanism by which users select the content they wish to consume. The labels are signalled using FIG type 1 and/or FIG type 2. It is essential that these labels are signalled and decoded. FIG x/1 shall be used for all audio services and FIG x/5 shall be used for all data services that may be user selected. FIG x/4 shall be used for all secondary service components that may be user selected.

In order to help devices to properly maintain the service list, broadcasters need to correctly signal the current service organization and all reconfigurations. In addition to the MCI and SI, the SCI, carried in FIG 0/20, shall be used to signal all changes to service elements in advance. In this way, the addition and removal of service elements, both continuous and part-time, the change to the SId of a service, or the transfer of a service to another ensemble, can be properly managed in receivers, including update of internal memory and presets.

In the subsequent clauses 6.3 to 6.5, the required signalling and receiver behaviours are described.

6.2.2 What is the service list

The service list is a feature of a digital radio receiver and the primary means for accessing, viewing and selecting all available services. An available service can be received at the place of the receiver with clear undisturbed audio. As a minimum the service list shall show all available service elements and provide a means to select any service element for consumption. Selecting a service element results in the tuner accessing the service element's ensemble and presenting the associated service element's content.

The service list consists of service list entries. Each service list entry represents one service element. A service list entry consists at least of a label associated with the service element; it should display the full length label with 16 characters. Each service list entry is associated with the service element identity (**SId**, **SCIdS**) and relevant tuning information, e.g. **EId**(s) and frequency. Additionally service list entries may show further information such as ensemble label, audio quality or programme type.

The service list shall show all service elements for which full MCI and labels has been received. It shall additionally show all service elements that are signalled as part-time, even when off-air. The service list shall not show service elements that the receiver cannot decode or present.

A receiver may provide optional means to mark a service list entry as "tentative", "end-of-life" or "part-time". Such marking may be shown in the service list to help the user understand the dynamic nature and current status of service schedules:

- **Tentative marking:** a service element that will commence operation in the future is signalled in the SCI as to be added (**Change flags** set to 01). A receiver may show the service element in the service list based on this information if it can mark it suitably to indicate the service is not available at present but will commence service in the future. The receiver should show the provided date-time to inform the user when the service element will start. A receiver shall not show a service element before it has commenced operation in the service list unless marked as tentative. When a service list entry marked as tentative is selected by the user, the receiver shall attempt to tune to the service element, and, if not available, update the SCI status and date-time from the on-air signal.

- **End-of-life marking:** a service element about to cease operation in the future is signalled in the SCI (with **Change flags** set to 10 or 11) as to be removed. A receiver may mark the service element as end of life to indicate it will cease operation in the future. A service list entry marked as such should show the date-time of the service element ceasing. When a service list entry marked as end-of-life is selected, the receiver shall update the SCI status and date-time from the on-air signal; if transfer information is given, the alternative service shall be preferred for tuning.

- **Part-time marking:** a service element with a regular, cyclic on-air/off-air schedule is signalled as part-time in the SCI (**Change flags** set to 01 or 11, **Part-time flag** set to 1). It shall be represented by a service list entry at all times, even during off-air periods. A receiver should mark the service list entry as to indicate the current on-air status, it should also show the date-time (if available) of the next change of status. When a service element marked as part-time is selected by the user, the receiver shall attempt to tune it, during the off-air period the receiver shall update the SCI information and date-time from the on-air signal.

It is recommended that a receiver supports part-time marking on service list entries.

For the purposes of describing service reconfigurations, a distinction shall be made between the service transport mechanism on a digital radio system and its audio content. This can be referred to as a container-content relationship. Throughout clause 6 the terms 'service' or 'service element' refer to the container, the term 'programme' refers to the content. In many cases 'service' and 'programme' can be used interchangeably, as a service element always carries only one programme and each programme is only carried in one service element. However under specific circumstances, different service elements may carry the same programme, e.g. for part-time regional programmes. In the use cases described in clause 6.5 a service may change its identity, while carrying a single, unchanged programme.

6.2.3 Establishment of the service list

When a receiver is switched on for the first time, it has no entries in the service list. Generally, in this condition, the receiver will begin an automatic scan of all the DAB channels in the receiver's tuning range and look for DAB ensembles. As each ensemble is found, the receiver will discover new service elements which it will add to its service list.

In some cases a service element will already have been found on another frequency or another ensemble: in this case the receiver should add the new frequency and/or new ensemble to the stored information as this represents an alternate source of the same service.

Receivers shall limit the service list to those service elements that are supported for user presentation; for example, a receiver that only supports audio service elements shall not include data service elements in the service list.

A receiver shall provide a function to update the service list upon a user action. This function shall perform the same full scan of all DAB channels as in the initial scan. In this process a clean list shall be created of all service elements that can be received at the time of scanning, including part-time service elements in off-air state.

6.2.4 Maintenance of the service list

After the initial establishment of the service list, maintenance processes are required to ensure the user is able to select all available services and is not confused by services that are no longer broadcast. As the user tunes to different services, the receiver shall check its service list against the received MCI and SI FIGs and make appropriate updates.

Receivers with multiple tuners may be able to check the status of other ensembles, even when the user is consuming a programme. Otherwise, a rescan of all DAB channels should be performed periodically in order to discover newly launched ensembles. It is recommended that receivers with single tuners should also check known ensembles and perform a rescan of all DAB channels on an occasional basis at a time that the receiver is not in use.

A receiver can only discover a change to the configuration of an ensemble when it has a tuner that is tuned to the ensemble. This applies to single tuner and multiple tuner receivers alike. Changes to the configuration of an ensemble that happen when the receiver is turned off, out of the coverage area or tuned to another ensemble cannot be discovered, but the SCI provides advance information which shall be decoded and cached for later use.

Changes to the configuration of an ensemble may be discovered in the following ways:

- a tuner detects a reconfiguration of the ensemble by the decoding of FIG 0/0 and the corresponding "next" MCI;
- a tuner changes the tune frequency, either by user action or in an automatic process, or re-selects a previously selected ensemble after a resume from a power cycle or signal loss and discovers changes of configuration by comparing the "current" MCI against the ensemble configuration stored in the device memory.

At these points, referred to as "discovery points", the receiver shall use the MCI and labels, in combination with the SCI, to immediately update its service list information for all services in the ensemble, including those that are not selected. The SCI and associated label provides the necessary information to maintain a part-time service element in the service list even when it is off-air.

The SCI also provides information that allows a receiver to check whether it already has an entry in its service list that corresponds to a service that has changed its identity or moved ensembles. The detailed behaviour required for combining the MCI and the SCI is given in clauses 6.3 to 6.5. If there is no SCI information for a service element then the service list is updated according to the MCI alone.

Generally, SCI is used to provide information on pending ensemble reconfigurations ahead of time. Receivers should make use of advance information for user information, e.g. an upcoming service being added to the service list with a date and time of its coming into operation. SCI can also provide information on ensemble reconfigurations that have occurred already, in some cases this information is essential to a receiver to discover the change after the fact.

In some cases, SCI provides the only means of creating or maintaining a service list entry, e.g. part-time service elements during off-air periods or services that change their identity. In these cases the receiver shall use SCI for maintenance of the service list, of service memory such as presets and for general tuning behaviour. In particular, usage of SCI is required to minimize service interruptions in the case of a service changing its identity or moving to another ensemble.

As SCI provides a date-time that may change at any time and that is only an indication of a time an ensemble reconfiguration will occur, the receiver should apply special care when updating user controls by SCI alone. The event of an ensemble reconfiguration and the updated MCI establishes the facts about the changes in the ensemble. It is only after updated MCI has been received and evaluated that a receiver can make reliable changes to the operational controls.

In order to maintain the cache of SCI events, receivers shall decode the SCI information and store it. Receivers may optimize the storage, e.g. by not storing information for data service elements if the receiver is not able to decode the indicated data application.

Receivers shall limit the service list to those service elements that are supported for user presentation; for example, a receiver that only supports audio service elements shall not include data service elements in the service list. Changes to labels are only required to be transferred to the user controls at one of the discovery points.

6.2.5 Signalling conventions

SCI provides a means to provide information on ensemble changes ahead of time.

For service elements being introduced or off-air temporarily, full MCI and labels is not available. Then SCI is the only means to discover a service element and it shall always be accompanied by an associated label. This will allow receivers to create and maintain a proper service list entry. For service elements being removed or changing source or identity, SCI is provided in parallel with full MCI. In these cases the service component identification in the SCI refers to an existing service element in the same ensemble.

The **Date-time** field in the SCI indicates the future point in time at which the ensemble reconfiguration will occur. The **Date-time** field may also contain the special value to indicate a past or unknown future date-time depending on the value of the **Change flags** field.

Part-time services alternate between an on-air and an off-air state. During the on-air period the service carries full MCI and labels. It is distinguished from a full-time service by the constant presence of an SCI field with the **P-T flag** set to 1. The **Date-time** field and the **Change flags** follow the on-air/off-air cycle respectively. During an off-air period, while MCI is absent, the SCI and label allow receivers to create and maintain service list entries.

In those cases when at least one of the **SId flag** and **EId flag** is set to 1, SCI provides information to the receiver of recommended sources for continuing a service element in the case of the service element being ceased or changing source and/or identity. This information may be essential to a receiver when the same or a linked service is not available. The **Transfer SId** will represent a different programme for ceased services and the same programme when a service changes its identity.

Broadcasters are recommended to provide transfer information whenever possible. It should remain on air after the reconfiguration (with **Date-time** field set to the special value) for a minimum period of time. This is to allow receivers returning to a service after the reconfiguration to take note of the past change and use the transfer information for service list update and continued service presentation.

It is recommended to schedule planned ensemble reconfigurations affecting the service list to a time of reduced listening, e.g. early morning hours, to reduce the chance of service disruptions. This applies in particular to services changing source or identity.

The permissible time-window for the date-time information is 28 days, to allow for an unambiguous interpretation of the date in the receiver. Broadcasters may decide freely how long in advance SCI should be provided. It is recommended to use a minimum of 7 days to allow users to take note of impeding changes in receiver displays and for receivers to store provided information in memory.

NOTE: The limitation of the time window implies that service elements with a regular on-air/off-air schedule that is longer than 28 days need to be signalled as continuous service element additions and removals, leading to repeated service list updates.

The SCI signalling occurs in three phases:

- the **preparation phase** occurs in the period (of up to 28 days) before the reconfiguration is signalled;
- the **reconfiguration phase** occurs during the reconfiguration signalling;
- the **consolidation phase** occurs after the reconfiguration has completed.

6.3 Adding service elements to an ensemble

6.3.1 Introduction

A service element is added to an ensemble that has not carried the service element before. A receiver shall create a new service list entry or maintain an existing service list entry.

When a new service element is initially launched, there will usually be some promotion activity from the broadcaster, which may start some time before the actual introduction of the service element into the ensemble. Broadcasters shall provide advance information by using the SCI. The service introduction itself will require a multiplex reconfiguration to add the new service element and this is signalled appropriately.

Part-time service elements are repeatedly removed from and returned to the ensemble. In order to allow receivers to maintain the entry in the service list for such part-time elements, the SCI and labels shall be provided during off-air periods.

6.3.2 Signalling

6.3.2.1 Preparation phase

For up to 28 days before the introduction of a service element into the ensemble, the broadcaster shall provide advance information by the use of FIG 0/20. The respective service label or service component label shall also be signalled.

When a service with several service components is brought to life, each service component shall be added using a dedicated SCI field.

The fields of the SCI shall be used as follows:

- The **SId** and **SCIdS** fields shall be used to identify the service element.
- The **Change flags** shall be set to 01 to indicate the service element will be added to the ensemble.
- The **P-T flag** shall be set to 0 for the initial introduction of a service element, or 1 for a returning part-time service element.
- The **SC flag** shall be set to 1 since the SC description is required.
- The **SC description** shall be present and the fields set according to the needs of the service component.
- The **Date-time** field shall be used to indicate, to the nearest second in UTC, when the reconfiguration will occur that adds the service element to the ensemble; after the reconfiguration the **Date-time** field shall assume the special value to indicate the reconfiguration has occurred.
- The **SId flag** and **EId flag** shall be set to 0; **Transfer SId** and **Transfer EId** fields shall be absent.

The service label or service component label shall be signalled as appropriate. For data service elements, FIG 0/13 shall be signalled.

Part-time services

For a returning part-time service, when the **P-T flag** is set to 1, the **Date-time** field should be set to the date-time of the next reconfiguration when the service element commences service. When the **Date-time** field is set to the special value this date-time is unknown.

After the reconfiguration, the **Date-time** field should be set to the date-time of the beginning of the next off-air period and the **Change flags** shall be set to 11 (see clause 6.4).

NOTE: The initial launch of a part-time service element is signalled as a continuous service element, i.e. the **P-T flag** is set to 0.

6.3.2.2 Reconfiguration phase

During the six second period before the introduction of the service element into the ensemble, the broadcaster shall signal a reconfiguration. The necessary Next MCI for the service element shall be provided.

The SCI shall be signalled as for the preparation phase. The respective service label or service component label shall be signalled.

6.3.2.3 Consolidation phase

There is no consolidation phase.

6.3.3 Receiver behaviour

6.3.3.1 General

A receiver adds a new service to the service list typically when creating the service list or when it finds a new service element has been added to an ensemble that was previously tuned. In both cases the same set of rules applies.

A service shall be added the service list when the full MCI is detected to identify a service element and the receiver does not have a service list entry corresponding to that service element. An existing service list entry corresponding to that service element shall be updated with the newly received information. The receiver shall add service list entries for all service component types it supports. A service element for which incomplete MCI is provided, shall be ignored by the receiver.

An exception to this rule is a service element that is temporarily off-air. It lacks full MCI and is only represented by SCI given by FIG 0/20 and an associated label. A receiver shall add or maintain a service list entry for such service components, so that the service list entry is readily available when the service resumes at a later time.

A receiver may create a service list entry ahead of time, when SCI is detected for a new service. During the preparation phase the receiver should clearly mark the service list entry for the user to understand the service list entry cannot be selected. As the SCI also provides date-time information, this may be shown to the user. However as the signalling of date-time information in the SCI might be subject to change, the receiver should attempt to keep this information updated.

A service list entry shall at least be made of the label. For primary service components, the service label shall be used. For secondary service components the associated service component label shall be used. The sorting of service list entries shall be consistent and shall not change without user intervention to explicitly change the sorting order of the service list.

For the identification of a service element only the ECC and SId/SCIdS fields in MCI shall be used; other means, such as labels shall not be used. Service elements with different ECC and/or SId/SCIdSs, but the same label, shall be treated as different service elements with separate service list entries. A service element with different labels (i.e. from more than one source) shall be represented by a single service list entry; the receiver chooses one of the provided labels as appropriate.

6.3.3.2 Response to signalling

When the receiver decodes the SCI with the **Change flags** set to 01, it is informed that a service element is to be added to the ensemble.

When the **P-T flag** is set to 0 it indicates that a new service element will be added at the date-time indicated. The receiver should create a tentative service list entry with the information provided by the SCI and the associated label; it should further indicate the date-time of the service element coming to life in the service list entry. The service list entry shall be created at or after the ensemble configuration when the receiver discovers MCI for the inserted service element for the first time.

When the **P-T flag** set to 1 it indicates that a service element that has been on-air previously is currently off-air and will return at the date-time indicated. The receiver shall create or maintain the service list entry with the information provided by the SCI and the associated label. It may additionally indicate the date-time of the service element coming to life again. It is recommended that a receiver marks a temporarily off-air service element in the list to indicate its part-time nature.

The **SC description** field provides information on the service element to enable a receiver to determine if it can decode the content (FIG 0/13 is also required for a data service element). Receivers shall not present to the user service elements that cannot be decoded.

6.4 Removing service elements from an ensemble

6.4.1 Introduction

A service element is removed from one of several ensembles on which it is broadcast or it is removed from all ensembles. A receiver shall make sure it removes the associated service list entry unless it continues to receive the service element.

When a service element will no longer be broadcast, then it shall be removed from the ensemble. If there is a requirement to change the Service Identifier (SId) of a service, or if the service is moved to a new ensemble, then clause 6.5 shall be followed instead.

Since the user may not be tuned to the ensemble at the time the service element is removed, advance information shall be provided. If an alternative service element is available, the broadcaster shall also indicate the service that the receiver shall select if the user is tuned to the service when it is removed.

Broadcasters shall provide advance information by using FIG 0/20. The removal itself will require a multiplex reconfiguration.

Part-time service elements are repeatedly removed from and returned to the ensemble. In order to allow receivers to maintain the entry in the service list for such part-time elements, the SCI and labels shall be provided during off-air periods.

NOTE: The final removal of a part-time service element is signalled as the removal of a continuous service element.

6.4.2 Signalling

6.4.2.1 Preparation phase

For up to 28 days before the removal of the service element from the ensemble, the broadcaster shall provide advance information by the use of FIG 0/20.

The fields of the SCI shall be used as follows:

- The **SId** and **SCIdS** fields shall be used to identify the service element; when the **SCIdS** is 0 for the primary service component, the entire service with all its service components is being removed. When the **SCIdS** refers to a secondary service component, only this service component is removed.
- The **Change flags** shall be set to 10 to indicate the service element will be removed from the ensemble (but will remain available from other ensembles, local removal), or to 11 to indicate that the service element will be removed from **all** ensembles (global removal).
- The **P-T flag** shall be set to 0 for the final removal of a service element, or 1 for a part-time service element that is to return later.
- The **SC flag** shall be set to 0 since the SC description is not required.
- The **SC description** shall be absent.
- The **Date-time** field shall be used to indicate, to the nearest second in UTC, when the reconfiguration will occur that removes the service element from the ensemble; if the **P-T flag** is set to 1 and **Date-time** field is set to the special value, the date-time of the service element going off-air is unknown. The service element will go off-air within 28 days from the start of the on-air period.
- When the **P-T** flag is set to 0, the **SId flag**, **EId flag, Transfer SId** and **Transfer EId** fields shall be used as indicated in table 2.
- When the **P-T flag** is set to 1, the **SId flag** and the **EId flag** shall be set to 0.

Table 2: Settings for selecting post-reconfiguration service

Condition	SId flag	EId flag	Transfer SId field	Transfer EId field	Local removal	Global removal
No transfer service indicated	0	0	Absent	Absent	No information	
Transfer to same service (same SId) in another ensemble	0	1	Absent	EId of other ensemble	Same service exists on other ensemble	Not used
Transfer to another service in same ensemble	1	0	SId of other service	Absent	Other service in this ensemble	
Transfer to another service in another ensemble	1	1	SId of other service	EId of other ensemble	Other service in other ensemble	

For part-time service elements (**P-T flag** set to 1), the **Change flags** shall be set to 11, indicating a global removal.

In the case of a local removal (**Change flags** are set to 10), the service element continues to exist on other ensembles.

A **Transfer SId** may be provided to point to a service on another ensemble to allow those receivers to continue the service presentation that cannot receive an ensemble on which the service continues. The **Transfer SId** identifies a different programme in this case.

When only a **Transfer EId** is provided, transfer shall be made to the same service on another ensemble.

A local removal shall only be signalled in situations where a service is available on more than one ensemble but the removal does not affect all ensembles. When a service element is removed from all its ensembles at the same time or when a service elements exists only in one ensemble, a global removal is signalled.

6.4.2.2 Reconfiguration phase

During the six second period before the removal of the service element from the ensemble, the broadcaster shall signal a reconfiguration. The next MCI shall not include the service element that is being removed.

The SCI shall be signalled as for the preparation phase. The service label or service component label shall be signalled.

6.4.2.3 Consolidation phase

There is no consolidation phase for a temporary removal (P-T flag is set to 1) since the service element will resume later (see clause 6.3).

After the reconfiguration has occurred for a permanent removal (**P-T flag** is set to 0), the broadcaster may provide confirmation information by the use of the SCI. It is recommended that the SCI remains on-air for at least 7 days after the reconfiguration.

The SCI shall be signalled as for the preparation phase, except that the **Date-time** field shall carry the special value to indicate that the change has occurred.

The service label or service component label shall not be signalled.

NOTE: Keeping the SCI on-air after the permanent removal of a service element allows receivers attempting to return to the removed service to take note of the configuration change and take appropriate action for service continuation.

6.4.3 Receiver behaviour

6.4.3.1 General

When a service element is removed from an ensemble this could affect the service presentation and the service list entry of the service component. When the selected service component is removed, the continuation of service is a primary concern. Reflecting a removed service component properly in the service list helps keep the service list up-to-date without the need for a full re-scan.

Updating the service list

A service component that is removed from an ensemble may continue to exist on another ensemble. Before deleting a service list entry, the receiver should ensure that no other known ensemble carries the service component being removed.

If a service component is removed with the SCI indicating **Change flags** set to '11', this implies that the service component is removed from all ensembles and the service list entry can safely by deleted in the receiver.

Continuing the service presentation

A service component shall be counted as removed if no or incomplete MCI is given and no SCI is given indicating a temporary off-air period. Service presentation shall stop immediately and the receiver should start a process to continue service presentation using available alternatives in the following order:

1) For a removed secondary service component, the service shall continue with the primary service component.

2) The selected service element on another ensemble. The SCI may contain a **Transfer EId** without a **Transfer SId** to support this selection. An EId provided in OE Services (FIG 0/24) may also be used.

3) A service that is linked by an active hard link. The linked service may be on FM.

4) In SCI for the removed service component, the broadcaster may indicate a 'preferred alternative' by means of a **Transfer SId**. This alternative represents a different service, and leads to a change of the audio programme. If a **Transfer EId** is given, the alternative is on another ensemble, a change of service requires re-tuning.

5) A service that is linked by an active soft link.

If by the above methods a continuation of service cannot be established, the receiver shall stop the service. It should display a 'service ceased' message as appropriate. When a service element is removed and the SCI has the change flags set to 11, the above process omits step 2.

When changing to another service in step 4. the audio programme will change, a receiver shall present a service referred to by a **Transfer SId** in the same way as a soft-linked service to the listener, prior to changing the audio programme (see also clause 5.6.3.4).

6.4.3.2 Response to signalling

When the receiver decodes the SCI with the **Change flags** set to 10, it is informed that a service element is to be removed from the tuned ensemble, *but continues on other ensembles*.

When the receiver decodes the SCI with the **Change flags** set to 11, it is informed that a service element is to be removed from the tuned ensemble *and from all other ensembles.*

If the **P-T flag** is set to 0, a permanent removal is signalled; after the reconfiguration no MCI is present for the removed service. The receiver shall update the service list at the reconfiguration as applicable. As the same service element may continue to exist on other ensembles, the receiver shall not remove a service entry when the **Change flags** are set to 10, when it is aware of the service on another ensemble.

If the **P-T flag** is set to 1, a temporary removal is signalled; after the reconfiguration no MCI is present for the removed service, but an updated SCI will be present indicating the next service resumption. The receiver shall not modify the service list. It may update the date-time information in the display to the time the service element goes off-air.

If the **P-T flag** is set to 1, transfer information shall be ignored.

If the **Date-time** field contains the special value, the date-time of the service element going off-air is unknown. The service element goes off-air at an unknown time within 28 days after the beginning of the on-air period.

6.5 Changing the identity or source of a service

6.5.1 Introduction

A service element undergoes a technical reconfiguration for administrative purposes. A service list entry shall not be modified. A receiver shall minimize any service disruption that may occur to the selected service element.

When a service requires to change its Service Identifier (SId), usually due to regulatory reasons, or to move from one ensemble to another, usually for commercial reasons, broadcasters wish to complete the change without disruption to their audience.

To allow receivers to store or display information for later use, advance information shall be provided.

Advance information shall be provided by using FIG 0/20. The change itself will require a multiplex reconfiguration. In the case where a service element moves to another ensemble, SCI shall also be provided in the other ensemble to indicate the change of source and possibly identity of the service element. A receiver can discern the source and destination ensembles by the presence of MCI for the referenced service element.

6.5.2 Signalling

6.5.2.1 Preparation phase

For up to 28 days before the change of SId or move of the service to another ensemble, the broadcaster shall provide advance information by the use of FIG 0/20.

The fields of the SCI shall be used as follows:

- The **SId** and **SCIdS** fields shall be used to identify the service. The **SCIdS** field is set to 0 since the change affects the whole service.
- The **Change flags** shall be set to 00 to indicate the service will receive a new identity or be moved to another ensemble.
- The **P-T flag** shall be set to 0 for a continuous service or 1 for a part-time service.
- The **SC flag** shall be set to 0.
- The **SC description** shall be absent.
- The **Date-time** field shall be used to indicate, to the nearest second in UTC, when the reconfiguration will occur that changes the SId of the service or moves the service to another ensemble.
- The **SId flag**, **EId flag, Transfer SId** and **Transfer EId** fields shall be used as indicated in table 3.

Table 3: Settings for changed SId or service moved to another ensemble

Condition	SId flag	EId flag	Transfer SId field	Transfer EId field
Not used	0	0	Absent	Absent
Service is moved to another ensemble (SId is unchanged)	0	1	Absent	EId of destination ensemble
SId of service is changed (same ensemble)	1	0	New SId for service	Absent
Service is moved to another ensemble and changes SId	1	1	New SId for service	EId of destination ensemble

The service label or service component label shall be signalled as appropriate. The label shall appear for the new SId or on the other ensemble at the same time as the SCI with **SId flag** and/or **EId flag** set to 1 respectively.

Part-time services

For a part-time service element the **P-T flag** is set to 1, the change of SId and/or EId shall take place at the end of an off-air period. This is to minimize possible service disruptions. The reconfigured service will resume its service in the reconfiguration under the new identity and/or source. Therefore the SCI group with **Change flags** set to 00 replaces the SCI group with the **Change flags** set to 01, which would be signalled during the cyclic off-air period normally.

When the **Change flags** are set to 00 while the **P-T flag** is set to 1, the signalling is for a permanent change of service configuration. Services shall not use **Change flags** set to 00 for cyclic reconfigurations.

Signalling on the other ensemble

When the **EId flag** is set to 1, the service will move from the tuned ensemble to another ensemble. For this, SCI shall be present on both ensembles with the **SId** and **SCIdS** fields, **Change flags**, **P-T flag**, **SId flag** and **EId flag** all set to the same values. The **Date-time** field shall be set as required on each ensemble. The SCI shall appear at the same time on both ensembles. If the respective removal from the originating ensemble and the insertion at the destination ensemble do not occur at the same time, then the insertion shall occur first.

6.5.2.2 Reconfiguration phase

During the six second period before the change to the SId of the service, or the service being removed from the ensemble, the broadcaster shall signal a reconfiguration.

For a service that changes its SId, the next MCI shall not include the existing SId and shall include the new SId. For a service that is moving to another ensemble, the next MCI shall not include the service that is being moved.

The SCI shall be signalled as for the preparation phase.

The service label shall be signalled for the service. For a service that is changing its SId, the service label shall also be signalled for the new SId. For a service that is moving to a new ensemble, the service label shall be signalled on the new ensemble.

6.5.2.3 Consolidation phase

After the reconfiguration has occurred, the broadcaster may provide confirmation information by the use of FIG 0/20. For a service that has changed its SId, the current MCI shall be signalled using the new SId. For a service that has moved to another ensemble, no MCI will be present for the service.

The SCI shall be signalled as for the preparation phase, except that the **Date-time** field shall carry a special value to indicate that the change has occurred. The **Date-time** field shall be set to the special value.

For part-time services in the on-air period following the source and/or identity change, SCI for the new SId shall be present (in the new ensemble in case of a source change) after the reconfiguration with date-time of the next off-air period and the **Change flags** set to 11 (see clause 6.4 for details).

SCI with **Change flags** set to 00 should remain on-air (in the old ensemble in case of a source change) for at least 7 days after the reconfiguration to allow receivers attempting to return to the removed service to take note of the configuration change and take appropriate action for service continuation. This applies in particular if transfer information is provided.

6.5.3 Receiver behaviour

6.5.3.1 General

The evolution of broadcast networks may from time to time cause purely administrative changes to ensembles. A service may change its identity, but continue as the same programme. A service may be moved from one ensemble to another, thus changing transmit frequency and possible coverage area, but otherwise remain the same programme. A service may also undergo an identity change, while being moved between ensembles.

These situations have in common a complex service and ensemble reconfiguration that shall not cause an update of the service list and should be handled by the receiver in an automated process that minimizes service interruptions.

Continuing the service presentation

While other means to continue the service presentation may exist, the transfer information in the SCI shall take precedence. Therefore the process of continuing the service presentation here is different from the case where a service is removed from an ensemble. The **Transfer SId** field points to the same audio programme. It is recommended that a receiver stores transfer information for later use. The process shall take the following order:

1) Continue service presentation with new identity and/or source as given in **Transfer SId** and/or **Transfer EId** fields respectively. Retuning may be required.

2) A service that is linked to the selected service by an active hard link. The linked service may be on the same or another ensemble. The linked service may be on FM.

3) A service element that is linked by an active soft link. The linked service element may be on the same or on another ensemble.

Generally the receiver shall not use linkage information instead of transfer information, unless the service moves to a new source that cannot be received by the receiver. The change of source and/or identity affects all service components of the changed service in the same way. Therefore continuation of service presentation shall occur with respect to the selected service component. Receiver memory shall be updated for all affected service elements.

6.5.3.2 Response to signalling

When the receiver decodes the SCI with the **Change flags** set to 00, it is informed that a service will change its Service Identifier (SId) and/or move to another ensemble. The change will affect all service components of the service in the same way. This setting may occur in a variety of broadcast scenarios, determined by the value of the subsequent fields of the SCI. All these scenarios have in common that the service content remains the same and that new service list entries shall not be created.

When the **SId flag** is set to 1, the **Transfer SId** field contains the new SId of the service. The receiver shall use the **Transfer SId** to update its internal memory for the service list, so that existing service list entries will point to the new identity. From the time indicated in the SCI, the new identity shall take precedence when service selection is performed. The receiver should also update associated internal memory such as presets accordingly.

When the **EId flag** is set to 1, the **Transfer EId** field contains the EId of the ensemble the service is transferred to. This indicates the service changes ensembles and will eventually cease to exist on the originating ensemble. Receivers shall update the internal tuning memory so that the new source takes precedence in tuning, when service selection is performed.

The **Date-time** field indicates the time point at which the reconfiguration takes place. When the receiver detects SCI with **Change flags** set to 00 for a future time point, when selecting a service element from that service it should attempt to select the new identity immediately. If the **Date-time** field contains the special value, indicating the change has already occurred, the receiver when selecting a service element from the affected service shall attempt to select the new identity at the new source immediately. If MCI can be detected for the new identity, the receiver shall update internal memory and the new identity and source shall take precedence for service selection.

Table 4: Interpretation of transfer information

Condition	SId flag	EId flag	Interpretation
Not used	0	0	-
Transferred from another ensemble, same SId	0	1	The service changes it source, it is transferred between ensembles.
Change of SId within an ensemble	1	0	An administrative change of SId takes place, but the service continues as the same audio programme.
Transferred from another ensemble, different SId	1	1	The service changes it source while going through an identity change.

When updating existing service list entries, all receiver memory should be updated, so that presets created by the user are reassigned.

If the **P-T flag** is set to 1 and the **Change flags** are set to 00, the service element is currently off-air and will resume service under the new source and/or identity. The receiver shall use the **Date-time** field as the date-time of the beginning of the next on-air period. In this special case, the receiver shall additionally use the SCI in the same way as it would when the **Change flags** are set to 01 and **P-T flag** set to 1.

6.6 Changing the label of a service element

6.6.1 Introduction

For listeners, the service label or service component label is used to select the content they desire. Therefore, broadcasters should be very careful about changing these labels. Labels may be changed at any time and do not require a multiplex reconfiguration, but good practice is to retain the same label for as long as possible and only change it occasionally. It is likely that promotion of the new label and on-air advice will be needed. To ensure consistent behaviour in receivers, label changes will not always be presented to users immediately.

6.6.2 Signalling

The new service label or service component label shall be signalled. No other changed signalling is needed.

6.6.3 Receiver behaviour

The receiver shall update the internal memory with new label information, when a changed label is detected (see clause 6.2.4). When an updated label has been detected all user controls shall be updated.

When a receiver detects different labels for a single service element on different ensembles, only one service list entry shall be created. The receiver shall use one of the available labels to mark this service list entry.

7 Announcements

7.1 Introduction

An announcement is a period of elevated interest within an audio programme. It is typically a spoken audio message, often with a lead-in and lead-out audio pattern (for example, a musical jingle). It may refer to various types of information such as traffic, news, sports and others.

The signalling of announcements is to allow a receiver to provide the user with an announcement mode, including specific functions such as raising a reduced volume during the announcement message or switching from another playback source to the radio programme for the announcement message and other features. The receiver resumes the original state and playback function after the end of the announcement.

Regular announcements are signalled as one of general information, such as traffic, news or weather (see ETSI TS 101 756 [2]). Alarm announcements signal that an emergency message is broadcast which has a higher priority than regular announcements. Alarm announcements are treated separately in the present document, as both the signalling and the expected receiver behaviour are different. The DAB Emergency Warning System (EWS) [i.2] provides significantly enhanced functionality over the alarm announcement and it is recommended that EWS is used in preference to alarm announcements.

An announcement may occur during a service in the tuned ensemble, but may also occur during a service in another ensemble.

A broadcaster indicates his authorization to interrupt a service by signalling that the service supports announcements. The announcement support signalling reflects an editorial decision that shall be respected by the receiver. Only authorized services with matching announcement support information may be interrupted by signalled announcements.

Within the tuned ensemble, the receiver changes the audio service from the selected service component to another service component within the same ensemble; a re-tuning step is not required. Often the announcement will occur on the selected service component; in this case the receiver activates the announcement mode without changing the selected service component.

For an announcement on another ensemble, the receiver performs a search for the new tuning frequency, performs the re-tuning step and selects the target audio service component. As the receiver may not be in the service area of the targeted ensemble, careful attention should be given to processing the signalling for announcements on other ensembles.

At the end of an announcement, the receiver returns to the status and playback function it had before the start of the announcement.

7.2 Signalling conventions

7.2.1 What is a cluster?

An announcement cluster is a group of services that share the same announcement target. Announcement support information defines to which clusters a service is a member.

A service can be a member of up to seven clusters and a member of up to four OE clusters. Announcement switching information triggers an announcement for a particular cluster.

7.2.2 Management of Cluster Ids and OE Cluster Ids

A tuned ensemble announcement cluster is identified within an ensemble by a **Cluster Id**. An OE announcement cluster is identified within an ensemble by an **OE Cluster Id**. The **Cluster Id** (see ETSI EN 300 401 [1], clause 8.1.6.1) and **OE Cluster Id** (see ETSI EN 300 401 [1], clause 8.1.6.3) are only valid within the ensemble on which they are signalled. **Cluster Id**s and **OE Cluster Id**s are separate namespaces and share no common interpretation; there is no relation between a **Cluster Id** and an **OE Cluster Id** that signal the same announcement on different ensembles.

7.2.3 Signalling announcement support

Announcement support (ASu) signalling in FIG 0/18 and FIG 0/25 declares that a service carries regular announcements of the supported types and defines an interruption authorization for that particular service.

A service carrying ASu information may be interrupted only by announcements to the clusters it is a member of and it may be interrupted only by the announcement types it supports. A service shall not be interrupted by announcements that it does not support. Therefore announcement support information represents a broadcaster's authorization for a service to be interrupted by a clearly defined set of announcements.

A service that signals support for a certain announcement type shall receive announcement switching to that announcement type regularly; no unused announcement types shall be signalled as supported. Announcement support shall not be switched on and off for short periods as users may not understand a temporary suspension or activation of announcement support.

Announcement support information carries the entire set of data in an **Announcement support (ASu)** field or **OE ASu** field. The **SId** is the reference to this information; there shall be at most one **ASu** field and at most one **OE ASu** field present for any service.

All announcement support information shall be static, as the signalling does not support a change mechanism. For recurring changes to the announcement configuration, a service shall be assigned to two or more clusters where each represents one particular configuration.

7.2.4 Enabling alarms

It is recommended that EWS [i.2] is used in preference to alarm announcements.

Announcement support for alarms is signalled by setting the **Alarm flag** in FIG 0/0 to 1. This indicates that an ensemble will carry alarm announcements so that in the event of an emergency that affects the ensemble's audience, an alarm announcement will be made to inform listeners. Ensembles shall not set the **Alarm flag** to 1 unless the necessary mechanisms have been put in place to carry emergency messages on the ensemble (signalled using FIG 0/19) or on another ensemble (signalled using FIG 0/26).

NOTE: Announcement support for alarms is identical for tuned and other ensemble alarms.

Ensembles that have the **Alarm flag** set shall also provide ensemble configuration information in FIG 0/7 which implies that the ensemble is compliant to ETSI EN 300 401 [1], (see ETSI TS 101 756 [2], annex D).

7.2.5 Signalling announcement switching

Announcement switching information triggers one announcement type for a single cluster of services.

For any cluster, only one active announcement may be present at any one time. For any active announcement only one announcement type may be signalled, i.e. only one bit shall be set in the **ASw flag** field of FIG 0/19 (see ETSI EN 300 401 [1], clause 8.1.6.2) and FIG 0/26 (see ETSI EN 300 401 [1], clause 8.1.6.4).

The signalling of the announcement type in the announcement switching information - the value of the **ASw flags** field - shall not change during an announcement. The only change allowed is to signal the end of the announcement, when the value of the entire **ASw flag** field is set to 0.

Announcement switching information is a temporary signalling that may only be present for the purpose of signalling a particular announcement; that is during the announcement and short periods before and after the announcement.

The sub-channel indicated in the **SubChId** field of FIG 0/19 shall be a sub-channel that belongs to an audio service component in the ensemble. The service component indicated by the **SId** and **SCIdS** fields of FIG 0/26 shall be an audio service component present on a broadcast ensemble. Service components that do not provide full current MCI and labels shall not serve as targets for announcements. If target sub-channels and service components are created only for the duration of the announcement, then the reconfiguration of the ensemble shall be completed before the announcement switching signalling starts.

The start of an announcement is signalled with a burst of announcement switching information for 5 seconds at a nominal repetition rate of 10 FIGs/second. At the end of the announcement a burst of announcement switching information occurs for 2 seconds at a nominal repetition rate of 10 FIGs/second. The ASw field in the burst of FIGs terminating an announcement shall have the **ASw flags** field set to 0. The bursts are to make reception of ASw information more robust and reliable.

The continuation of the announcement is signalled with announcement switching information at a repetition rate of once per second. It is recommended to signal the ASw fields for all announcements in the ensemble in a single FIG.

7.2.6 Announcement concurrency

A service may have several concurrent announcements signalled to it, via the clusters, at any time. Concurrent announcements shall occur on different clusters and shall use different announcement types. For any given cluster, only one announcement may be active at any time.

Parallel announcements of different types should generally not be signalled to a single target audio service component.

The initial lead time of one announcement may overlap the end of another announcement provided by the same target service component to allow for back-to-back announcements (for example, to signal different segments in a news block).

7.2.7 Synchronization requirements

7.2.7.1 General

As announcement switching information and the associated announcement occur on different transmission channels of a DAB ensemble, or even on different ensembles with differing latency, ensemble operators and broadcasters shall apply appropriate delay compensation between FIC and MSC and between FIC on different ensembles.

NOTE: For the purposes of synchronization, the announcement switching signalling in FIG 0/19 and FIG 0/26 is most relevant. Therefore definitions and requirements in clause 7.2.7 apply equally to regular and alarm announcements.

7.2.7.2 User requirements

In principle, all delay compensation is to ensure the best user experience. Transmissions and receivers shall implement appropriate measures for this. Two conflicting problem cases shall be considered:

- **"Announcement truncation":** a delayed start and a premature end of the announcement cut off parts of the announcement message, which defeats the purpose of the announcement function and shall be avoided.
- **"Audio leakage":** a premature start and a delayed end of the announcement present the user audio outside the announcement. As the user will typically be switched into a different audio from the selected service, this is considered an inconvenience and should be minimized.

Perfect delay compensation can only be achieved in theory; in practice "announcement truncation" shall be avoided at the cost of "audio leakage". Ensembles should slightly over-protect against "announcement truncation" by eliminating the average time offset between FIC and MSC and then adding an audio protection time offset to cover up random jitter.

7.2.7.3 Timing issues

In order to avoid "announcement truncation", the announcement switching signalling on the tuned ensemble shall lead the audio at the start of announcement by 1 second and it shall trail the audio at the end of the announcement by 1 second. If OE announcement switching signalling is provided for the same announcement, then it shall lead the announcement switching signalling of the tuned ensemble by 1 second.

In order to apply these timing offsets, broadcasters shall determine the timing offset between the FIC and the MSC for tuned ensemble announcements and between the FIC of the source ensemble and the MSC of the target ensemble for other ensemble announcements. The broadcast control system shall be so designed to compensate for all internal timing offsets and so ensure that the required timing offsets are achieved.

For tuned ensemble announcements, this process ensures that announcement switching signalling starts 1 second before the announcement audio starts and that the announcement switching signalling continues for 1 second after the end of the announcement audio.

For other ensemble announcements, this process ensures that OE announcement switching signalling starts 2 seconds before the announcement audio starts (in the other ensemble) and that the OE announcement switching signalling continues until the end of the announcement audio (in the other ensemble). Receivers shall determine the target ensemble and shall wait for at least 2 seconds to receive the (tuned ensemble) announcement switching information. Receivers able to monitor the FIC on the target ensemble in the background may delay switching to the target service component until the (tuned ensemble) announcement switching signalling is present on the target ensemble.

7.2.8 New flag

The **New flag** in the **Announcement Switching (ASw)** field (FIG 0/19) and the **OE ASw** field (FIG 0/26) shall be set to 1.

7.3 Receiver response

7.3.1 Announcement mode

The announcement mode is a set of receiver functions that are collectively activated during an announcement. This may include various measures that improve audibility and comprehensibility of the audio announcement. The precise definition of this set of receiver functions is implementation dependent, except for the following requirements:

- A receiver shall present the audio of the target sub-channel or service component for the full time required by the announcement switching signalling.

- A receiver shall indicate on its display (as far as applicable) that an announcement is active during the announcement.
- A receiver shall restore the same playback status as before the announcement upon announcement termination.

7.3.2 Support for alarms

Support for tuned ensemble alarms in the receiver is mandatory, support for other ensemble alarms is optional. Functions of the announcement mode to achieve the best user experience with alarm audio messages are implementation dependent.

7.3.3 Monitoring the signalling

A receiver that supports announcements monitors the two-part signalling for announcement support and announcement switching with dedicated processes. The signal monitoring for regular announcements may be subject to the announcement function being activated by the user.

Announcement support for alarms is signalled by the **Alarm flag** in FIG 0/0 and applies equally for all services in the ensemble. Receivers shall not monitor or respond to alarm announcement signalling unless configuration information (FIG 0/7) is present in the ensemble.

Regular announcement support signalling (FIG 0/18, FIG 0/25) is detected at service acquisition, re-acquisition and ensemble reconfigurations, the so-called "discovery points" (see also clause 6.2.4). At the discovery point the receiver filters the announcement support information available on the ensemble for the currently selected service.

At most one FIG 0/18 and one FIG 0/25 will be evaluated for any service. If announcement support is signalled for the selected service to any of the announcement types the receiver supports, it shall store in memory all **Cluster Ids** and **OE Cluster Id**s signalled.

Monitoring announcement switching information is a real-time process. The receiver shall monitor all FIG 0/19, and FIG 0/26 if OE announcements are supported, for any of the **Cluster Id**s and **OE Cluster Id**s stored for the currently selected service.

If the **Alarm flag** is set, the alarm **Cluster Id** and the alarm **OE Cluster Id** shall be added to the internal memory for signal monitoring. Alarm announcements may interrupt tuned ensemble regular announcements at any time.

When monitored ASw information has been detected (i.e. matching the clusters stored against the selected service), the receiver shall respond to the first ASw field it receives at the start and end of the announcement.

7.3.4 Filtering of announcements

Receiver response to announcement switching information for regular announcements shall be subject to announcement support information for the selected service and may be subject to a user setting. The user may activate any of the announcement types that the receiver supports; in practice a user will turn on or off a particular announcement type.

When an announcement switching event is detected, by way of a FIG 0/19 or FIG 0/26, the receiver first matches the **Cluster Id** or **OE Cluster Id** against the stored Cluster Ids and OE Cluster Ids, respectively, for the selected service. Only if a match is found shall the receiver go on to match announcement types; otherwise the announcement process shall be terminated.

Next, the receiver matches the **ASw flags** against the supported and activated announcement types: that is, both the **ASu flags** field and the **ASw flags** field shall have a common bit set to 1 and the respective announcement type shall be activated in the receiver. Only if this three-way match is positive shall the announcement mode be activated.

For other ensemble announcements, the receiver shall perform a second (two-way) matching of the FIG 0/26 **ASw flags** field in the source ensemble against the FIG 0/19 **ASw flags** field in the target ensemble. Only if the **ASw flags** fields match, the announcement mode shall be activated; otherwise the receiver shall return to the source ensemble, without activating the announcement mode.

If the **Cluster Id** or **OE Cluster Id** is set to "1111 1111" (0xFF), the **ASw flags** field shall have only the bit b0 set to 1; receivers shall verify this setting before activating the announcement mode for an alarm announcement.

If the **ASw flags** field has set bit b0 to 1 with a **Cluster Id** or **OE Cluster Id** other than "1111 1111" (0xFF), the **ASw** field shall be ignored.

Receiver response to alarm announcements shall not be subject to a user setting.

7.3.5 Prioritization of announcements

As announcements may be signalled to different **Cluster Id**s or **OE Cluster Id**s concurrently, receivers shall perform prioritization.

When an alarm announcement is detected, a receiver shall immediately terminate any ongoing regular announcement and switch to the signalled alarm announcement.

A receiver shall not interrupt the active announcement mode by another regular announcement; therefore the first detected announcement takes precedence and shall be presented in its entirety.

NOTE: Overlapping announcements may be activated differently by receivers depending on tuning status or announcement activation.

A receiver shall not interrupt an ongoing OE announcement or alarm, i.e. monitoring of alarm signalling shall not be active while tuning an announcement target ensemble.

7.3.6 Target selection process

7.3.6.1 General

When the announcement mode is being activated for a regular announcement or for an alarm announcement the receiver performs a target selection process to identify the audio which carries the announcement message.

For tuned ensemble announcements, the target, given in FIG 0/19, is a sub-channel in the tuned ensemble.

For other ensemble announcements, the target, given in FIG 0/26, is a service component in another ensemble. For OE regular announcements, the associated ensemble identifier is given in the FIG 0/25 information; for OE alarms, the associated ensemble identifier is given in the FIG 0/24 information (see also clause 5.3) and information retrieved from pre-tuning memory.

When implementing support for OE announcements and alarms in a receiver, care should be taken to minimize service interruptions due to signalling of announcements or alarms on ensembles that are not available for tuning. Receivers that cannot test a tuning frequency for a valid signal without interrupting the user selected audio should generally only attempt switching to announcements or alarms on ensembles that have been tuned to before.

When retuning to another ensemble for an announcement, receivers shall not evaluate more than 4 different frequencies to find a valid signal. The process to select the target service component shall be terminated if it does not succeed within 2 seconds.

7.3.6.2 Tuned ensemble announcements and alarms

On the tuned ensemble, the receiver finds the target sub-channel in the FIG 0/19, changes to the new sub-channel (if needed) and activates the announcement mode. If the sub-channel signalled in the FIG 0/19 corresponds to the sub-channel of the currently selected service then no change of sub-channel is needed for the activation of the announcement mode.

7.3.6.3 Other ensemble announcements

For OE regular announcements, the receiver resolves target ensemble and service component from FIG 0/25 and FIG 0/26 information respectively in the tuned ensemble.

To identify available signals for the target ensemble, the receiver inspects pre-tuning memory and FIG 0/21 Frequency information, if available, to find a valid tuning frequency for the target ensemble.

NOTE: As the announcement target service component may be created dynamically, pre-tuning memory in a receiver may be inaccurate in associating the target service component with an ensemble. Receivers only use pre-tuning memory to find a frequency for the target ensemble.

Once the target ensemble is tuned, it uses the **SId** and **SCIdS** provided in the FIG 0/26 to resolve the target sub-channel from the FIG 0/2 MCI in the target ensemble. It finds the FIG 0/19 for the target sub-channel to activate the announcement mode and uses the **Cluster Id** for monitoring of the announcement signalling.

Upon termination of the OE announcement, the receiver shall deactivate the announcement mode and return to the source ensemble and service component.

7.3.6.4 Other ensemble alarms

For OE alarm announcements, only a target service component is provided in FIG 0/26. The receiver shall use OE Services information (FIG 0/24) to resolve a target ensemble for the announcement. If a target ensemble can be identified, a valid tuning frequency shall be found from Frequency information (FIG 0/21) and pre-tuning memory.

NOTE: As the alarm target service component may be created dynamically, pre-tuning memory in a receiver may be unable to associate the target service component with an ensemble and the target EId can only be found from OE Services information (FIG 0/24).

Once the target ensemble is tuned, the receiver finds the target sub-channel from current MCI (FIG 0/2). It also finds an **ASw** field in FIG 0/19 using the settings for alarms. If the correct sub-channel is signalled the announcement mode is activated.

Upon termination of the OE alarm, the receiver shall deactivate the announcement mode and return to the source ensemble and service component.

7.3.7 Terminating an announcement

An announcement is terminated when the received FIG 0/19 **ASw flag** field corresponding to the announcement is set to 0. The announcement mode shall be deactivated and the receiver shall restore the functional status and audio source from before the beginning of the announcement.

NOTE: While the announcement is ongoing, the source service and/or source ensemble may undergo reconfigurations; upon announcement termination the receiver inspects the respective MCI to properly resume the source service component.

When terminating an alarm announcement that interrupted a regular announcement, the announcement mode shall be deactivated and the receiver shall restore the functional status and audio source from before the beginning of the (later interrupted) regular announcement.

7.3.8 Performance aspects

Manufacturers shall ensure good synchronization between information received in the FIC and events occurring in the audio of the selected service.

Receivers shall accurately account for an average time offset between FIC response and audio presentation; receivers that perform audio buffering are recommended to also account for state dependent time offset variations between FIC response and audio presentation that exceed 100 ms.

Receivers shall respond to the first **ASw** field that is detected at the start and end of an announcement to activate and de-activate the announcement mode.

Receivers implementing an optional "networked stand-by" mode shall not respond to announcement signalling unless the wake-up time to audio on the target sub-channel does not exceed sub-channel switching time by 20 % on the same receiver.

7.3.9 Audio codec and user application handling

FIG 0/19 only signals the sub-channel that carries the announcement. To determine whether the audio is coded as DAB audio or DAB+ audio it is necessary to find the **SubChId** in the FIG 0/2 MCI. If the announcement audio carries PAD applications (for example, dynamic label or SlideShow), or the service includes data service components, these should be presented according to the capabilities of the receiver.

It is recommended that receivers present non-audio service components contained in the announcement service in the same way as on the source service. Receivers shall not present such non-audio service components from the source service during the announcement.

7.3.10 Interpretation of the 'New Flag'

Receivers shall ignore the **New flag** provided in FIG 0/19 and FIG 0/26.

7.4 Tuned ensemble regular announcements

7.4.1 Introduction

A service provider signals that regular announcements, provided within the ensemble, may interrupt his service when certain conditions, identified by the announcement support information in FIG 0/18, are met. This is a permanent signalling.

A service provider signals the presence of a regular announcement, with its parameters, using the announcement switching information in FIG 0/19. The announcement switching information shall be properly synchronized to the audio in the target sub-channel to avoid "announcement truncation" and minimize "audio leakage".

Receiver support for tuned ensemble regular announcements is optional, however a receiver that supports this feature shall meet all requirements of clause 7.4. It is recommended that automotive receivers support this feature.

7.4.2 Announcement support

7.4.2.1 Signalling

For every service in the ensemble one **ASu** field in FIG 0/18 may be present. It defines the announcement interruption privileges for the service. A service may have at most one **ASu** field present at any time.

The **SId** field identifies the service; when the service has more than one audio service component, the announcement support information applies equally to all audio service components. Announcements shall not apply to data service components. The **SId** field shall be set to an SId present in the ensemble.

The **ASu flags** field defines the announcement types that the service supports; only regular announcement types shall be signalled. A service may support any number of announcement types; for any supported announcement type the respective bit flag shall be set to 1 in the **ASu flags** field, all other bit flags shall be set to 0.

Announcement support shall be signalled only to announcement types that actually occur within the service schedule.

The list of **Cluster Id**s defines a set of clusters to which the service belongs. A **Cluster Id** shall be unique within the ensemble. **Cluster Id**s ="0000 0000" and "1111 1111" are reserved and shall not be used. Services within a cluster need not share the same announcement support types.

7.4.2.2 Receiver response

A receiver monitors the announcement support signalling for the selected service.

The receiver shall discover changes to the announcement support information at least at one of the discovery points (see clause 6.2.4). At the time of announcement switching, the receiver shall respond according to the most recently detected announcement support information.

A receiver shall filter ASu information by the **SId** of the selected audio service; any other announcement support information shall be ignored. The type of announcement to be enabled in the receiver may be subject to a user setting. A setting of the **ASu flags** field that does not match any of the activated announcement types in the receiver shall be ignored.

If a valid **ASu** field has been detected, all signalled **Cluster Id**s shall be stored in memory and monitoring of announcement switching to these **Cluster Id**s shall be started.

7.4.3 Announcement switching

7.4.3.1 Signalling

An audio message in the tuned ensemble is signalled by announcement switching information in FIG 0/19 so that receivers can activate an announcement mode to give it elevated significance. The announcement switching signalling shall be synchronized to the audio timing with delay compensation (see clause 7.2.7).

The **Cluster Id** field identifies the cluster of services that shall respond to the announcement. Only one **ASw** field shall be present for a single **Cluster Id** at any time. Unused **Cluster Id**s, not present in any **ASu** field signalled for services in the ensemble, shall not be used.

The **ASw flags** field shall have one bit (bit flags b1 to b15) set to 1 corresponding to the announcement type while the announcement is active; when the announcement has been terminated, the **ASw flag** field shall be set to 0.

The **New flag** shall be set to 1.

The **SubChId** defines the target sub-channel for the announcement. It shall identify a sub-channel that is present in the ensemble.

7.4.3.2 Receiver response

If announcement support for the selected service has been detected, the receiver shall monitor continuously the announcement switching information in FIG 0/19 for stored **Cluster Id**s so that an announcement is detected immediately. Upon detection of applicable ASw information, the receiver shall perform filtering of the announcement types and target selection.

An **ASw flag** field with more than one bit flag set to 1 is invalid and shall be ignored. The bit flag set in the **ASw flags** field shall match a set bit flag in the **ASu flags** field. The bit flag set in the **ASw flags** field shall match an announcement type the receiver has activated. If both conditions are met, the receiver selects the target sub-channel and activates the announcement mode.

During the announcement the receiver shall monitor ASw information for only the active **Cluster Id**, including the alarm **Cluster Id** if applicable; any other ASw information shall be ignored. If **Cluster Id** = "1111 1111" is detected an Alarm is signalled, special rules apply, as described in clause 7.6. When the **ASw flag** field is set to 0, the termination of the announcement is detected, the announcement mode is deactivated and the source service component is selected.

The **New flag** shall be ignored.

For the detection of start and end of the announcement, the receiver shall not rely on a particular repetition rate of the ASw information. The receiver shall respond to the first **ASw** field that is detected at the beginning and the end of the announcement.

7.5 Other ensemble regular announcements

7.5.1 Introduction

A service provider signals that regular announcements, provided in another ensemble, may interrupt his service when certain conditions, identified by the announcement support information in FIG 0/25, are met. This is a permanent signalling.

A service provider signals the presence of a regular announcement on another ensemble, with its parameters, using the announcement switching information in FIG 0/26. The announcement switching information shall be properly synchronized to the audio in the target service component to avoid "announcement truncation" and minimize "audio leakage". The receiver uses provided service information and pre-tuning memory to identify a tuning frequency that carries the target service component.

The correct functioning of other ensembles announcements requires that both ensembles provide valid announcements signalling: on the source ensemble, other ensemble announcement signalling (as detailed in clause 7.5) is required; on the target ensemble, tuned ensemble announcements switching is required (as detailed in clause 7.4.3).

Receiver support for other ensemble regular announcements is optional, however a receiver that supports this feature shall meet all requirements of clause 7.5.

7.5.2 Announcement support

7.5.2.1 Signalling

For every service in the ensemble one **OE ASu** field in FIG 0/25 may be present. It defines the OE announcement interruption privileges for the service. A service may have at most one **OE ASu** field present at any time.

The **SId** field identifies the service; when the service has more than one audio service component, the announcement support information applies equally to all audio service components. Announcements shall not apply to data service components. The **SId** field shall be set to an SId present in the ensemble.

The **ASu flags** field defines the announcement types that the service supports; only regular announcement types shall be signalled. A service may support any number of announcement types; for any supported announcement type the respective bit flag shall be set to 1 in the **ASu flags** field, all other bit flags shall be set to 0.

Announcement support shall be signalled only to announcement types that actually occur within the service schedule.

Up to four announcement targets may be provided in the target list. The **Num targets** field indicates the length of the target list reduced by one. Each **Target** field provides an **OE Cluster Id** and associates an **EId** to it. The **OE Cluster Id**s define the set of clusters to which the service belongs. An **OE Cluster Id** shall be unique within the ensemble; it bears no meaning beyond the ensemble. **Cluster Id** = "1111 1111" is reserved and shall not be used. The target **EId** shall represent a valid ensemble. Services within a cluster need not share the same announcement support types.

It is recommended that FIG 0/21 Frequency information (see clause 5.4) is provided for all target ensembles.

7.5.2.2 Receiver response

A receiver monitors the OE announcement support information for the selected service.

The receiver shall discover changes to the announcement support information at least at one of the discovery points (see clause 6.2.4). At the time of announcement switching the receiver shall respond according to the most recently detected announcement support information.

A receiver shall filter ASu information by the **SId** of the selected audio service; any other announcement support information shall be ignored. The type of announcement to be enabled in the receiver may be subject to a user setting. A setting of the **ASu flags** field that does not match any of the activated announcement types in the receiver shall be ignored.

If a valid **OE ASu** field has been detected, all signalled **OE Cluster Id**s shall be stored in memory with associated **EId** and monitoring of OE announcement switching to these **OE Cluster Id**s shall be started.

A receiver supporting OE announcements should monitor FIG 0/21 Frequency information for target ensembles. This information should be available for fast retrieval at the time of announcement switching.

7.5.3 Announcement switching

7.5.3.1 Signalling

An audio message on another ensemble is signalled by OE announcement switching information in FIG 0/26 so that receivers can retune to provide the audio message. The activation of the announcement mode depends on the signalling on the target ensemble. The announcement switching signalling on source and target ensembles shall be synchronized with each other and the timing shall be offset (see clause 7.2.7).

NOTE: The signalling of an announcement message in the target ensemble follows all rules for a tuned ensemble announcement, as described in clause 7.4.

The **OE Cluster Id** field identifies the cluster of services that shall respond to the announcement. Only one **OE ASw** field shall be present for a single **OE Cluster Id** at any time. Unused **OE Cluster Id**s, not present in any **OE ASu** field signalled for services in the ensemble, shall not be used.

The **ASw flags** field shall have one bit (bit flags b1 to b15) set to 1 corresponding to the announcement type while the announcement is active; when the announcement has been terminated, the **ASw flags** field shall be set to 0.

The **New flag** shall be set to 1.

The combination of **SId** and **SCIdS** defines the target service component for the announcement. It shall identify a service component present in the target ensemble defined in the OE announcement support information.

7.5.3.2 Receiver response

If OE announcement support for the selected service has been detected, the receiver shall monitor continuously the OE announcement switching information in FIG 0/26 for stored **OE Cluster Id**s so that an OE announcement is detected immediately. Upon detection of applicable ASw information, the receiver shall perform filtering of the announcement types and target selection.

An **ASw flags** field with more than one bit flag set to 1 is invalid and shall be ignored. The bit flag set in the **ASw flags** field shall match a set bit flag in the **ASu flags** field. The bit flag set in the **ASw flags** field shall match an announcement type the receiver has activated. If both conditions are met, the receiver starts the target selection.

The receiver identifies a valid tuning frequency for the target ensemble using pre-tuning memory and Frequency information (FIG 0/21) as far as available. If the target ensemble is transmitted on several frequencies, the receiver may attempt to re-tune to up to four frequencies. The process shall be designed such that the re-tuning process can be completed within 1 second or less.

NOTE: Receivers quickly completing the retuning process to the target ensemble may have to wait for a limited period for the (tuned ensemble) announcement switching information to appear. See clause 7.2.7.

Once the re-tuning has completed, the receiver reads the MCI on the target ensemble to verify the signalled service component exists in the target ensemble and to resolve the sub-channel for the target service component. It finds the applicable announcement switching information for the target sub-channel by comparing the sub-channel of the target service component with the sub-channel signalled in FIG 0/19; this provides the **Cluster Id** for the ongoing announcement. If several FIG 0/19 **ASw** fields are present, the receiver shall attempt to match the sub-channel in all signalled clusters. The first **Cluster Id** that has the sub-channel match shall be used for the announcement and is stored in memory.

The receiver shall verify the announcement type in the **ASw flags** field against the **ASw flags** field provided in the source ensemble, and if the test fails, terminate the announcement process. If the test succeeds, the receiver switches to the target sub-channel and activates the announcement mode.

During the announcement the receiver shall monitor FIG 0/19 ASw information for only the stored **Cluster Id**; any other ASw information including alarm signalling shall be ignored. When the **ASw flags** field is set to 0, the termination of the announcement is detected, the announcement mode shall be deactivated and retuning to the source ensemble and service component is performed.

The **New flag** shall be ignored.

For the detection of start and end of the announcement, the receiver shall not rely on a particular repetition rate of the ASw information. The receiver shall respond to the first **ASw** field that is detected at the beginning and the end of the announcement in FIG 0/26 and FIG 0/19 respectively.

Upon return to the source ensemble, the receiver shall ignore any potentially ongoing announcement switching information in the terminated **OE Cluster Id** for 5 seconds.

7.6 Tuned ensemble alarm announcements

7.6.1 Introduction

A ensemble provider signals that alarm announcements may interrupt all services in the ensemble by setting the **Alarm flag** in FIG 0/0. This is a permanent signalling.

A service provider signals the presence of an alarm announcement using the announcement switching information in FIG 0/19 with the pre-defined alarm cluster. The announcement switching information shall be properly synchronized to the audio in the target sub-channel to avoid "announcement truncation" and minimize "audio leakage".

Only one alarm announcement shall be signalled at any given time.

Receiver support for tuned ensemble alarm announcements is mandatory and receivers shall meet all requirements of clause 7.6.

7.6.2 Announcement support

7.6.2.1 Signalling of the Alarm flag

The setting of the **Alarm flag** in FIG 0/0 signals announcement support for all services in the ensemble. When the **Alarm flag** is set to 1 it is active and an alarm announcement may interrupt any service in the ensemble.

7.6.2.2 Receiver response

A receiver that detects an active **Alarm flag** in FIG 0/0 shall activate signal monitoring of FIG 0/19 for the special alarm cluster.

A receiver shall detect a change to the state of the **Alarm flag** at least at one of the discovery points (see clause 6.2.4). At the time of an alarm announcement the receiver shall respond according to the most recently received **Alarm flag** setting.

7.6.3 Announcement switching

7.6.3.1 Signalling

An alarm message in the tuned ensemble is signalled by announcement switching information in FIG 0/19 so that receivers can activate the announcement mode to give it elevated significance. The announcement switching signalling shall be synchronized to the audio timing with delay compensation (see clause 7.2.7).

The **Cluster Id** shall be set to "1111 1111" (0xFF).

The **ASw flags** field shall have only bit b0 set to 1 while the announcement is active; when the alarm announcement has been terminated, the **ASw flags** field shall be set to 0.

The **New flag** shall be set to 1.

The **SubChId** defines the target sub-channel for the alarm announcement. It shall identify a sub-channel that is present in the ensemble.

7.6.3.2 Receiver response

Once signal monitoring for alarm announcements has been activated, a receiver shall respond immediately to any **ASw** field that has the bit flag b0 set to 1 and the **Cluster Id** set to "1111 1111". Alarm announcements shall take precedence over any other announcement types.

The response to alarm announcements shall not be subject to any user setting; receivers shall switch to the target sub-channel unconditionally, any active regular announcements shall be terminated. Receivers may implement specific functions in the announcement mode that are only activated for alarm announcements with the primary goal to enhance audibility and comprehensibility of alarm announcements.

The receiver shall verify that the **ASw flags** field has only bit b0 set to 1. If more than one bit is set the ASw information is invalid and shall be ignored. If the test succeeds the receiver switches to the target sub-channel and activates the announcement mode.

During the announcement the receiver shall monitor ASw information for only the alarm **Cluster Id**. When the **ASw flags** field is set to 0, the termination of the announcement is detected, the announcement mode shall be deactivated and the source sub-channel is selected.

The **New flag** shall be ignored.

For the detection of start and end of the announcement, the receiver shall not rely on a particular repetition rate of the ASw information. The receiver shall respond to the first **ASw** field that is detected at the beginning and the end of the announcement.

7.7 Other ensemble alarm announcements

7.7.1 Introduction

A ensemble provider signals that alarm announcements may interrupt all services in the ensemble by setting the **Alarm flag** in FIG 0/0. This is a permanent signalling.

A service provider signals the presence of an alarm announcement on another ensemble using the announcement switching information in FIG 0/26 with the pre-defined alarm cluster. The announcement switching information shall be properly synchronized to the audio in the target sub-channel to avoid "announcement truncation" and minimize "audio leakage". The receiver uses provided service information and pre-tuning memory to identify a tuning frequency that carries the target service component.

The correct functioning of other ensemble alarms requires that both ensembles provide valid announcements signalling: on the source ensemble, other ensemble alarms signalling (as detailed in clause 7.7) is required; on the target ensemble, tuned ensemble announcement switching is required (as detailed in clause 7.6.3).

Only one OE alarm announcement shall be signalled at any given time. It is recommended that all OE alarm announcements occur on the same target service component.

Receiver support for other ensemble alarm announcements is optional, however a receiver that supports this feature shall meet all requirements of clause 7.7.

7.7.2 Announcement support

7.7.2.1 Signalling of the Alarm flag

The setting of the **Alarm flag** in FIG 0/0 signals announcement support for all services in the ensemble. When the **Alarm flag** is set to 1 it is active and an OE alarm announcement may interrupt any service in the ensemble.

FIG 0/24 OE Services information (see clause 5.3) shall be signalled for target alarm announcements to identify the ensemble(s) those service components are carried in. It is recommended that FIG 0/21 Frequency information (see clause 5.4) is also signalled for those ensembles.

7.7.2.2 Receiver response

A receiver that detects an active **Alarm flag** in FIG 0/0 shall activate signal monitoring of FIG 0/26 for the special alarm cluster.

A receiver shall detect a change to the state of the **Alarm flag** at least at one of the discovery points (see clause 6.2.4). At the time of an alarm announcement the receiver shall respond according to the most recently received **Alarm flag** setting.

A receiver supporting OE Alarm announcements shall monitor FIG 0/24 OE Services information and should monitor FIG 0/21 Frequency information. This information should be available for fast retrieval at the time of announcement switching.

7.7.3 Announcement switching

7.7.3.1 Signalling

An alarm message on another ensemble is signalled by OE announcement switching information in FIG 0/26 so that receivers can retune to the target service. The activation of the announcement mode depends on the signalling on the target ensemble. The announcement switching signalling on source and target ensembles shall be synchronized with each other and the timing shall be offset (see clause 7.2.7).

NOTE: The signalling of an alarm message in the target ensemble follows all rules for a tuned ensemble alarm announcement, as described in clause 7.6.

The **OE Cluster Id** shall be set to "1111 1111" (0xFF).

The **ASw flags** field shall have only bit b0 set to 1 while the announcement is active; when the alarm announcement has been terminated, the **ASw flags** field shall be set to 0.

The **New flag** shall be set to 1.

The combination of **SId** and **SCIdS** defines the target service component for the alarm announcement.

7.7.3.2 Receiver response

Once signal monitoring for alarm announcements has been activated, a receiver shall respond immediately to any **OE ASw** field that has the bit flag b0 set to 1 and the **OE Cluster Id** set to "1111 1111". Alarm announcements shall take precedence over any other announcement types.

The response to OE alarm announcements shall not be subject to any user setting; receivers shall execute the alarm announcement unconditionally, any active regular announcements shall be terminated. Receivers may implement specific functions in the announcement mode that are only activated for alarm announcements with the primary goal to enhance audibility and comprehensibility of alarm announcements.

If an announcement in the alarm cluster is detected, the receiver resolves a target ensemble and associated tuning frequency for the target service component. It shall use OE Services information, and Frequency information and pre-tuning memory as far as available. If the target ensemble is transmitted on several frequencies, the receiver may attempt to re-tune to up to four frequencies. The process shall be designed such that the re-tuning process can be completed within 1 second or less.

NOTE: Receivers quickly completing the retuning process to the target ensemble may have to wait for a limited period for the (tuned ensemble) announcement switching information to appear. See clause 7.2.7.

Once the re-tuning has completed, the receiver reads the MCI on the target ensemble to verify the signalled service component exists in the target ensemble and to resolve the sub-channel for the target service component. It finds the applicable announcement switching information for the target sub-channel by comparing the sub-channel of the target service component with the sub-channel signalled in FIG 0/19; the **Cluster Id** should be "1111 1111".

The receiver shall verify that the **ASw flags** field has only bit b0 set to 1. If more than one bit is set the ASw information is invalid and shall be ignored. If the test succeeds, the receiver switches to the target sub-channel and activates the announcement mode.

During the announcement the receiver shall monitor FIG 0/19 ASw information for only the alarm **Cluster Id**. When the **ASw flags** field is set to 0, the termination of the announcement is detected, the announcement mode shall be deactivated and retuning to the source ensemble and service component is performed.

The **New flag** shall be ignored.

For the detection of start and end of the announcement, the receiver shall not rely on a particular repetition rate of the any of the ASw information. The receiver shall respond to the first **ASw** field that is detected at the beginning and the end of the announcement in FIG 0/26 and FIG 0/19 respectively.

Upon return to the source ensemble, the receiver shall ignore any potentially ongoing announcement switching information in the alarm cluster for 5 seconds.

8 Text labels

8.1 Introduction

The identification of available service elements for selection by the user is provided by the human readable SI labels, carried in the various extensions of FIG type 1 and FIG type 2. SI labels comprise all FIG type 1 and FIG type 2 labels: ensemble labels, programme service labels, service component labels, data service labels and X-PAD User Application labels. On more advanced receivers, these text labels may be supplemented or replaced by logos, but the SI label is the core method for identification. As such, SI labels need to be readily comprehensible so that users can confidently select the correct service.

In addition to SI labels, the other text-based core service information feature is the dynamic label. This is typically used to convey general news, music meta-data (artist/title), programme identification and more, and is the most common programme associated data feature in use.

Since the DAB system was first introduced, significant changes have occurred in the understanding and provision of text representation and presentation. Therefore, no further 8-bit character sets are envisioned and the use of FIG type 1 labels is strongly recommended to be restricted to carrying SI labels using the EBU-Latin character set only. For all other scripts, FIG type 2 labels should be used, since they allow 16-character labels to be provided.

The changes to displays on products also means that the simple formatting characters formerly provided for the dynamic label are no longer appropriate, and shall not be used. This is reflected in the reservation of the code points 0x0A, 0x0B and 0x1F in the Complete EBU Latin-based repertoire character set (see ETSI TS 101 756 [2], annex C).

In locations where SI labels are required using both Latin and other scripts, a mixture of FIG type 1 (for Latin labels) and FIG type 2 (for labels in other scripts) may be used in an ensemble. When an SI label has a mixture of Latin and other scripts included, it is necessary to use a FIG type 2 label.

The choice of characters is essentially unlimited, but receivers can only display characters for which they have glyphs. The correct presentation of characters is non-trivial when going beyond basic ASCII: script direction, contextual forms, combining characters and so on make the correct presentation of labels a complex task. This clause 8 specifies additional signalling to allow key complexity features of labels to be conveyed to receivers and to allow regional profiles to be created (see ETSI TS 101 756 [2], annex E). The content of regional profiles, which includes the sets of characters required to support the labels and the presentation capabilities needed, should be determined collaboratively by broadcasters, regulators and equipment suppliers - if more capabilities are specified than needed, then products may be overly expensive; if too few capabilities are specified, users will be frustrated.

Different receiver architectures will have an effect on how labels are decoded and processed in receivers: an embedded receiver will need to examine and process the label in detail to match its capability, whereas a device using a sophisticated external display may only need to provide reassembly and forwarding.

8.2 Unicode

Most scripts are written and read in a Left-To-Right (LTR) direction, whilst other scripts use a Right-To-Left (RTL) direction; examples are Latin (LTR) and Arabic (RTL). In order to deal with this situation, the Unicode standard [3] defines methods to determine which direction the text should be presented, and rules for when LTR and RTL text are present in the same block of characters, the Unicode bidirectional algorithm, or Bidi for short. Bidi requires that the base direction is specified - that is the direction that should be used when the direction cannot be derived solely from the code points of the text block.

Unicode defines the logical order of the characters as the order in which they are typed. SI labels and dynamic labels are encoded in the Unicode logical order. That means that for LTR only text, the first character will appear left-most in the display, and for RTL only text, the first character will appear right-most on the display. For mixed direction text, the position of the first character of the label is determined by the base direction and the content of the first text block of the label.

EXAMPLE 1: Label (logical order): english words ARABIC WORDS mots français

Base direction	Display shows
LTR	english words SDROW CIBARA mots français
RTL	mots français SDROW CIBARA english words

The first text block is LTR script (shown in lower case), the second is RTL script (shown in UPPER CASE), the third is LTR script: when the base direction is LTR the first text block (LTR) is presented at the left of the display, followed by the second block (RTL), followed by the third block (LTR); when the base direction is RTL the first text block (LTR) is presented at the right of the display, followed by the second block (RTL), followed by the third block (LTR). Each text block is presented in the direction of the script, therefore LTR characters are always presented with the first character of the block on the left, and RTL characters are always presented with the first character of the block on the right.

Numerals, regardless of script, are always presented LTR even when the script direction is RTL. Some characters are designated mirrored characters, such as brackets, quotation marks, etc., whereby a pair of glyphs are mirror images of each other and act as opening and closing marks for a phrase of text. The presentation of these characters is dependent on the script direction.

EXAMPLE 2: Label (logical order): ARABIC 1234 (ABC} WORDS english 1234 [abc» words

Base direction	Display shows
LTR	SDROW {CBA) 1234 CIBARA english 1234 [abc» words
RTL	english 1234 [abc» words SDROW {CBA) 1234 CIBARA

The brackets are mirrored in the RTL text block but not in the LTR text block. The numbers are always presented LTR.

Unicode provides code points for characters but it does not define the glyph that should be used when presenting the character on screen. In many cases the code point can be mapped to a single glyph for a particular font, but in many others, the glyph to be displayed is selected according to the context of the character by analysis of the code points of the surrounding characters. This is particularly true for cursive scripts, such as Arabic, although presentation code point ranges may also be available. These scripts typically have characters that can have markedly different display widths.

Presentation code points are used in the Complete EBU Latin-based repertoire character set: code points exist for base characters and for the same character with particular diacritics. Unicode permits the same character to be represented by following a base character with a combining character, for example e + ˆ= ê (U+0065 followed by U+0302 produces the same glyph as U+00EA). This type of combining character is called a non-spacing combining character because it does not change the width of the base character. In Indic scripts, there are also spacing combining characters which can extend the width of the character to the left, to the right or to both, and in these scripts the only way to display the script is to create the final glyph by combining glyph parts - presentation forms are not available.

8.3 Text control field

8.3.1 Introduction

For non-Latin scripts, or text requirements which go beyond the coverage provided by the Complete EBU Latin-based repertoire character set (see ETSI TS 101 756 [2], annex C), the encoding that uses the fewest bytes to represent the label should be used, either UTF-8 or UTF-16.

NOTE: UTF-16 is an extension of UCS-2, which is specified in ETSI EN 300 401 [1]; in DAB the code range is restricted to 2-byte characters (i.e. basic multilingual plane only).

The correct presentation of characters is non-trivial when going beyond basic ASCII: script direction, contextual forms, combining characters and so on make the correct presentation of labels a complex task.

Additional signalling is provided for FIG type 2 labels and dynamic labels in the form of the **text control field** to provide an indication of the complexity of the label and which allows receivers to better determine if they have the necessary capabilities to correctly present the label.

No provision is made to carry the text control field for FIG type 1 labels because their use is restricted to the Complete EBU Latin-based repertoire character set.

8.3.2 Encoding of the text control field

The structure of the text control field is shown in figure 5.

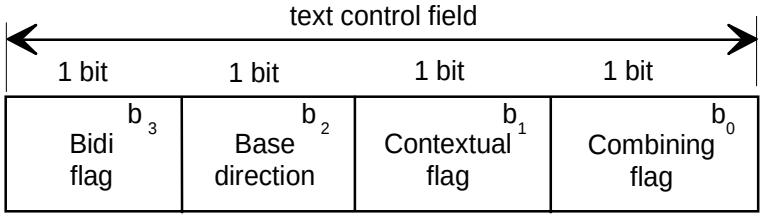


Figure 5: Structure of the text control field

Bidi flag: this 1-bit flag shall indicate whether the label contains bidirectional text (excluding numerals) as follows:

0: bidirectional text is not present;

1: bidirectional text is present.

Base direction: this 1-bit flag shall define the Unicode base direction of the label as follows:

0: left-to-right (LTR);

1: right-to-left (RTL).

Contextual flag: this 1-bit flag shall indicate whether contextual characters are used in the label as follows:

0: contextual characters are not present (presentation characters only);

1: contextual characters are present.

Combining flag: this 1-bit flag shall indicate whether combining characters are used in the label as follows:

0: combining characters are not present;

1: combining characters are present.

The bidi flag and base direction are used to indicate the directional features of the label. They provide information to the receiver about the complexity of the label content.

When the bidi flag is set to 0, it indicates that the label contains text in only one writing direction, either LTR or RTL, as indicated by the base direction flag. It indicates that the label can be presented correctly without the Unicode bidirectional algorithm being implemented. However, RTL labels may include blocks of numeric characters that need to be presented LTR and mirrored characters that need to be presented correctly and so analysis and processing of RTL labels is always required. Labels with the bidi flag set to 0 can also be processed by receivers that implement the Unicode bidirectional algorithm.

When the bidi flag is set to 1, it indicates that the label contains text blocks in both writing directions. To correctly display these labels, receivers need to implement the Unicode bidirectional algorithm.

The base direction flag indicates the default writing direction, regardless of the setting of the bidi flag. It shall be set according to the desired writing direction.

The contextual flag and combining flag are used to indicate whether the label contains characters that change their glyph according to context, and whether the label contains characters that combine to produce a single glyph, respectively. They provide information to the receiver about the complexity of the label content.

When the contextual flag is set to 0 it indicates that only presentational code points are present in the label, that is, that there is a 1:1 relationship between a code point and the required glyph (taking account of writing direction for mirrored characters).

When the contextual flag is set to 1 it indicates that code points are present in the label that require the displayed glyph to be chosen according to the surrounding characters. This may be according to the character position in the word (for example the isolated, initial, medial or finial forms of Arabic characters) or rendering styles (for example the repeated consonants in south Asian scripts).

When the combining flag is set to 0, it indicates no combining characters are present in the label and therefore the receiver is not required to compose the glyph from glyph parts.

When the combining flag is set to 1, it indicates that combining characters are present in the label and therefore the receiver is required to be able to compose the glyph from glyph parts. Combining characters can be non-spacing, in which case the width of the base character is unaltered, or spacing, in which case the width of the character may spread to the left, to the right or both. More than one combining character may follow a base character.

8.3.3 Transport of the text control field

8.3.3.1 FIG type 2 labels

ETSI EN 300 401 [1], clause 5.2.2.3.1 defines the FIG type 2 extended label. As defined, FIG type 2 extended labels carry a flag field to allow the creation of a shortened label. Since this feature is no longer required, and the capacity of the FIC is limited, the **Rfu** bit that is present between the **Segment index** and **Extension** fields is used to modify the definition. When this bit is set to 0, the extended label data field is exactly as defined in [1], clause 5.2.2.3.2. However, when the **Rfu** bit is set to 1, it indicates that the 16-bit **Character flag field** is absent from the extended label data field and that the 4-bit **Rfa field** is replaced by the **text control field**, see figure 6.

NOTE: The next revision to ETSI EN 300 401 [1] will include these changes.

Transmission equipment shall set the **Rfu** bit to 1. Receivers shall check the **Rfu** bit. They are only required to decode the FIG type 2 label when the **Rfu** bit is set to 1. When the **Encoding flag** is set to 1 it signifies UTF-16 character encoding, restricted to 2 bytes per character, big endian byte order (i.e. basic multilingual plane only).

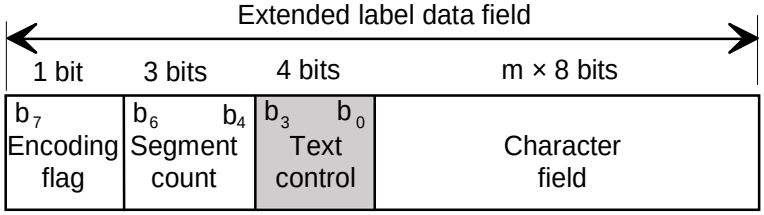


Figure 6: Structure of the extended label data field (Rfu bit = 1)

8.3.3.2 Dynamic labels

For dynamic labels, the **text control field** replaces the 4-bit **Rfa field** in the control field when the **C flag** = 0 and the **first flag** = 1, see ETSI EN 300 401, clause 7.4.5.2 [1], and figure 7.

NOTE: The next revision to ETSI EN 300 401 [1] will include this change.

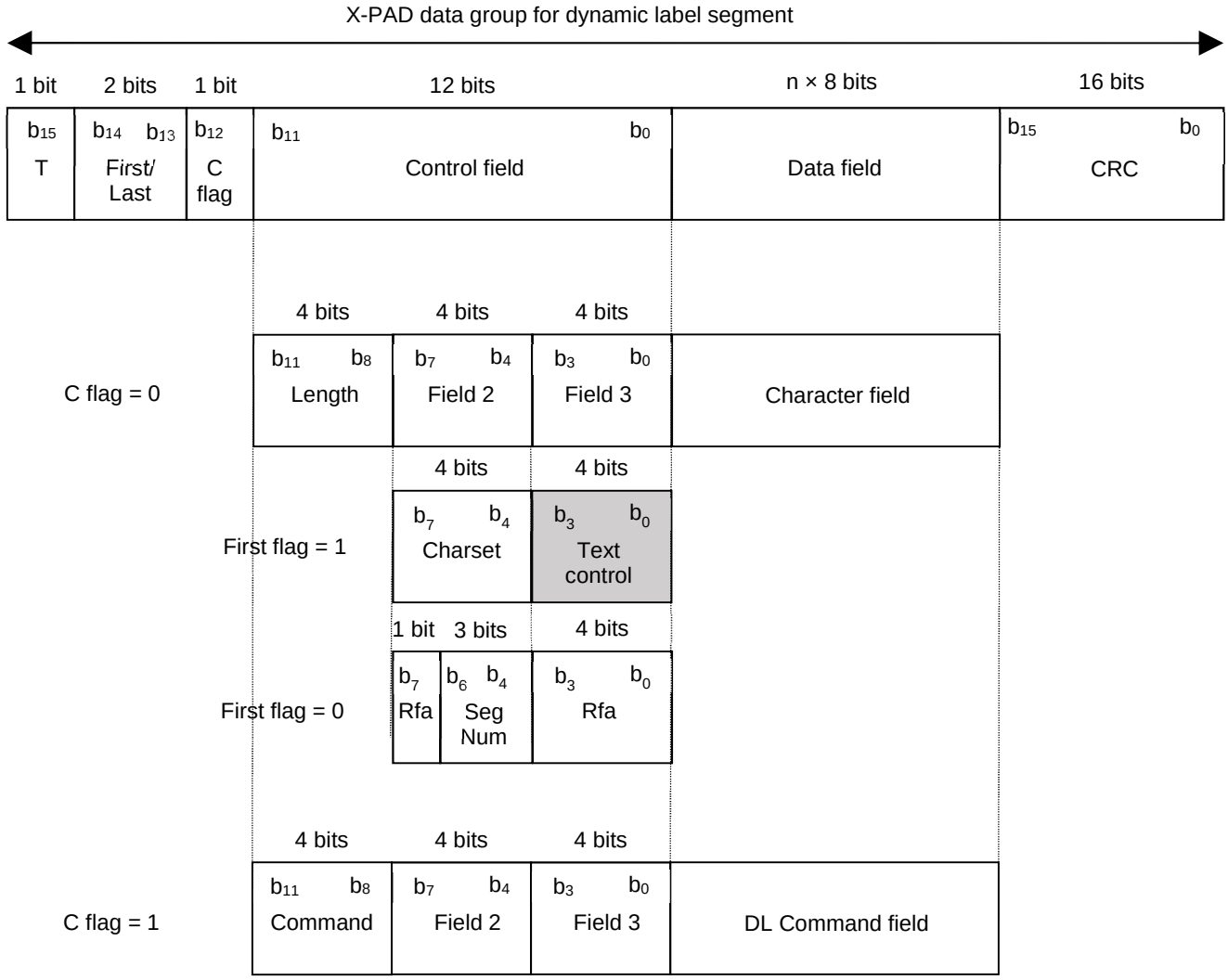


Figure 7: Structure of the X-PAD data group for each dynamic label segment

8.4 Decoding and presenting text labels

Different geographical regions have different requirements for text support, based on the languages in common use by the broadcasters of the region. The capability of receivers to decode and present text labels is therefore an important factor in making a successful digital radio launch and deployment.

For any region that requires greater language support than provided by the Complete EBU Latin-based repertoire [2], receivers shall include decoding of FIG 2 labels and the **text control field** in both FIG 2 labels and dynamic labels. The regional profiles specified in ETSI TS 101 756 [2], annex E are designed to reflect the minimum capability that is necessary within the region; receivers with multi-profile capabilities are also encouraged. Receivers that do not meet the needs of the relevant regional profile should not be marketed in that region as they will not provide the proper digital radio experience.

FIG type 2 labels and dynamic labels are composed of one or more segments. The general sequence for label handling is as follows:

- The label shall be fully reassembled; if segments are lost due to reception errors the complete label shall be discarded.
- The **text control field** shall be analysed: each of the flags when set requires a certain capability in the receiver to present the label correctly.
- The label shall be processed according to the flag settings, receiver capabilities, display capabilities, etc.

The actual processes required are dependent on the architecture of the receiver.

The **text control field** provides flags which allow receivers to know which rendering capabilities they need to display the label correctly. If a label requires capabilities that the receiver does not have then the label cannot be presented. For such labels, a strategy to provide alternative labels to allow each service element to be selected by the user shall be implemented. This situation will only occur if a receiver designed for a non-compatible regional profile is brought to a location that requires a different regional profile. Receivers brought to the market in regions where a dedicated regional profile exists shall support all requirements of that regional profile.

8.5 Display of RTL and bi-directional labels

8.5.1 Introduction

Labels which contain RTL text will have the **base direction** set to RTL and/or the **bidi flag** set to 1. These flags allow a receiver to determine whether it has the capabilities to display the label. Table 5 provides an overview of the required capabilities.

Table 5: Receiver capability requirements

Bidi flag	Base direction	Label content	Receiver capability needed for presentation
0	0	LTR script(s) only	LTR presentation
0	1	RTL script(s) only	RTL presentation
1	0	LTR script(s) with RTL script block(s)	Bidirectional presentation
1	1	RTL script(s) with LTR script block(s)	Bidirectional presentation

Receivers that implement the Unicode bidirectional algorithm are able to process all labels regardless of the settings of the **bidi flag** and **base direction** (although presentation depends on the settings of the **contextual flag** and **combining flag** and access to glyphs for all contained code points).

Receivers that present RTL scripts shall implement either the Unicode bidirectional algorithm [4] or the minimum RTL implementation (see clause 8.5.2) in order to achieve correct presentation. When the Unicode bidirectional algorithm is implemented, it is recommended to enclose the label with directional isolates (if implemented, or embedding if not) to take account of the setting of the **base direction**.

Receivers that present bidirectional text (i.e. text with the **bidi flag** set to 1) shall implement either the Unicode bidirectional algorithm [4] or the simplified bidirectional algorithm (see clause 8.5.3), the latter requiring the text to contain explicit direction control using directional isolates for proper display.

Receivers that do not implement either the Unicode bidirectional algorithm or the simplified bidirectional algorithm shall not present labels with the **bidi flag** set to 1.

Receivers that scroll the display for dynamic label presentation shall take account of the **base direction** and particular care in the presentation of bidirectional labels.

8.5.2 Minimum RTL implementation

Receivers that do not implement the Unicode bidirectional algorithm may choose to implement the minimum RTL implementation. This implementation allows labels signalled as **bidi flag** = 0, **base direction** = 1 to be correctly formatted (although the settings of the **contextual flag** and **combining flag** may still mean that the label cannot be correctly displayed). In respect of the Unicode bidirectional algorithm [4], when **bidi flag** = 0, **base direction** = 1 the label will not contain any character of bidi class L.

Numeric digits, and their associated characters, are always displayed LTR, even in RTL scripts. The minimum RTL implementation therefore requires numeric digits in the label to be identified, and processing applied in order to present them correctly. For embedded displays, this can be done by presenting the characters in identified numeric blocks in reverse order. Numeric blocks are sequences of characters that include at least one numeric digit, as defined in table 6. The blocks shall be determined by searching for a numeric digit and then including each character before and after until a space character, U+0020, is found. If the space character has a numeric digit on both sides, then the block shall be extended until the next space character is found, and so on.

Table 6: Numeric digits

Numeral type	Characters	Unicode codepoints
European digits	0 1 2 3 4 5 6 7 8 9	U+0030 to U+0039
Arabic-Indic digits	٠ ١ ٢ ٣ ٤ ٥ ٦ ٧ ٨ ٩	U+0660 to U+0669
Eastern Arabic-Indic digits	۰ ۱ ۲ ۳ ۴ ۵ ۶ ۷ ۸ ۹	U+06F0 to U+06F9

NOTE 1: This simple algorithm generally produces similar output to the Unicode bidirectional algorithm for text labels when **bidi flag** = 0, **base direction** = 1. However, it cannot correctly deal with all situations.

EXAMPLE: "123", "4 567", "$10.99", "66%", "11/03/2018", "15:20", "14€30", "85°F", "١٢٣٫٤٥٦", "٠٫٧٨٩" are all numeric blocks.

Some characters are designated mirrored characters whereby a pair of glyphs are mirror images of each other and act as opening and closing marks for a phrase of text. The presentation of these characters is dependent on the writing direction. The minimum RTL implementation therefore requires the identification of the following mirrored characters and for processing to be applied to present them correctly: ( ) [ ] { } < > « ».

NOTE 2: In the Unicode bidirectional algorithm, the *pairs* of mirrored characters are identified as they may be nested and enclose mixed direction scripts. However, since only labels with the **bidi flag** set to 0 (or text blocks isolated by the simplified bidirectional algorithm) are processed by the minimum RTL implementation, it is not required to find the pairs since the text is assumed to be well formed. Unicode defines more mirrored characters than those required in the minimum RTL implementation; if these additional characters are present in the text block they will not be mirrored by this algorithm.

8.5.3 Simplified bidirectional algorithm

Receivers that do not implement the Unicode bidirectional algorithm may choose to implement the simplified bidirectional algorithm in order to achieve basic support for bidirectional text created by service providers according to the rules of the simplified bidirectional algorithm. This implementation allows labels signalled as **bidi flag** = 1 to be correctly formatted (although the settings of the **contextual flag** and **combining flag** may still mean that the label cannot be correctly displayed). The simplified bidirectional algorithm requires the implementation of the minimum RTL implementation (see clause 8.5.2).

Labels encoded using the simplified bidirectional algorithm shall explicitly signal text blocks which are not in the same direction as that indicated by the **base direction**. In this way, receivers are able to detect which parts of the label shall be presented LTR and which shall be presented RTL.

Bidirectional labels with the **base direction** indicating LTR shall insert the Right-to-Left-Isolate (RLI, U+2067) control character before an RTL text block and shall insert the Pop-Directional-Isolate (PDI, U+2069) control character at the end of the RTL text block. No other directional control codes shall be used. One or more RTL text blocks may be signalled.

Bidirectional labels with the **base direction** indicating RTL shall insert the Left-to-Right-Isolate (LRI, U+2066) control character before the LTR text block and shall insert the Pop-Directional-Isolate (PDI, U+2069) control character at the end of the LTR text block. No other directional control codes shall be used. One or more LTR text blocks may be signalled.

For the examples of bidirectional text provided in clause 8.2, the label will be modified as follows when encoded according to the simplified bidirectional algorithm:

EXAMPLE 1: If **base direction** = LTR
Label: english words <RLI>ARABIC WORDS<PDI> mots français
If **base direction** = RTL
Label: <LRI>english words<PDI> ARABIC WORDS <LRI>mots français<PDI>

Base direction	Display shows
LTR	english words SDROW CIBARA mots français
RTL	mots français SDROW CIBARA english words

The position and type of isolates are determined by the base direction of the complete label. The first text block is LTR script (shown in lower case), the second is RTL script (shown in UPPER CASE), the third is LTR script. The control characters are shown as <LRI>, <RLI> and <PDI>. The presentation is identical whether processed by the simple bidirectional algorithm or the Unicode bidirectional algorithm.

EXAMPLE 2: If **base direction** = LTR
Label: <RLI>ARABIC 1234 (ABC} WORDS<PDI> english 1234 [abc» words
If **base direction** = RTL
Label: ARABIC 1234 (ABC} WORDS <LRI>english 1234 [abc» words<PDI>

Base direction	Display shows
LTR	SDROW {CBA) 1234 CIBARA english 1234 [abc» words
RTL	english 1234 [abc» words SDROW {CBA) 1234 CIBARA

The isolates relate to the text blocks and so no isolates are provided for the numeric block within the RTL text block since the minimum RTL algorithm is part of the simplified bidirectional algorithm.

A receiver that implements the simplified bidirectional algorithm shall analyse each label with the **bidi flag** set to 1 and determine whether it contains at least one RLI control code if the **base direction** = LTR or at least one LRI control code if the **base direction** = RTL; if so, the text can be processed, if no such control code is found, then the text cannot be correctly presented since it requires the full Unicode bidirectional algorithm.

When the **base direction** is LTR, the label will be presented from left to right on the receiver display. Providing the first character is not an RLI control code, the first text block is LTR and shall be displayed LTR until the RLI control code is reached. The RLI, at whatever position in the text it occurs, indicates the start of an RTL text block, and the receiver shall determine how much horizontal space is required for this RTL text block by searching for the PDI control code, and use the minimum RTL implementation to correctly deal with numerical blocks and mirrored characters. Characters following the PDI, if any, are LTR and the process repeats until the entire text is displayed.

When the **base direction** is RTL, the label will be presented from right to left on the receiver display. Providing the first character is not an LRI control code, the first text block is RTL and shall be displayed RTL until the LRI control code is reached, using the minimum RTL implementation to correctly deal with numerical blocks and mirrored characters. The LRI, at whatever position in the text it occurs, indicates the start of an LTR text block, and the receiver shall determine how much horizontal space is required for this LTR text block by searching for the PDI control code. Characters following the PDI, if any, are RTL and the process repeats until the entire text is displayed.

NOTE: Receivers that implement the Unicode bidirectional algorithm will correctly process labels encoded using the simple bidirectional algorithm and so do not need to follow the simplified bidirectional algorithm search process.

Annex A (informative): Service following - use case examples

A.1 DAB to DAB in Multi-Frequency Networks

A.1.1 Problem Description

The same ensemble is transmitted on channel 5A and channel 5C (174 928 kHz and 178 352 kHz). Both areas are adjacent. The transmission mode is I. The signals are co-timed and synchronized.

Users may expect almost seamless switching between the different frequencies when the ensembles are identical and synchronized.

Synchronized means that the frame start (Null symbol) is sent at precisely the same moment in time, the interleaving is identical and the guard interval is respected. A synchronized MFN is managed like an SFN, the only difference being that the signal is modulated on more than one frequency.

A.1.2 Concept

A dual tuner device will find the alternative by scanning in the background.

A single tuner/decoder device should try to monitor the signal strength at all the frequencies indicated by the frequency information in FIG 0/21. If a signal is available on another frequency it might try to retune the tuner while keeping the decoder pipeline in operation. Confirming the EId of the chosen frequency before switching reception of the MSC should provide for seamless switching.

A.1.3 Signalling

A.1.3.1 FIG 0/6 Service linking information

Not required for this use case.

A.1.3.2 FIG 0/24 OE Services

Not required for this use case.

A.1.3.3 FIG 0/21 Frequency Information

Table A.1

Field	Value	Description
C/N	0	Start of database
OE	0	Tuned ensemble
P/D	0	Not used for Frequency Information
Length of FI list	9	Length in bytes of the following FI list
Id field (Identifier field)	0xD201	EId
R&M	0b0000	Range and Modulation, DAB ensemble.
Continuity flag	1	The ensemble is co-timed and synchronized
Length of Freq. list	6	Length in bytes of the following Freq list
Control field	0b00010	Adjacent area, transmission mode I
Freq	0x02AB5	The multiplier of 16 kHz of the centre frequency 174 928 kHz
Control field	0b00010	Adjacent area, transmission mode I
Freq	0x02B8B	The multiplier of 16 kHz of the centre frequency 178 352 kHz

A.1.4 Receiver Behaviour

A.1.4.1 Single tuner device

When the signal quality is too low, the receiver attempts to recover the ensemble from the alternate frequency. The EId is checked to ensure that the same ensemble has been found. If the switching occurs before the null symbol, it may be possible to keep the decoder pipeline in operation and so avoid audio loss.

If the ensemble is not found on the alternate frequency, the receiver may monitor both frequencies to determine when signal recovery is possible.

A.1.4.2 Multiple tuner device

The second tuner should be used to monitor the signal quality of the alternate frequency. As the signal quality of the two signals varies, the receiver can choose which provides the best user experience. In general, the decoder pipeline can be kept in operation and so no audio loss will be experienced when switching between tuners.

A.2 Same service on different ensembles

A.2.1 Problem Description

The same programme service (same SId) is available on multiple ensembles (differing EIds), and those ensembles are on a variety of frequencies.

The following example represents a case in Switzerland in 2011.

Figure A.1 illustrates the coverage areas of three different ensembles. The example explains service following for one service that is available in all ensembles. The bold line illustrates a possible travelling route between the different coverage areas.

The ensemble with EId 0x4001 is geographically adjacent to the ensemble with EId 0x4041 and some areas have coverage from both ensembles, for example, when travelling from point A to point B. The ensemble with EId 0x4081 is surrounded by high mountains and has no overlapping coverage areas with any of the other ensembles. This region can only be reached via tunnels or passes. So travelling from any of the other ensembles to the coverage area of ensemble 0x4081 will result in lost DAB reception.

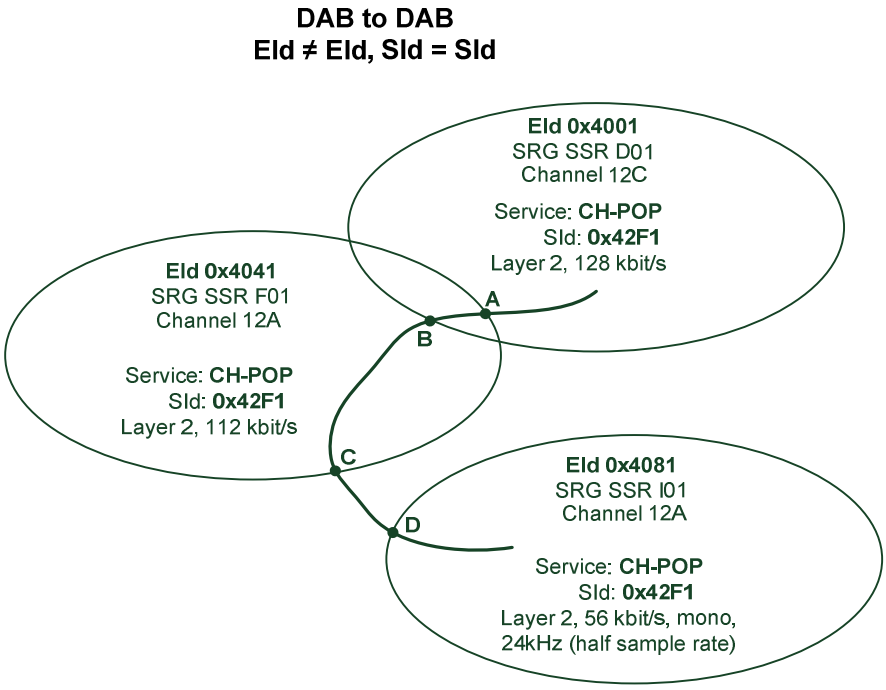


Figure A.1: Ensemble coverage schematic

A.2.2 Concept

Signalling of FIG 0/24 will help a single tuner to identify alternative ensembles carrying the same service.

Signalling of FIG 0/21 will help a single tuner to find the frequencies of the alternative ensembles.

A dual tuner may detect the same service on a different ensemble and frequency by periodic analysis of received signals.

A.2.3 Signalling

A.2.3.1 FIG 0/6 Service linking information

Not required for this use case.

A.2.3.2 FIG 0/24 OE Services

Signalled in all ensembles, EIds in the same order.

Table A.2

Field	Value	Description
C/N	0	Start of database
OE	0	Tuned ensemble (the service exists in the tuned Ensemble)
P/D	0	Programme service
SId	0x42F1	Service for which ensemble information is being signalled
CAId	0	No Conditional Access applicable
Number of EIds	3	Number of Ensemble Ids to follow
Ensemble Identifier	0x4001	Ensemble Id of an ensemble carrying the Service
Ensemble Identifier	0x4041	Ensemble Id of an ensemble carrying the Service
Ensemble Identifier	0x4081	Ensemble Id of an ensemble carrying the Service

A.2.3.3 FIG 0/21 Frequency Information

Signalled *differently* in each ensemble since this is the OE frequency information. Signalled in Ensemble **0x4001** as in table A.3.

Table A.3

Field	Value	Description
C/N	0	Start of database
OE	1	Other Ensemble (the frequencies belong to other ensembles)
P/D	0	Not used
Length of FI list	12	Length in bytes of the following FI list
Id field (Identifier field)	0x4041	EId of the other ensemble
R&M	0b0000	Range and Modulation, DAB ensemble
Continuity flag	0	Continuous audio output not expected
Length of Freq. list	3	Length in bytes of the following Freq list
Control field	0b00010	Adjacent area, transmission mode 1
Freq	0x036AC	223 936 kHz
Id field (Identifier field)	0x4081	EId of the other ensemble
R&M	0b0000	Range and Modulation, DAB ensemble
Continuity flag	0	Continuous audio output not expected
Length of Freq. list	3	Length in bytes of the following Freq list
Control field	0b00011	Not adjacent area, transmission mode 1
Freq	0x036AC	223 936 kHz

A.2.4 Receiver Behaviour

A receiver should decide a point at which the currently received signal is inadequate. For a single tuner device this may be through use of a fixed reference point, and for a dual tuner by comparison with the quality of signals being received from other frequencies.

The receiver should attempt to locate the current service on an alternative ensemble. It may use the OE services and frequency information provided by the broadcaster.

If this fails, or no OE services information is being signalled, the receiver will attempt to locate the current service by scanning all possible frequencies and analysing all the services on the ensembles found.

For example, travelling from point A to C in figure A.1: between A and B, both ensembles are available and in both ensembles FIG 0/24 and FIG 0/21 are signalled. Switching between the ensembles may occur at any time.

A.3 Linking regional variations of a service on different ensembles

A.3.1 Problem Description

A programme service is broadcast across multiple regions, on different ensembles. At certain times the programme service is identical on all ensembles, and at other times the content is different. As the programme content is not always identical on all ensembles at all times, each regional variation has a different SId code. The periods of non-identical content may be of only a few minutes duration, such as commercial breaks or news bulletins, or may be for the duration of a programme segment.

At times when the programme service is identical, the broadcaster wants the listener to experience continuous coverage, as if the service was transmitted with the same SId code on all ensembles.

At times when the programme service is not identical, the broadcaster does not want the listener to be switched between regions automatically by the receiver.

The broadcaster can control whether linking is active or inactive, and can do so quickly and often.

In figure A.2 the four related services are carried on four ensembles. On weekdays the following schedule applies: from midnight until 06:00, the four services carry the same audio; from 06:00 until 10:00, the four services are all separate; from 10:00 until 16:00 the four services again have identical audio; from 16:00 until 19:00 the two services in the west share one programme and the two services in the east share a different programme; from 19:00 until midnight all four services are again identical. On Saturday, the schedule has all four services carrying the same audio except between 14:30 and 17:00 when the two services in the north carry one football match, whilst the two services in the south each carry their own football match. On Sundays, all services carry the same audio.

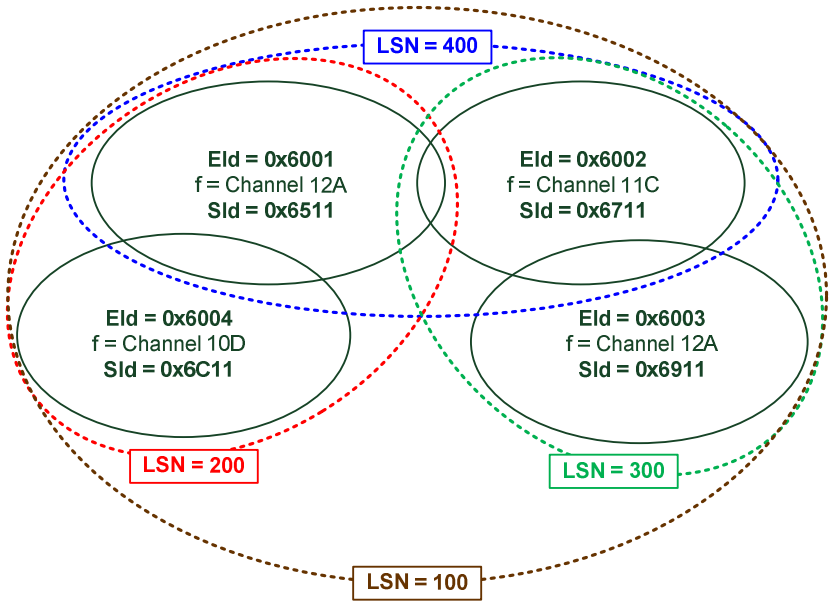


Figure A.2: Service and ensemble configuration

A.3.2 Concept

There are four network configurations with services that need to be linked and each is represented by a linkage set. The linkage sets are signalled in each ensemble that has a service that is linked, so all ensembles include the linkage set for all four services linked together and the two western ensembles also have the linkage set for the two western services linked whilst the two eastern ensembles also have the linkage set for the two eastern services linked. In addition, the two northern ensembles have the linkage set for the two northern services linked. FIG 0/6 in the long form is used to repeat these definitions throughout the day. According to the time of day, the appropriate linkage set is activated or deactivated using FIG 0/6 in the short form.

To assist single tuner devices, the OE Services (FIG 0/24) and frequency information (FIG 0/21) may be signalled. OE Services links the SId and EId and frequency information provides the centre frequency of the ensembles.

A dual tuner may detect the linked services on other ensembles by periodic analysis of received signals.

A.3.3 Signalling

A.3.3.1 FIG 0/6 Service linking information

The information describing the linkage set for all services linked together (LSN = 100) is signalled on all four ensembles. The information describing the linkage set when the western services are linked together (LSN = 200) is signalled on the two western ensembles. Similarly, for the eastern (LSN = 300) and northern (LSN = 400) linkage sets. Therefore each ensemble will signal three linkage sets at all times. The state of the link actuators will always be set in accordance with the linkage schedule both in the slower cyclic repetition of the long form database definition FIGs and the faster cyclic repetition of the short form activation state FIGs. When the activation states change then a burst of short form FIGs are sent.

The following information is sent on ensemble 0x6001; the linkage actuator settings are according to the time of day. This snapshot was taken at 18:00 on a Wednesday when the "western" linkage set is activated.

Table A.4

Field	Value	Description
C/N	0	Start of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x100	All services linkage set
IdLQ	0b00	Each Id represents a DAB SId
IdLP	0	The services in the list have a normal preference level
Number of Ids	4	Number of Ids in the list
Id1	0x6511	SId of service
Id2	0x6711	SId of service
Id3	0x6911	SId of service
Id4	0x6C11	SId of service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x200	Western linkage set
IdLQ	0b00	Each Id represents a DAB SId
IdLP	0	The services in the list have a normal preference level
Number of Ids	2	Number of Ids in the list
Id1	0x6511	SId of service
Id2	0x6C11	SId of service
Id List Flag	1	Long form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x400	Northern linkage set
IdLQ	0b00	Each Id represents a DAB SId
IdLP	0	The services in the list have a normal preference level
Number of Ids	2	Number of Ids in the list
Id1	0x6511	SId of service
Id2	0x6711	SId of service

The activation state of all the linkage sets in the ensemble is sent regularly. Some examples are given below, again for a Wednesday.

At 05:07, 11:13, 14:58, 20:03; 23:45, the following information was sent.

Table A.5

Field	Value	Description
C/N	1	Control function - activation state
OE	0	Rfu
P/D	0	Programme service
Id List Flag	0	Short form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x100	All services linkage set
Id List Flag	0	Short form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x200	Western linkage set
Id List Flag	0	Short form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x400	Northern linkage set

At 06:27, 07:53, 09:59 the following information was sent.

Table A.6

Field	Value	Description
C/N	1	Control function - activation state
OE	0	Rfu
P/D	0	Programme service
Id List Flag	0	Short form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x100	All services linkage set
Id List Flag	0	Short form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x200	Western linkage set
Id List Flag	0	Short form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x400	Northern linkage set

At 16:00, 17:13, 18:54 the following information was sent.

Table A.7

Field	Value	Description
C/N	1	Control function - activation state
OE	0	Rfu
P/D	0	Programme service
Id List Flag	0	Short form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x100	All services linkage set
Id List Flag	0	Short form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x200	Western linkage set
Id List Flag	0	Short form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x400	Northern linkage set

At 19:00:00 the network configuration changes from the western services being linked to all services being linked. The following change information was sent in a short burst.

Table A.8

Field	Value	Description
C/N	1	Control function - activation state
OE	0	Rfu
P/D	0	Programme service
Id List Flag	0	Short form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x200	Western linkage set
Id List Flag	0	Short form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x100	All services linkage set
NOTE: The deactivated linkage set is signalled first, the activated linkage set is signalled second.		

A.3.3.2 FIG 0/24 OE Services

The information describing the OE Services is signalled on each of the four ensembles for the three related services. The information is different on each ensemble because each has a different set of three other ensemble services.

The following information is sent on ensemble 0x6001.

Table A.9

Field	Value	Description
C/N	0	Start of database
OE	1	Other Ensemble (the service is carried in other ensembles)
P/D	0	Programme service
SId	0x6711	Service for which ensemble information is being signalled
CAId	0	No Conditional Access applicable
Number of EIds	1	Number of Ensemble Ids to follow
Ensemble Identifier	0x6002	Ensemble Id of an ensemble carrying the Service
SId	0x6911	Service for which ensemble information is being signalled
CAId	0	No Conditional Access applicable
Number of EIds	1	Number of Ensemble Ids to follow
Ensemble Identifier	0x6003	Ensemble Id of an ensemble carrying the Service
SId	0x6C11	Service for which ensemble information is being signalled
CAId	0	No Conditional Access applicable
Number of EIds	1	Number of Ensemble Ids to follow
Ensemble Identifier	0x6004	Ensemble Id of an ensemble carrying the Service

A.3.3.3 FIG 0/21 Frequency Information

The information describing the Frequency information is signalled on each of the four ensembles for the three related ensembles. The information is different on each ensemble because each has a different set of three other ensembles.

The following information is sent on ensemble 0x6001.

Table A.10

Field	Value	Description
C/N	0	Start of database
OE	1	Other Ensemble (the service is carried in other ensembles)
P/D	0	not used for Frequency Information
Length of FI list	18	Length in bytes of the following FI list
Id field (Identifier field)	0x6002	EId
R&M	0b0000	Range and Modulation, DAB ensemble
Continuity flag	0	The ensemble is not co-timed and synchronized
Length of Freq. list	3	Length in bytes of the following Freq list
Control field	0b00010	Adjacent area, transmission mode I
Freq	0x035CC	The multiplier of 16 kHz of the centre frequency 220 352 kHz
Id field (Identifier field)	0x6003	EId
R&M	0b0000	Range and Modulation, DAB ensemble.
Continuity flag	0	The ensemble is not co-timed and synchronized
Length of Freq. list	3	Length in bytes of the following Freq list
Control field	0b00011	Non-adjacent area, transmission mode I
Freq	0x036AC	The multiplier of 16 kHz of the centre frequency 223 936 kHz
Id field (Identifier field)	0x6004	EId
R&M	0b0000	Range and Modulation, DAB ensemble.
Continuity flag	0	The ensemble is not co-timed and synchronized
Length of Freq. list	3	Length in bytes of the following Freq list
Control field	0b00010	Adjacent area, transmission mode I
Freq	0x03482	The multiplier of 16 kHz of the centre frequency 215 072 kHz

A.3.4 Receiver Behaviour

If the receiver has determined that it needs to switch to an alternative service, and has failed to find the service following stage 1, it should refer to the FIG 0/6 information.

In each linkage set where the current SId is present in the Id list, and where the linkage set is activated (LA = 1), there is a list of alternative SId codes for the current service.

In conjunction with OE Services (FIG 0/24) and Frequency information (FIG 0/21) (where provided), the receiver should try the alternative SId codes in any sequence to locate the alternative service on a different ensemble, and with an acceptable signal quality.

The receiver should minimize the time taken to locate the alternative service.

If the receiver has current knowledge of which alternative ensembles and services can be received, for instance by using a dual tuner, then it should switch immediately to the alternative service with the best signal quality.

If ensemble information and frequency information is provided for the alternative services, then this should be used to initially try only frequencies where one or more ensembles carrying the alternative services may be received. In this case, multiple alternative services on multiple alternative ensembles may be checked on a fewer number of frequencies.

If an alternative service cannot be found on the ensembles and frequencies provided in FIG 0/24 and FIG 0/21, the receiver tries all possible frequencies to locate an alternative service.

On locating an alternative service, and concluding that the signal quality is better than the current service and all other alternatives, the receiver switches automatically and without user intervention.

After switching to the alternative service on a different ensemble, the receiver uses service linking information received from the new ensemble for further linking.

A.4 Linking technology variations of a service on different ensembles

A.4.1 Problem Description

The very same audio content may be distributed in different ensembles (different EId) with a different SId because on some ensembles the audio is coded as MPEG layer 2 (DAB), whilst on other ensembles the audio is coded as AAC (DAB+). Those ensembles may or may not have overlapping coverage areas.

The following example represents a case in Switzerland in 2011.

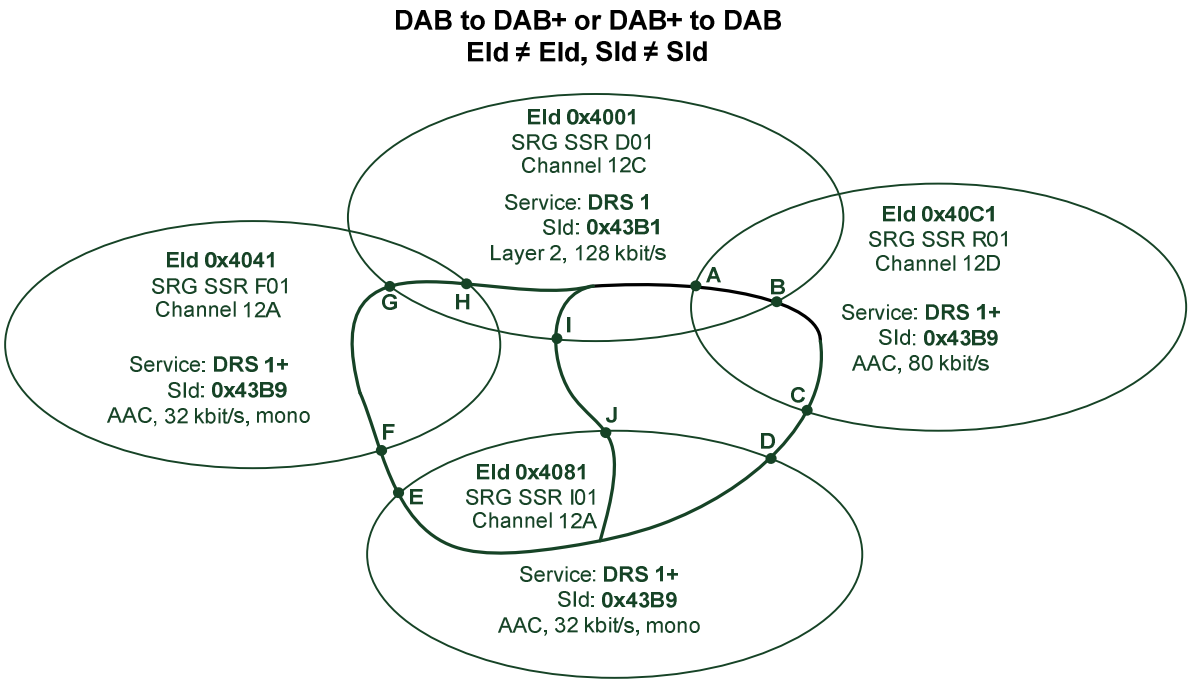


Figure A.3: Ensemble coverage schematic

A.4.2 Concept

Signalling of FIG 0/6 in both/all ensembles is be provided to link those services together.

Signalling of FIG 0/21 will help a single tuner device to find the alternative frequency.

Signalling of FIG 0/24 will help a single tuner to identify the ensemble carrying the linked service.

A dual tuner device may find the alternative frequency by scanning all frequencies in the background.

A dual tuner may also detect the linked service in one of the scanned frequencies.

A.4.3 Signalling

A.4.3.1 FIG 0/6 Service linking information

The same signalling is provided in all ensembles, but the order of the Ids will be different since the first Id is the SId of a service in the ensemble. This is the signalling in ensemble 0x4001.

Table A.11

Field	Value	Description
C/N	0	Start of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x3B1	Linkage Set Number
IdLQ	0b00	Each Id represents a DAB SId
IdLP	0	The services in the list have a normal preference level
Number of Ids	2	Number of Ids in the list
Id1	0x43B1	SId
Id2	0x43B9	SId

A.4.3.2 FIG 0/24 OE Services

In ensemble **0x4001.**

Table A.12

Field	Value	Description
C/N	0	Start of database
OE	1	Other Ensemble (the service is carried in other ensembles)
P/D	0	Programme service
SId	0x43B9	Service for which ensemble information is being signalled
CAId	0	No Conditional Access applicable
Number of EIds	3	Number of Ensemble Ids to follow
Ensemble Identifier	0x4041	Ensemble Id of an ensemble carrying the Service
Ensemble Identifier	0x4081	Ensemble Id of an ensemble carrying the Service
Ensemble Identifier	0x40C1	Ensemble Id of an ensemble carrying the Service

In the other ensembles.

Table A.13

Field	Value	Description
C/N	0	Start of database
OE	1	Other Ensemble (the service is carried in other ensembles)
P/D	0	Programme service
SId	0x43B1	Service for which ensemble information is being signalled
CAId	0	No Conditional Access applicable
Number of EIds	1	Number of Ensemble Ids to follow
Ensemble Identifier	0x4001	Ensemble Id of an ensemble carrying the service

A.4.3.3 FIG 0/21 Frequency Information

In Ensemble **0x4001.**

Table A.14

Field	Value	Description
C/N	0	Start of database
OE	1	Other Ensemble (the frequencies belong to other ensembles)
P/D	0	Not used for Frequency Information
Length of FI list	18	Length in bytes of the following FI list
Id field (Identifier field)	0x4041	EId of the other ensemble
R&M	0b0000	DAB ensemble
Continuity flag	0	Continuous audio output not expected
Length of Freq. list	3	Length in bytes of the following Freq list
Control field	0b00010	Geographically adjacent area, transmission mode 1
Freq	0x036AC	223 936 kHz
Id field (Identifier field)	0x4081	EId of the other ensemble
R&M	0b0000	DAB ensemble
Continuity flag	0	Continuous audio output not expected
Length of Freq. list	3	Length in bytes of the following Freq list
Control field	0b00011	Geographically not adjacent area, transmission mode 1
Freq	0x036AC	223 936 kHz
Id field (Identifier field)	0x40C1	EId of the other ensemble
R&M	0b0000	DAB ensemble
Continuity flag	0	Continuous audio output not expected
Length of Freq. list	3	Length in bytes of the following Freq list
Control field	0b00010	Geographically adjacent area, transmission mode 1
Freq	0x037ED	229 072 kHz

In each of the other three ensembles, the frequencies of the ensembles will be provided in the same way. The order of the frequencies in the list(s) does not have a meaning.

A.4.4 Receiver Behaviour

If the receiver has determined that it needs to switch to an alternative service, and has failed to find the service following stage 1, it should refer to the FIG 0/6 information.

In each Linkage Set where the current SId is present in the Id list, and where the link set is active (LA = 1), there is a list of alternative SId codes for the current service.

In conjunction with OE Services (FIG 0/24) and frequency information (FIG 0/21) (where provided), the receiver should try the alternative SId code to locate the alternative service on a different ensemble, and with an acceptable signal quality.

For example, travelling from point A to C in figure A.3: between A and B, two ensembles are available and in both ensembles FIG 0/24 and FIG 0/21 are signalled. Switching between the ensembles may occur at any time.

The receiver should minimize the time taken to locate the alternative service.

If the receiver has current knowledge of which alternative ensembles and services can be received, for instance by using a dual tuner, then it should switch immediately to the alternative service with the best signal quality.

If OE Services and frequency information is provided for the alternative services, then this should be used to initially try only frequencies where one or more ensembles carrying the alternative services may be received. In this case, multiple alternative services on multiple alternative ensembles may be checked on a fewer number of frequencies.

On locating an alternative service, and concluding that the signal quality is better than the current service and all other alternatives, the receiver switches automatically and without user intervention.

If an alternative service cannot be found on the ensembles and frequencies provided in FIG 0/24 and FIG 0/21, the receiver tries all possible frequencies to locate an alternative service.

If travelling from B to D in the example above, crossing an uncovered area, the receiver starts scanning at point C but the last received FIG 0/6, FIG 0/24 and FIG 0/21 remain in the memory and is used when entering the other ensemble at point D.

After switching to the alternative service on a different ensemble, the receiver uses service linking information received from the new ensemble for further linking.

A.5 Soft linking of services

A.5.1 Problem Description

The broadcaster with a number of regional variations of a programme service illustrated in clause A.3 also wishes to indicate to his listeners when a *related* service is available during the times when the services are *not* hard linked together. He may do this by creating a linkage set that includes all his DAB services and his services on other bearers too, like FM-RDS, and signalling this as a *soft* link.

At certain times, some of the regional variations carry identical content, and are linked together using hard links, whilst others are either carrying exclusive content or linked to other regional variants. The regional variants with identical content will be present in one activated Hard linkage set, but they may also be present in one activated Soft linkage set at the same time. Therefore, when a receiver cannot find the user selected service or any alternative hard linked services, it may offer the listener a soft linked service that is available.

Figure A.4 illustrates the DAB services available from the example in clause A.3 and also adds the FM-RDS services. Note that LSN = 100 has again been chosen to represent the set of all services, but this is distinguishable from the linkage set in clause A.3 because of the associated Soft/Hard flag.

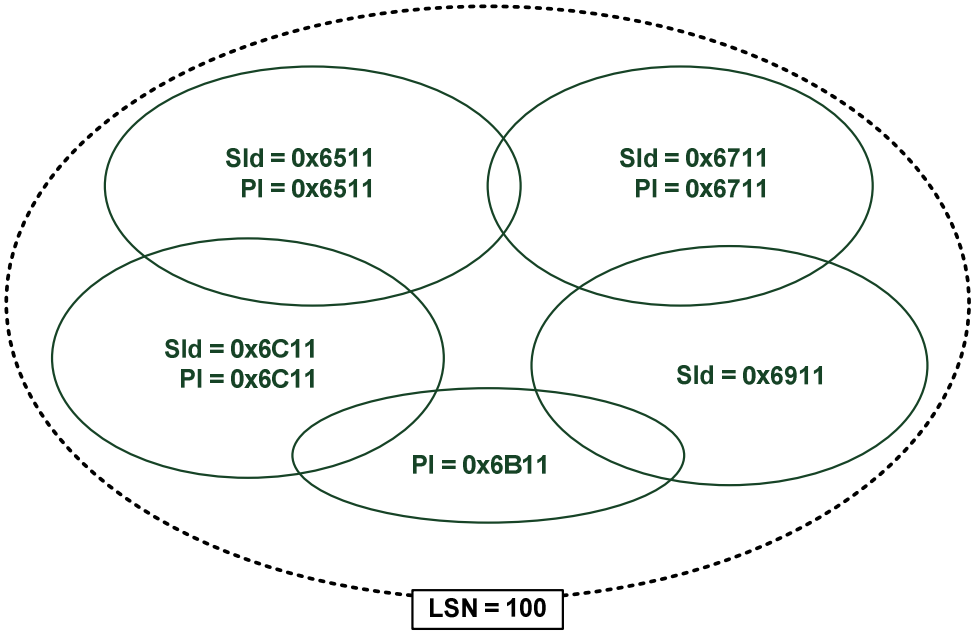


Figure A.4: Service configuration

A.5.2 Concept

All the regional variations of the programme service are contained in a single Linkage Set, with the S/H flag set to 0, indicating a soft linkage set. Some or all of the regional variations may also be in hard linkage sets, which are activated only when the programme content is identical.

The linkage set is transmitted by all the ensembles carrying the regional variations. The soft linkage set is always active.

To assist single tuner devices, the OE Services (FIG 0/24) and frequency information (FIG 0/21) may be signalled. OE Services connects the SId and EId and frequency information provides the centre frequency of the ensembles and the frequency of the FM-RDS services.

A dual tuner may detect the linked services on other ensembles by periodic analysis of received signals.

A.5.3 Signalling

A.5.3.1 FIG 0/6 Service linking information

The FIG 0/6 information describes all the similar regional variations of the programme service, both carried on DAB and FM-RDS. Each ensemble signals this linkage set at all times. Note that the same LSN is used in this example for the soft and hard linkage sets that represent all the services, but that these linkage sets are distinct because the one in clause A.3 is a hard linkage set and the one in this annex is a soft linkage set. However, different LSNs could also have been allocated.

Since the soft linkage set is always active no change FIGs will be sent. However, the linkage information will be sent every two minutes and the state of the linkage actuator will be sent every 10 seconds.

Table A.15

Field	Value	Description
C/N	0	Start of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	0	Soft link
ILS	0	National link
LSN	0x100	Linkage Set Number for all regional variants
IdLQ	0b00	Each Id represents a DAB SId
IdLP	0	The services in the list have a normal preference level
Number of Ids	4	Number of Ids in the list
Id1	0x6511	SId of Service
Id2	0x6711	SId of Service
Id3	0x6911	SId of Service
Id4	0x6C11	SId of Service

Table A.16

Field	Value	Description
C/N	1	Continuation of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	0	Soft link
ILS	0	National link
LSN	0x100	Linkage Set Number for all regional variants
IdLQ	0b01	Each Id represents an FM-RDS PI code
IdLP	0	The services in the list have a normal preference level
Number of Ids	4	Number of Ids in the list
Id1	0x6511	PI code of Service
Id2	0x6711	PI code of Service
Id3	0x6B11	PI code of Service
Id4	0x6C11	PI code of Service

A.5.3.2 FIG 0/24 OE Services

See clause A.3.

A.5.3.3 FIG 0/21 Frequency Information

See clause A.3. Frequency information could also be provided for the RDS services.

A.5.4 Receiver Behaviour

If the receiver has determined that it needs to switch to an alternative service, and has already exhausted all possibilities to link to an identical programme service, it should refer to FIG 0/6 information where the S/H flag is set to 0, indicating soft links.

The receiver may use the soft linking information to switch to a regional variant when moved from one ensemble coverage area to another. Mobile receivers should try to ensure that switching back and forth between regional variants does not occur too frequently and cause user dissatisfaction.

A.6 Linkage of DAB and FM-RDS services

A.6.1 Problem Description

In an area where DAB signal quality is too low, and the listened service is not available on another frequency signalled as use case 1, 2, 3, 4, a receiver may switch to FM.

In an area where DAB signal quality becomes receivable corresponding to a listened FM service, a receiver may switch to DAB.

The following example represents a case in Switzerland in 2011.

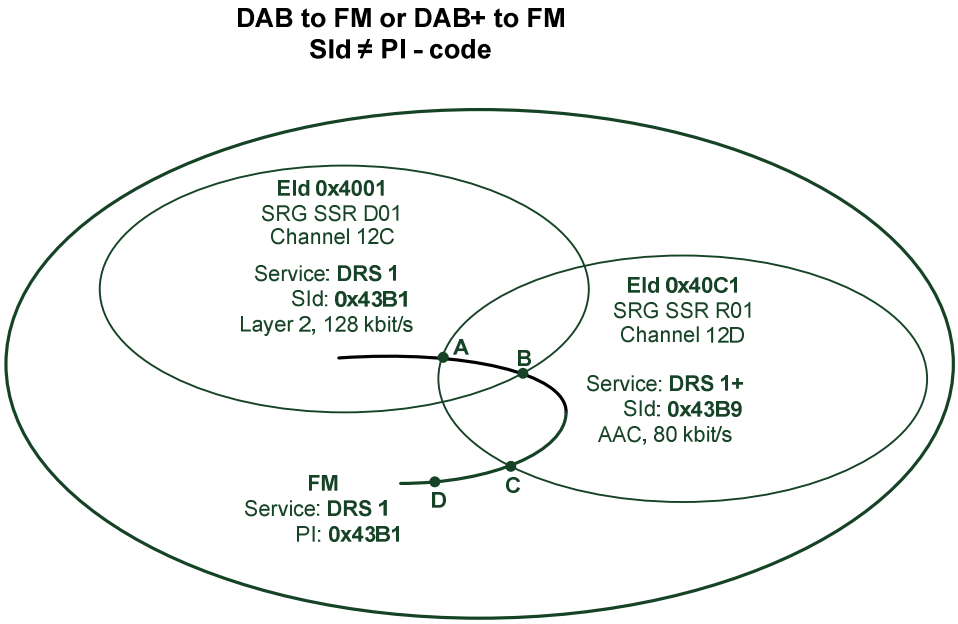


Figure A.5: Ensemble coverage schematic

A.6.2 Concept

In case of direct association PI/SId no additional signalling is needed be transmitted. FIG 0/21 may be used to inform receiver about FM frequencies. Implicit linking is considered to be a hard link.

In the example above this applies to the Ensemble 0x4001 where SId and PI code are equal.

Other FIG 0/6 may be used to create a link (hard or soft) between an SId and a PI code which are not the same.

In the example above this applies to the Ensemble 0x40C1 where SId and PI code are unequal.

Only one hard link may be activated at any given time.

A.6.3 Signalling

A.6.3.1 FIG 0/6 Service linking information

In Ensemble **0x40C1**, example above.

Table A.17

Field	Value	Description
C/N	0	Start of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x123	Linkage Set Number
IdLQ	0b01	Each Id represents an RDS PI code
IdLP	0	The services in the list have a normal preference level
Number of Ids	2	Number of Ids in the list
Id1	0x43B9	SId of the service (key service)
Id2	0x43B1	PI code of FM services to link with
NOTE: This example shows the way that the key service is a DAB SId despite the IdLQ indicating that the Ids are RDS PI codes. This allows a contraction of the signalling when there is just one DAB SId in the linkage set.		

A.6.3.2 FIG 0/24 OE Services

Not required for this use case.

A.6.3.3 FIG 0/21 Frequency Information

Frequency Information may be provided but not mandatory for this case. It helps speed up tuning to the FM alternative on a single tuner device. In large FM networks one can end up in a very long list of FM frequencies.

The following list represents the signalling of just one frequency of the example above for the Ensemble 0x40C1.

Table A.18

Field	Value	Description
C/N	0	Start of database
OE	1	Other Ensemble (the frequencies belong to other ensembles)
P/D	0	Not used for Frequency Information
Length of FI list	4	Length in bytes of FI list
Id field (Identifier field)	0x43B1	RDS PI code
R&M	0b1000	FM with RDS
Continuity flag	0	Continuous audio output not expected (audio not co-timed)
Length of Freq. list	1	Length in bytes of the following Freq list
Freq	0x3F	93,8 MHz

A.6.4 Receiver Behaviour

Travelling from point B to D in the example above:

- When leaving point C the DAB reception fails and no other DAB Ensemble can be tuned to.
- If the receiver has determined that it needs to switch to an alternative service, it should refer to the FIG 0/6 information.

- In conjunction with Frequency (FIG 0/21) information the receiver should try the alternative FM frequency to locate the alternative service, and with an acceptable signal quality.
- The receiver should minimize the time taken to locate the alternative service on FM.
- The receiver should continue to follow the DAB service linking rules even when it has switched to an alternative on FM.

A.7 Linking regional variations using explicit and implicit linking

A.7.1 Problem Description

In one ensemble (EId 0x4001) there are 7 (DAB+) services carrying the same content most of the time, but different content at some times. Some, not all, of these services are also available in other ensembles. During the time of identical audio, there is one FM service available in the same coverage areas using the PI code 0x43B1. During the time of non-identical audio, the broadcaster changes the PI code to regional and there are 7 regional FM services available (PI codes from 0x44B1 to 0x4AB1).

At times when the audio content is identical, the broadcaster wants the listener to experience continuous coverage as he moves around the combined coverage area.

At times when the audio content is regionalized, the broadcaster wants the listener to only be switched between sources of the same regional variant.

A.7.2 Concept

There are 8 network configurations and so 8 linkage sets with services that need to be linked, but only one configuration is represented by a hard linkage set that includes all 7 SIds and 1 PI code. When the network is regionalized, implicit linking is sufficient.

During the time when the content is identical, the broadcaster activates the hard linkage set and sets the PI codes of all the FM services to 0x43B1. During the time when the content is regionalized, the broadcaster deactivates the hard linkage set and changes the PI codes of all the FM services to the regional values. In this case every regional DAB service is linked with one regional FM service by implicit linking (SId == PI code).

The hard linkage set is signalled 24 hours a day in each ensemble that has a service that is linked. The transmission order list is determined for each ensemble.

To assist single tuner devices, the OE Services (FIG 0/24) and frequency information (FIG 0/21) may be signalled. A dual tuner may detect the linked services on other ensembles by periodic analysis of received signals.

A.7.3 Signalling

A.7.3.1 FIG 0/6 Service linking information

The linkage set (LSN = 001) is signalled on four ensembles but the transmission order list is not the same on all four:

Ensemble 0x4001 (all 7 SIds are carried in the ensemble; any one of them could be the first Id)

SIds: 0x4AB1, 0x44B1, 0x45B1, 0x46B1, 0x47B1, 0x48B1, 0x49B1; PI codes: 0x43B1

Ensemble 0x4041 (only SId 0x44B1 is carried in the ensemble; therefore it is the first Id)

SIds: 0x44B1, 0x45B1, 0x46B1, 0x47B1, 0x48B1, 0x49B1, 0x4AB1; PI codes: 0x43B1

Ensembles 0x4081, 0x40C1 (only SId 0x4AB1 is carried in these ensembles therefore it is the first Id)

SIds: 0x4AB1, 0x44B1, 0x45B1, 0x46B1, 0x47B1, 0x48B1, 0x49B1; PI codes: 0x43B1

The following information is sent on ensemble 0x4001; the linkage actuator setting is according to the regional status. This snapshot was taken during non-regional time.

Table A.19

Field	Value	Description
C/N	0	Start of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x001	Linkage Set Number
IdLQ	0b00	Each Id represents a DAB SId
IdLP	0	The services in the list have a normal preference level
Number of Ids	7	Number of Ids in the list
Id1	0x4AB1	SId of Key service
Id2	0x44B1	SId
Id3	0x45B1	SId
Id4	0x46B1	SId
Id5	0x47B1	SId
Id6	0x48B1	SId
Id7	0x49B1	SId

RDS linkage field is transmitted in another FIG 0/6 using C/N=1.

Table A.20

Field	Value	Description
C/N	1	Continuation of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x001	Linkage Set Number
IdLQ	0b01	Each Id represents an RDS PI code
IdLP	0	The services in the list have a normal preference level
Number of Ids	1	Number of Ids in the list
Id1	0x43B1	PI code

The activation state of all the linkage sets in the ensemble is sent regularly. The example is given below, again for a non-regional time.

Table A.21

Field	Value	Description
C/N	1	Control function - activation state
OE	0	Rfu
P/D	0	Programme service
Id List Flag	0	Short form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x001	Set number

For regional time the state of Linkage Actuator is signalled as 0 in both the long and short forms.

A.7.3.2 FIG 0/24 OE Services

Not required for this use case.

A.7.3.3 FIG 0/21 Frequency Information

Not required for this use case.

A.7.4 Receiver Behaviour

If the receiver has determined that it needs to switch to an alternative service, it tries to find the same SId in other ensemble following stage 1.

If failed, the receiver analyses the FIG 0/6 information:

a) If LSN 001 is **active**, the receiver checks availability of alternative SIds and then PI codes from the Id List.

b) If LSN 001 is **inactive**, the receiver tries to perform implicit linking to PI code == SId.

A.8 Preventing implicit linkage to FM-RDS

A.8.1 Problem Description

When a service is broadcast on both DAB and FM-RDS and the SId and PI code are identical, no service linking information is needed because implicit linking provides the connection. However, if the content varies between the two bearers then the connection needs to be disabled.

A.8.2 Concept

The "dead link" mechanism is used and the linkage set is activated during the times when the content is not identical, thus overriding the implicit link.

A.8.3 Signalling

A.8.3.1 FIG 0/6 Service linking information

The linkage set is broadcast continually. The Linkage Actuator is set to 0 (deactivated) when the content is the same and to 1 (activated) when the content is different. Since this linkage set is small, it is recommended to repeat the definition every 10 seconds which means that both the definition and activation state are indicated at the required rate.

Table A.22

Field	Value	Description
C/N	0	Start of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	0 or 1	Linkage set is activated or deactivated according to requirements
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x19F	Linkage Set Number
IdLQ	0b01	Each Id represents an RDS PI code
IdLP	0	The services in the list have a normal preference level
Number of Ids	1	Only SId is given
Id1	0xC19F	SId of service that is being 'dead linked' (key service)
NOTE: This example shows the way that the key service is a DAB SId despite the IdLQ indicating that the Ids are RDS PI codes. This allows a contraction of the signalling when there is just one DAB SId in the linkage set.		

A.8.3.2 FIG 0/24 OE Services

Not required for this use case.

A.8.3.3 FIG 0/21 Frequency Information

Not required for this use case.

A.8.4 Receiver Behaviour

Receivers decoding this information disable implicit linking from this service to FM when the linkage set is activated.

A.9 Linking to another DAB service, but preventing implicit linkage to FM-RDS

A.9.1 Problem Description

Normally identifiers for DAB and FM-RDS services are drawn from the same pool of codes and regulatory authorities ensure that DAB and FM-RDS services do not have the same identifier unless they are exactly the same service. However, there may be occasions when this is not the case and a broadcaster wants to prevent receivers from service following to FM. In this case the DAB service with SId = 0xC19F carries different audio to the FM-RDS service with the same PI code and the DAB service provider wishes to prevent implicit linking from DAB to FM.

At the same time the DAB service provider wishes to signal a hard link to the DAB service with SId = 0xC19E and to the DAB service with SId = 0xC19D. For all three services, linking to FM is disabled.

A.9.2 Concept

The "dead link" mechanism is used.

A.9.3 Signalling

A.9.3.1 FIG 0/6 Service linking information

Table A.23

Field	Value	Description
C/N	0	Start of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x19F	Linkage Set Number
IdLQ	0b00	Each Id represents a DAB SId
IdLP	0	The services in the list have a normal preference level
Number of Ids	3	Number of Ids in the list
Id1	0xC19F	SId of Service
Id2	0xC19E	SId of Service
Id3	0xC19D	SId of Service

Table A.24

Field	Value	Description
C/N	1	Continuation of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x19F	Linkage Set Number
IdLQ	0b01	Each Id represents an FM-RDS PI code
IdLP	0	The services in the list have a normal preference level
Number of Ids	0	Number of Ids in the list, indicating 'dead link'

A.9.3.2 FIG 0/24 OE Services

Not required for this use case.

A.9.3.3 FIG 0/21 Frequency Information

Not required for this use case.

A.9.4 Receiver Behaviour

As general hard linking behaviour - receiver will try all provided DAB SIds, but implicit linking from these services to FM is disabled.

A.10 Linkage of DAB and DRM data services

A.10.1 Problem Description

In an area where DAB signal quality is too low, and the selected service is not available on another frequency, a receiver may switch to DRM.

In an area where DAB signal quality becomes receivable corresponding to a selected DRM service, a receiver may switch to DAB.

A.10.2 Concept

FIG 0/6 is used to signal a hard link from a DAB data service to a DRM data service. FIG 0/21 may be used to inform receivers about DRM frequencies.

A.10.3 Signalling

A.10.3.1 FIG 0/6 Service linking information

Table A.25

Field	Value	Description
C/N	0	Start of database
OE	0	Rfu
P/D	1	Data service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	1	International link
LSN	0x123	Linkage Set Number
IdLQ	0b11	Each Id represents a DRM Service Identifier
IdLP	0	The services in the list have a normal preference level
Number of Ids	2	Number of Ids in the list
Id1	0xE01243B9	SId of DAB service (key service)
Id2	0x00E0D567	SId of DRM service (the most significant byte is set to 0x00)
NOTE: This example shows the way that the key service is a DAB SId despite the IdLQ indicating that the Ids are DRM SIds. This allows a contraction of the signalling when there is just one DAB SId in the linkage set.		

A.10.3.2 FIG 0/24 OE Services

Not required for this use case.

A.10.3.3 FIG 0/21 Frequency Information

Frequency Information is recommended but not mandatory for this case. It helps speed up tuning to the DRM alternative.

Table A.26

Field	Value	Description
C/N	0	Start of database
OE	1	Other Ensemble (the frequencies belong to other ensembles)
P/D	0	Not used for Frequency Information
Length of FI list	6	Length in bytes of FI list
Id field (Identifier field)	0xD567	Two least significant bytes of DRM service identifier 0xE0D567
R&M	0b0110	DRM
Continuity flag	0	Continuous audio output not expected (audio not co-timed)
Length of Freq. list	3	Length in bytes of the following Freq list
Id field 2	0xE0	Most significant bytes of DRM service identifier 0xE0D567
Freq	0xA710	100 MHz (DRM robustness mode E)

A.10.4 Receiver Behaviour

As general hard linking behaviour - receiver will also try the provided DRM SId if the DAB reception fails.

A.11 International links

A.11.1 Problem Description

DAB audio services with different extended country codes (ECC) may be linked. Linkage sets may also contain FM-RDS services with different extended country codes (ECC) and DRM audio services. For these cases, an international link is needed (ILS = 1).

Since a particular DAB service can only be present in either an activated National linkage set or an activated International linkage set at any given time it is important to ensure that conflicts are not inadvertently signalled. In general, this is best avoided by using only linkage sets with ILS = 1 for such services.

The following example is based on a case in UK/B/NL border region in 2019.

Service "BBC WorldService" is transmitted in the UK on DAB (Ensemble EId 0xCE15, ECC 0xE1; Service SId 0xC238, ECC 0xE1), FM-RDS (PI-code 0xC218), in Belgium on DAB (EId 0x6002, ECC 0xE0; SId 0x6335, ECC 0xE0), in the Netherlands on DAB (EId 0x8181, ECC 0xE3; SId 0x6335, ECC 0xE0), and covering Europe from Cyprus on DRM (SId 0xE1C238, frequency 10 MHz).

NOTE: In fact only the DAB services mentioned were on-air in 2019 and no service following information was provided in any on-air ensemble for this service.

A.11.2 Concept

When programme services with different ECCs are present and/or DRM or AMSS services are present, the linkage set uses ILS = 1.

A.11.3 Signalling

A.11.3.1 FIG 0/6 Service linking information

Signalled in the UK ensemble:

Table A.27

Field	Value	Description
C/N	0	Start of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	1	International link
LSN	0x345	Linkage Set Number
IdLQ	0b00	Each Id represents a DAB Service Identifier
IdLP	0	The services in the list have a normal preference level
Number of Ids	2	Number of Ids in the list
Id1	0xE1C238	ECC and SId of service
Id2	0xE06335	ECC and SId of service

Table A.28

Field	Value	Description
C/N	1	Continuation of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	1	International link
LSN	0x345	Linkage Set Number
IdLQ	0b01	Each Id represents an FM-RDS PI code
IdLP	0	The services in the list have a normal preference level
Number of Ids	1	Number of Ids in the list
Id1	0xE1C218	ECC and PI code of service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	1	International link
LSN	0x345	Linkage Set Number
IdLQ	0b11	Each Id represents a DRM or AMSS SId
IdLP	0	The services in the list have a normal preference level
Number of Ids	1	Number of Ids in the list
Id1	0xE1C238	SId of service

Signalled in the B and NL ensembles:

Table A.29

Field	Value	Description
C/N	0	Start of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	1	International link
LSN	0x345	Linkage Set Number (example)
IdLQ	0b00	Each Id represents a DAB Service Identifier
IdLP	0	The services in the list have a normal preference level
Number of Ids	2	Number of Ids in the list
Id1	0xE06335	ECC and SId of service (carried in ensemble)
Id2	0xE1C238	ECC and SId of service

Table A.30

Field	Value	Description
C/N	1	Continuation of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	1	International link
LSN	0x345	Linkage Set Number
IdLQ	0b01	Each Id represents an FM-RDS PI code
IdLP	0	The services in the list have a normal preference level
Number of Ids	1	Number of Ids in the list
Id1	0xE1C218	ECC and PI code of service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	1	International link
LSN	0x345	Linkage Set Number
IdLQ	0b11	Each Id represents a DRM or AMSS SId
IdLP	0	The services in the list have a normal preference level
Number of Ids	1	Number of Ids in the list
Id1	0xE1C238	SId of service

Note that if the service provider wants to explicitly indicate that no FM-RDS services are equivalent (i.e. to disable implicit linking), then a "dead link" may be signalled by specifying an empty list of PI codes. International linkage sets may also contain "dead links".

A.11.3.2 FIG 0/24 OE Services

Not required for this use case.

A.11.3.3 FIG 0/21 Frequency Information

Frequency Information is recommended but not mandatory for this case. It helps speed up tuning to the DRM alternative.

Table A.31

Field	Value	Description
C/N	0	Start of database
OE	1	Other Ensemble (the frequencies belong to other ensembles)
P/D	0	Not used for Frequency Information
Length of FI list	6	Length in bytes of FI list
Id field (Identifier field)	0xC238	Two least significant bytes of DRM service identifier 0xE1C238
R&M	0b0110	DRM
Continuity flag	0	Continuous audio output not expected (audio not co-timed)
Length of Freq. list	3	Length in bytes of the following Freq list
Id field 2	0xE1	Most significant bytes of DRM service identifier 0xE1C238
Freq	0x2710	10 MHz (DRM robustness mode A to D)

A.11.3.4 FIG 0/9 Country, LTO and International Table

Signalled in the NL ensemble:

Table A.32

Field	Value	Description
C/N	0	Rfu
OE	0	Rfu
P/D	0	Rfu
Ext flag	1	Extension field present
Ens LTO	2	UTC + 1
Ens ECC	0xE3	Applies to all services in the ensemble not identified in the extension field
Inter Table Id	1	
Num of services	1	
ECC	0xE0	
Service Identifier	0x6335	Service with ECC 0xE0

A.11.4 Receiver Behaviour

As general hard linking behaviour - receiver will try the provided DAB, FM-RDS and DRM services if the DAB reception fails.

A.12 Same service across international border

A.12.1 Problem Description

The same programme service (same SId) is available on multiple ensembles (differing EIds), in different countries with different ECCs.

The following example is based on a case in South Tyrol (Italy) in 2018.

NOTE: This example does not replicate the actual transmission situation in all aspects.

Figure A.6 illustrates the coverage areas of two different ensembles. The example explains service following for one service that is available in both ensembles. The bold line illustrates a possible travelling route between the different coverage areas.

The ensemble with EId 0x4081 is in Switzerland and is geographically adjacent to the ensemble with EId 0x510D, which is in Italy; the coverage areas overlap. The ECC of the Swiss ensemble and the Swiss service is 0xE1; the ECC of the Italian Ensemble is 0xE0.

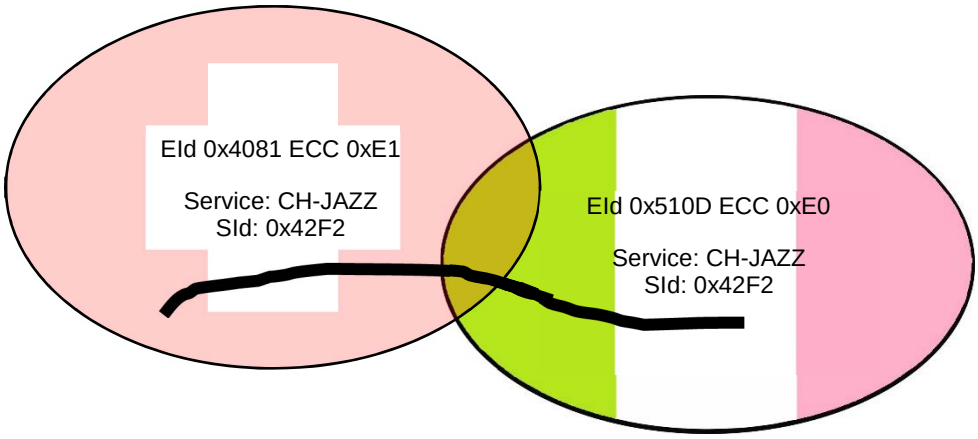


Figure A.6: Ensemble coverage schematic

A.12.2 Concept

Signalling of FIG 0/9 provides the ECCs of all services.

A dual tuner may detect the same service on a different ensemble and frequency by periodic analysis of received signals.

A.12.3 Signalling

A.12.3.1 FIG 0/6 Service linking information

Not required for this use case.

A.12.3.2 FIG 0/24 OE Services

Not required for this use case.

A.12.3.3 FIG 0/21 Frequency Information

Not required for this use case.

A.12.3.4 FIG 0/9 Country, LTO and International Table

Signalled in the Swiss ensemble:

Table A.33

Field	Value	Description
C/N	0	Rfu
OE	0	Rfu
P/D	0	Rfu
Ext flag	0	No extension field
Ens LTO	2	UTC + 1
Ens ECC	0xE1	Applies to all services in the ensemble
Inter Table Id	1	

Signalled in the Italian ensemble:

Table A.34

Field	Value	Description
C/N	0	Rfu
OE	0	Rfu
P/D	0	Rfu
Ext flag	1	Extension field present
Ens LTO	2	UTC + 1
Ens ECC	0xE0	Applies to all services in the ensemble not identified in the extension field
Inter Table Id	1	
Num of services	2	
ECC	0xE1	
Service Identifier	0x42F2	Service with ECC 0xE1
Service Identifier	0x42F3	Service with ECC 0xE1

A.12.4 Receiver Behaviour

A receiver should decide a point at which the currently received signal is inadequate. The receiver should attempt to locate the current service on an alternative ensemble The ECC is provided by FIG 0/9; the ECC of all services in the Swiss ensembles matches the ensemble ECC, but for the Swiss services in the Italian ensemble, the ECCs do not match and so the extension field is used to provide the correct ECC for those services.

If this fails, or no OE services information is being signalled, the receiver will attempt to locate the current service by scanning all possible frequencies and analysing all the services on the ensembles found.

A.13 Linking national and regional variations of a service on different ensembles

A.13.1 Problem Description

A group of related programme services are broadcast across multiple regions, on different DAB ensembles and on FM-RDS, and also on a nationwide DAB ensemble. At certain times the audio content is identical on all services, and at other times the content is different. As the programme content is not always identical at all times, each regional variation has a different SId code, which has a matching RDS PI code, and another SId is carried by the national variant. The periods of non-identical content may be of only a few minutes duration, such as commercial breaks or news bulletins, or may be for some hours, corresponding to the duration of a programme segment.

At times when the programme service is identical, the broadcaster wants the listener to experience continuous coverage, as if the service was transmitted with the same SId code on all ensembles, but also wants receivers to prefer the variant carried on the ensemble with the smallest coverage footprint that is available so that the most localised programming is available when audio differences occur.

At times when the programme service is not identical, the broadcaster does not want the listener to be switched between services automatically by the receiver; however, the broadcaster still wants receivers to prefer the most local content when tuning in.

The broadcaster can control whether linking is active or inactive, and can do so quickly and often.

Figure A.7 shows four related services carried on four ensembles. The national ensemble is indicated in red, the regional ensembles in other colours. The coverage of the national ensemble includes the areas covered by the regional ensembles. The services carried on the regional ensembles are also carried on FM-RDS with with the regional PI code identical to the regional DAB service identifier. Hard and soft linkage sets are used to link all the variants – the hard linkage set is active when the audio content is identical across all variants and the soft linkage set is active when the content is different. The figure shows a car journey from the green regional ensemble to the orange regional ensemble which passes through an area with coverage from the national ensemble only. At the start of the journey the audio content is identical across all ensembles but soon after the car reaches the orange region, the audio content becomes localised.

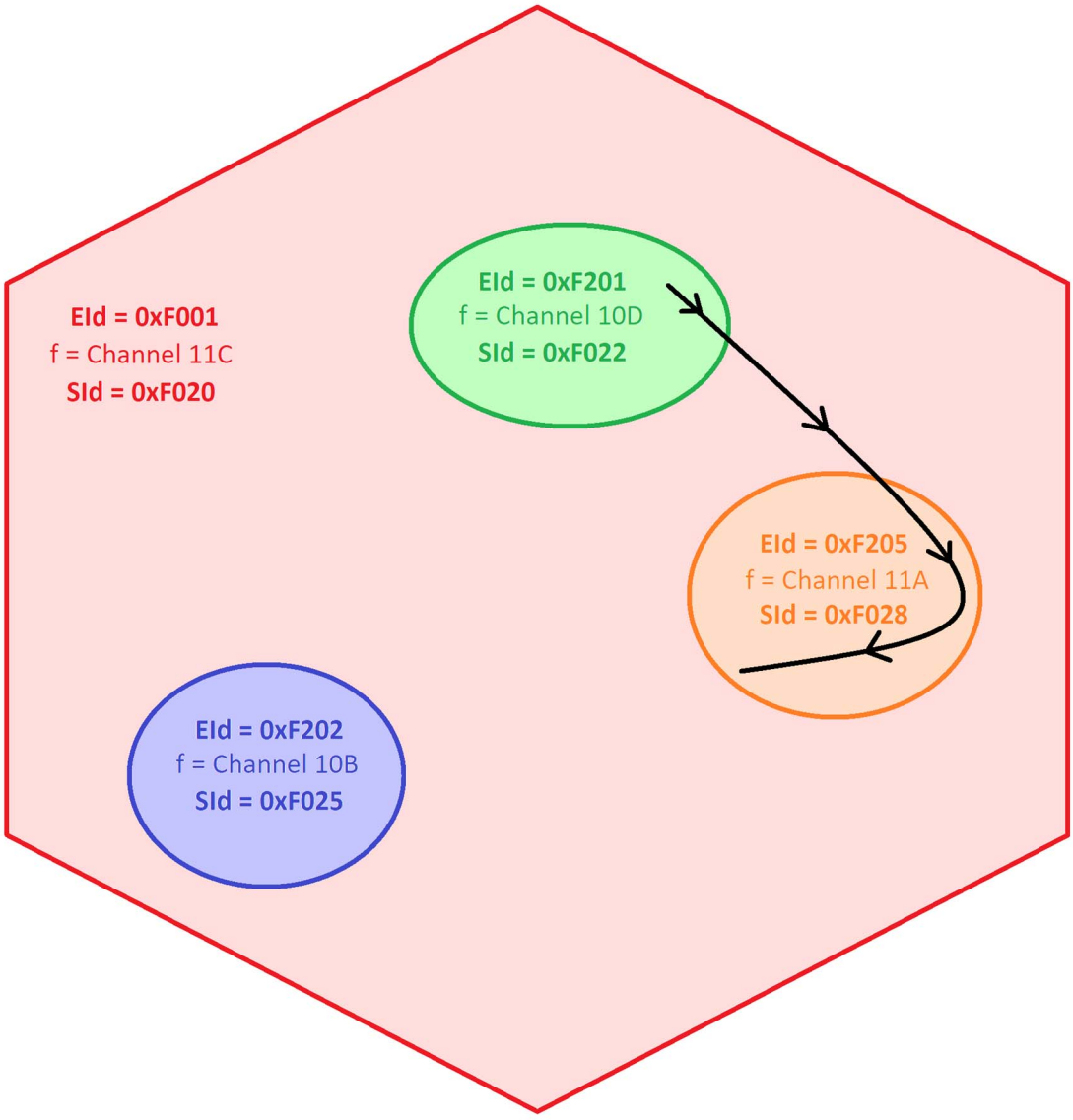


Figure A.7: Service and ensemble configuration

A.13.2 Concept

At the beginning of the journey, the driver selects his local service. As his journey progresses, he leaves the coverage provided by the green regional ensemble and has only the coverage provided by the national ensemble. The receiver should automatically retune to maintain his listening. As the car moves into the coverage provided by the orange regional ensemble, the driver would like to hear the local service for that city.

A.13.3 Signalling

A.13.3.1 FIG 0/6 Service linking information

Two network configurations exist, one with all services carrying the same audio – a hard linkage set – and one where the audio is related but localised – a soft linkage set. The configurations are mutually exclusive but contain the same set of services. Each ensemble signals both linkage sets at all times. The state of the link actuators will always be set in accordance with the audio content both in the slower cyclic repetition of the long form database definition FIGs and the faster cyclic repetition of the short form activation state FIGs when the network configuration changes. When the activation states change then a burst of short form FIGs are sent.

In each ensemble the IdLP flag is set to 1 for the national service and set to 0 for the regional services to indicate that the regional variants are preferred over the national service. The order of the Ids in each ensemble varies.

The following information is sent on ensemble 0xF001 when the audio content is identical on all four ensembles; the linkage actuator is activated in the hard linkage set and deactivated in the soft linkage set. The activation status will be reversed when the audio content is different on the different ensembles. As both the linkage sets contain 4 DAB SIds and 3 RDS PI codes, both a start of database FIG 0/6 and, due to the number of Ids, two continuation of database FIG 0/6s are signalled to fully define the two linkage sets on the national ensemble.

Table A.35

Field	Value	Description
C/N	0	Start of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x100	Linkage set number
IdLQ	0b00	Each Id represents a DAB SId
IdLP	1	The services in the list have a low preference level
Number of Ids	1	Number of Ids in the list
Id1	0xF020	SId of service (key service)
Id List Flag	1	Long form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	0	Soft link
ILS	0	National link
LSN	0x100	Linkage set number
IdLQ	0b00	Each Id represents a DAB SId
IdLP	1	The services in the list have a low preference level
Number of Ids	1	Number of Ids in the list
Id1	0xF020	SId of service (key service)

Table A.36

Field	Value	Description
C/N	1	Continuation of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x100	Linkage set number
IdLQ	0b00	Each Id represents a DAB SId
IdLP	0	The services in the list have a normal preference level
Number of Ids	3	Number of Ids in the list
Id1	0xF022	SId of service
Id2	0xF025	SId of service
Id3	0xF028	SId of service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x100	Linkage set number
IdLQ	0b01	Each Id represents an RDS PI code
IdLP	0	The services in the list have a normal preference level
Number of Ids	3	Number of Ids in the list
Id1	0xF022	PI of service
Id2	0xF025	PI of service
Id3	0xF028	PI of service
Id List Flag	1	Long form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	0	Soft link
ILS	0	National link
LSN	0x100	Linkage set number
IdLQ	0b00	Each Id represents a DAB SId
IdLP	0	The services in the list have a normal preference level
Number of Ids	3	Number of Ids in the list
Id1	0xF022	SId of service
Id2	0xF025	SId of service
Id3	0xF028	SId of service
C/N	1	Continuation of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	0	Soft link
ILS	0	National link
LSN	0x100	Linkage set number
IdLQ	0b01	Each Id represents an RDS PI code
IdLP	0	The services in the list have a normal preference level
Number of Ids	3	Number of Ids in the list
Id1	0xF022	PI of service
Id2	0xF025	PI of service
Id3	0xF028	PI of service

The activation state of both the linkage sets is sent regularly.

The following information is sent on regional ensemble 0xF205 when the audio content is different on all four ensembles; the linkage actuator is deactivated in the hard linkage set and activated in the soft linkage set. The activation status will be reversed when the audio content is the same on the different ensembles. As both the linkage sets contain 4 DAB SIds and 3 RDS PI codes, both a start of database FIG 0/6 and a continuation of database FIG 0/6 are signalled to fully define the two linkage sets on the regional ensembles.

Table A.37

Field	Value	Description
C/N	0	Start of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x100	Linkage set number
IdLQ	0b00	Each Id represents a DAB SId
IdLP	0	The services in the list have a normal preference level
Number of Ids	3	Number of SIds in the list
Id1	0xF028	SId of service (key service)
Id2	0xF022	SId of service
Id3	0xF025	SId of service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	0	Soft link
ILS	0	National link
LSN	0x100	Linkage set number
IdLQ	0b00	Each Id represents a DAB SId
IdLP	0	The services in the list have a normal preference level
Number of Ids	3	Number of SIds in the list
Id1	0xF028	SId of service (key service)
Id2	0xF022	SId of service
Id3	0xF025	SId of service

Table A.38

Field	Value	Description
C/N	1	Continuation of database
OE	0	Rfu
P/D	0	Programme service
Id List Flag	1	Long form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x100	Linkage set number
IdLQ	0b00	Each Id represents a DAB SId
IdLP	1	The services in the list have a low preference level
Number of Ids	1	Number of SIds in the list
Id1	0xF020	SId of service (national service)
Id List Flag	1	Long form
Linkage Actuator	0	Linkage set is deactivated
Soft/Hard	1	Hard link
ILS	0	National link
LSN	0x100	Linkage set number
IdLQ	0b01	Each Id represents an RDS PI code
IdLP	0	The services in the list have a normal preference level
Number of Ids	3	Number of Ids in the list
Id1	0xF028	PI of service
Id2	0xF025	PI of service
Id3	0xF022	PI of service
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	0	Soft link
ILS	0	National link
LSN	0x100	Linkage set number
IdLQ	0b00	Each Id represents a DAB SId
IdLP	1	The services in the list have a low preference level
Number of Ids	1	Number of SIds in the list
Id1	0xF020	SId of service (national service)
Id List Flag	1	Long form
Linkage Actuator	1	Linkage set is activated
Soft/Hard	0	Soft link
ILS	0	National link
LSN	0x100	Linkage set number
IdLQ	0b01	Each Id represents an RDS PI code
IdLP	0	The services in the list have a normal preference level
Number of Ids	3	Number of Ids in the list
Id1	0xF028	PI of service
Id2	0xF025	PI of service
Id3	0xF022	PI of service

The information in the other regional ensembles will be similar (the order of the regional Ids will be different).

A.13.4 Receiver Behaviour

At the beginning of the journey, the driver selects the regional variant, 0xF022. The receiver acquires the service linking information for this ensemble: the hard linage set is activated with 3 DAB alternatives, one of which is less preferred (the national variant) and 3 FM-RDS alternatives for the key service. The receiver is equipped with two tuners and so is able to check the availability and quality of the alternatives continuously.

As the journey progresses, the signal quality of the green regional ensemble deteriorates below the threshold and so the receiver retunes to the only alternative, which is the national variant, 0xF020, despite it being signalled as less preferred – the FM-RDS coverage has also deteriorated below the required quality. Since the IdLP flag is set to 1, indicating that the tuned service is not the preferred service, the receiver periodically checks whether one of the other services is available. As the car moves into the coverage provided by the orange regional ensemble, it finds the regional variant, 0xF028, and retunes. After 20 minutes, the audio content changes between the different variants and so the linkage actuators are reversed: the hard linkage set is now deactivated, the soft linkage set is activated. If the journey continued beyond the coverage area of the regional ensemble, the receiver could offer the national service as a related service, but since the audio is different between the national and regional services, the switch should not be automatic (unless a user option has been set).

When the hard linkage set is deactivated, meaning that the audio content is different on different variants, implicit linking between the DAB and FM-RDS regional services will provide an alternative when the signal quality falls below the required thereshold. When a journey means that the vehicle moves out of the regional coverage area, signal loss will occur and the receiver may then try to find an alternative by checking the active soft linkage set.

Another driver makes the identical journey but has a single tuner receiver. His experience will be different. When the car leaves the coverage area of the green regional ensemble and the signal quality of the green regional ensemble deteriorates below the threshold, the receiver will look for alternatives by retuning to each possible service. The receiver will find and retune to the national variant, 0xF020. Later, however, when the car enters the coverage area of the orange regional ensemble, it will not retune to the regional variant, 0xF028, because it has no means to detect other signals: whilst the signal quality of the national ensemble remains good, the receiver will stay tuned to the national ensemble until the end of the journey. If a new journey was begun whilst the regional and national services are carrying different audio, the receiver should choose the regional variant because the national variant has the IdLP flag set to 1 indicating that it is less preferred that one of the regional variants.

Annex B (informative): Service following - Transition from existing implementations

Prior to the adoption of the present document, service following implementations were created by various service providers and receiver manufacturers. These implementations may differ in small respects from the rules described in the present document. In general, these differences will not be significant, but in countries where service linking information has been provided for some time and which have a great number of DAB car receivers, special care needs to be taken when changing service following signalling. Some older receivers may not be able to follow all the rules defined in the present document, or may require additional information or information signalled in a different way. Therefore a service provider may choose to create a setup suitable for different generations of receivers for a transition period.

The following differences may occur:

- Some receivers follow a paradigm that if *any* service linking information is provided that *all* implicit linking is disabled. In this case the service provider can provide a linkage set with the SId and PI codes hard linked.
- Some receivers will always follow implicit links between DAB and FM-RDS. If this behaviour is not wanted then SIds should be allocated to avoid this problem.
- Some receivers require that linkage sets are defined with the same SId as the first Id in the list and as a subsequent Id in order to follow a service on different ensembles. Service providers can still provide this signalling under the rules defined in the present document, albeit with a slight signalling overhead.

Annex C (informative): Service lists - use case examples

C.1 New service on air

C.1.1 Use case description

A new audio service will be introduced to an existing ensemble. This service has never been on an ensemble before. After the insertion, the service is visible in the service list and can be selected by the listener. The service remains in the ensemble for the future. The broadcaster announces this service prior to the actual insertion to allow receivers not tuned to the ensemble at the time of the insertion to know when the new service will be available. Some receivers may display additional information.

C.1.2 Concept

To provide information about this service prior to the reconfiguration, FIG 0/20 is used. Using this signalling, the receiver knows the date and time of the insertion and can take action to enable selection of the service from that point onwards. Some receivers may display the date and time when the service will be available.

A reconfiguration of the ensemble is required to insert the service.

All the signalling can only be received if the receiver is tuned to that ensemble.

C.1.3 Signalling

C.1.3.1 Preparation phase

The Current MCI contains no reference to 0x1234.

The service label is signalled.

The FIG 0/20 contains a Service Component Information field with the following values.

Table C.1

Field	Value	Description
SId	0x1234	The Service Identifier of the new service.
SCIdS	0	The primary service component.
Change flags	01b	The service will be *added* to the ensemble.
P-T flag	0b	The new service will be broadcast *continuously*.
SC flag	1b	The SC description field is required and present.
SC desc.	CA flag: 0b A/S flag: 0b SCTy: 111111b	The CA flag is set to 0 (No access control). The A/D flag is set to 0 (An ASCTy follows). The SCTy indicates DAB+ audio.
Date-time	Date: 01100b Hour: 01101b Minute: 000000b Second: 000000b	The new service will be added to the ensemble on 2016-01-01 at 13:00:00 UTC. At the nearest second to this point of time the multiplex reconfiguration occurs.
SId flag	0b	The Transfer SId field is absent.
EId flag	0b	The Transfer EId field is absent.

C.1.3.2 Reconfiguration phase

The Current MCI contains no reference to 0x1234.

The Next MCI contains reference to 0x1234.

The service label is signalled.

The FIG 0/20 is signalled as for the preparation phase.

C.1.3.3 Consolidation phase

The Current MCI contains reference to 0x1234.

The service label is signalled.

No FIG 0/20 is signalled.

C.1.4 Receiver behaviour

Receivers will use the information provided in FIG 0/20 to prepare for the addition of the new service. Some receivers will display this information to the user in an appropriate way.

After the ensemble reconfiguration, all receivers tuning to the ensemble will display the new service in the service list. Some receivers will display the new service in the service list even before tuning to the ensemble, but mark the entry as "tentative".

C.2 Continuous service is removed (with transfer)

C.2.1 Use case description

A service will be removed from all ensembles on the 10th of March 2016 at 14:30 (UTC). The broadcaster would like listeners to be transferred to another service which is available on the same ensemble when the original service has been removed.

C.2.2 Concept

To provide information about the removal of this service prior to the reconfiguration, FIG 0/20 will be used. Using this signalling, the receiver can, if the removed service is being listened to at the point of removal, automatically switch to the service indicated by the broadcaster. This action is also enabled if the service was being listened to when the receiver was switched off and is switched on again after the service has been removed.

A reconfiguration of the ensemble is required to remove the service.

All the signalling can only be received while the receiver is tuned to that ensemble.

C.2.3 Signalling

C.2.3.1 Preparation phase

The Current MCI contains reference to 0x1234.

The service label is signalled.

The FIG 0/20 contains a Service Component Information field with the following values.

Table C.2

Field	Value	Description
SId	0x1234	The Service Identifier of the service.
SCIdS	0	The primary service component.
Change flags	11b	The service will be *removed* from all ensembles. (Due to SCIdS = 0 the whole service is concerned - all secondary components, if any, will also be removed.)
P-T flag	0b	The service is a *continuous* service.
SC flag	0b	The SC description field is not present.
Date-time	Date: 10010b Hour: 01110b Minute: 011110b Second: 000000b	The service will be removed from the ensemble on the given time relative to the currently received MJD. At the nearest second to this point of time the multiplex reconfiguration occurs.
SId flag	1b	The Transfer SId field is present.
EId flag	0b	The Transfer EId field is absent.
Transfer SId	0x1278	Preferred transfer service SId.

C.2.3.2 Reconfiguration phase

The Current MCI contains reference to 0x1234.

The Next MCI contains no reference to 0x1234.

The service label is signalled.

The FIG 0/20 is signalled as for the preparation phase.

C.2.3.3 Consolidation phase

The Current MCI contains no reference to 0x1234.

The service label is not signalled.

The FIG 0/20 contains a Service Component Information field with the following values, which is signalled for some days after the service removal.

Table C.3

Field	Value	Description
SId	0x1234	The Service Identifier of the service.
SCIdS	0	The primary service component.
Change flags	11b	The service will be *removed* from all ensembles. (Due to SCIdS = 0 the whole service is concerned - all secondary components were also be removed.)
P-T flag	0b	The service was a *continuous* service.
SC flag	0b	The SC description field is not present.
Date-time	Date: 11111b Hour: 11111b Minute: 111111b Second: 111111b	The service has now been removed.
SId flag	1b	The Transfer SId field is present.
EId flag	0b	The Transfer EId field is absent.
Transfer SId	0x1278	Preferred transfer service SId.

C.2.4 Receiver behaviour

Receivers will use the information provided in FIG 0/20 to prepare for the removal of the service. Some receivers will display this information to the user in an appropriate way.

After the ensemble reconfiguration, all receivers tuning to the ensemble will no longer display the service in the service list. Some receivers will mark the entry as "end of life" in the service list before tuning to the ensemble to confirm the removal.

C.3 New Part-time service on air

C.3.1 Use case description

A new part-time audio service will be introduced to an existing ensemble. This service has never been on an ensemble before. After the insertion, the service is visible in the service list and can be selected by the listener. The service remains in the service list even while it is temporarily off-air. The broadcaster announces this service prior to the actual insertion to allow receivers not tuned to the ensemble at the time of the insertion to know when the new service will be available. Some receivers may display additional information.

C.3.2 Concept

To provide information about this service prior to the reconfiguration, FIG 0/20 is used. Using this signalling, the receiver knows the date and time of the insertion and can take action to enable selection of the service from that point onwards. Some receivers may display the date and time when the service will be available.

A reconfiguration of the ensemble is required to insert the service.

All the signalling can only be received while the receiver is tuned to that ensemble.

C.3.3 Signalling

C.3.3.1 Preparation phase

The Current MCI contains no reference to 0x4444.

The service label is signalled.

The FIG 0/20 contains a Service Component Information field with the following values.

Table C.4

Field	Value	Description
SId	0x4444	The Service Identifier of the new service.
SCIdS	0	The primary service component.
Change flags	01b	The service will be *added* to the ensemble.
P-T flag	0b	The new service will be added permanently.
SC flag	1b	The SC description field is required and present.
SC desc.	CA flag: 0b A/S flag: 0b SCTy: 111111b	The CA flag is set to 0 (No access control). The A/D flag is set to 0 (An ASCTy follows). The SCTy indicates DAB+ audio.
Date-time	Date: 01100b Hour: 01101b Minute: 000000b Second: 000000b	The new service will be added to the ensemble on 2016-01-01 at 13:00:00 UTC. At the nearest second to this point of time the multiplex reconfiguration occurs.
SId flag	0b	The Transfer SId field is absent.
EId flag	0b	The Transfer EId field is absent.

C.3.3.2 Reconfiguration phase

The Current MCI contains no reference to 0x4444.

The Next MCI contains reference to 0x4444.

The service label is signalled.

The FIG 0/20 is signalled as for the preparation phase.

C.3.3.3 Consolidation phase

The Current MCI contains reference to 0x4444.

The service label is signalled.

No FIG 0/20 is signalled.

C.3.4 Receiver behaviour

Receivers will use the information provided in FIG 0/20 to prepare for the addition of the new service. Some receivers will display this information to the user in an appropriate way.

After the ensemble reconfiguration, all receivers tuning to the ensemble will display the new service in the service list. Some receivers will display the new service in the service list even before tuning to the ensemble, but mark the entry as "tentative".

C.4 Part-time service is on-air

C.4.1 Use case description

A part-time service is currently on-air. The service is shown in the service list. The broadcaster announces that the service is a part time service using FIG 0/20 and indicates the date-time when the part-time service will go off-air.

C.4.2 Concept

To provide information about when the part time service will go off-air, FIG 0/20 will be used. Using this signalling, the receiver can determine when the service is likely to be available, even when tuned to another ensemble.

A reconfiguration of the ensemble is required to temporarily remove the part-time service.

All the signalling can only be received while the receiver is tuned to that ensemble.

C.4.3 Signalling

C.4.3.1 Preparation phase

The Current MCI contains reference to 0x4444.

The service label is signalled.

The FIG 0/20 contains a Service Component Information field with the following values.

Table C.5

Field	Value	Description
SId	0x4444	The Service Identifier of the service.
SCIdS	0	The primary service component.
Change flags	11b	The part time service will go off-air in all ensembles.
P-T flag	1b	The service is a part-time service.
SC flag	0b	The SC description field is not required.
Date-time	Date: 01100b Hour: 10110b Minute: 000000b Second: 000000b	The part time service will resume on 2016-01-01 at 22:00:00 UTC. At the nearest second to this point of time the multiplex reconfiguration occurs.
SId flag	0b	The Transfer SId field is absent.
EId flag	0b	The Transfer EId field is absent.

C.4.3.2 Reconfiguration phase

The Current MCI contains reference to 0x4444.

The Next MCI contains no reference to 0x4444.

The service label is signalled.

The FIG 0/20 is signalled as for the preparation phase.

C.4.3.3 Consolidation phase

There is no consolidation phase. As soon as the part-time service is off-air, preparation phase signalling is required to indicate when the service will resume, see clause C.5.

C.4.4 Receiver behaviour

Receivers will use the information provided in FIG 0/20 to prepare for the temporary removal of the part-time service. Some receivers will display this information to the user in an appropriate way.

After the ensemble reconfiguration, all receivers will continue to display the service in the service list, even though it is no longer on-air and not signalled in the MCI.

C.5 Part-time service is off-air

C.5.1 Use case description

The A part-time service is currently on-air. The service is shown in the service list even while it is temporarily off-air. The broadcaster announces that the service is a part time service using FIG 0/20 and indicates the date-time when the part-time service will come on-air again.

C.5.2 Concept

To provide information about when the part time service will come on-air again, FIG 0/20 will be used. Using this signalling, the receiver can determine when the service is likely to be available, even when tuned to another ensemble.

A reconfiguration of the ensemble is required to temporarily remove the part-time service.

All the signalling can only be received while the receiver is tuned to that ensemble.

C.5.3 Signalling

C.5.3.1 Preparation phase

The Current MCI contains no reference to 0x4444.

The service label is signalled.

The FIG 0/20 contains a Service Component Information field with the following values.

Table C.6

Field	Value	Description
Sid	0x4444	The Service Identifier of the part-time service.
SCIdS	0	The primary service component.
Change flags	01b	The service will be *added* to the ensemble.
P-T flag	1b	The service is part-time.
SC flag	1b	The SC description field is required and set.
SC desc.	CA flag: 0b A/S flag: 0b SCTy: 111111b	The CA flag is set to 0 (No access control). The A/D flag is set to 0 (An ASCTy follows). The SCTy indicates DAB+ audio.
Date-time	Date: 01100b Hour: 10100b Minute: 000000b Second: 000000b	The part time service will resume on 2016-01-01 at 20:00:00 UTC. At the nearest second to this point of time the multiplex reconfiguration occurs.
SId flag	0b	The Transfer SId field is absent.
EId flag	0b	The Transfer EId field is absent.

C.5.3.2 Reconfiguration phase

The Current MCI contains no reference to 0x4444.

The Next MCI contains reference to 0x4444.

The service label is signalled.

The FIG 0/20 is signalled as for the preparation phase.

C.5.3.3 Consolidation phase

There is no consolidation phase. As soon as the part-time service is on-air, preparation phase signalling is required to indicate when the service will go off-air, see clause C.4.

C.5.4 Receiver behaviour

Receivers will use the information provided in FIG 0/20 to prepare for the return of the part-time service. Some receivers will display this information to the user in an appropriate way.

After the ensemble reconfiguration, all receivers will continue to display the service in the service list.

C.6 SId change within an ensemble

C.6.1 Use case description

The SId of a service needs to change for some reason. The service remains in the service list with only the SId changing. If the service is selected for playback it continues without interruption. The broadcaster announces the change of SId using FIG 0/20.

C.6.2 Concept

An administrative change is required but no impact should be experienced by users. FIG 0/20 is used to signal in advance that the SId will change to allow receivers to know that they should continue to present the service after the reconfiguration without user intervention and should reassign any presets, etc., so that the user is unaware of the change. The signalling continues after the reconfiguration to allow receivers that have not tuned to the ensemble for some time to still recognize that the "new" service is actually the same service with a new SId.

At the moment the SId changes, a reconfiguration of the ensemble is required.

All the signalling can only be received while the receiver is tuned to that ensemble.

C.6.3 Signalling

C.6.3.1 Preparation phase

The Current MCI contains reference to service 0xC123 and does not contain reference to 0xC177.

The service label is signalled for service 0xC123.

The FIG 0/20 contains a Service Component Information field with the following values.

Table C.7

Field	Value	Description
SId	0xC123	The Service Identifier of the service before the reconfiguration.
SCIdS	0	The whole service is affected: all service components receive the new SId.
Change flags	00b	The identity or source of the service is changing.
P-T flag	0b	A continuous service.
SC flag	0b	The SC description field is absent.
Date-time	Date: 01100b Hour: 10100b Minute: 000000b Second: 000000b	The SId will change on 2016-01-01 at 20:00:00 UTC. At the nearest second to this point of time the multiplex reconfiguration occurs.
SId flag	1b	The Transfer SId field is present.
EId flag	0b	The Transfer EId field is absent.
Transfer SId	0xC177	The Service Identifier for the service after the reconfiguration.

C.6.3.2 Reconfiguration phase

The Current MCI contains reference to service 0xC123 and does not contain reference to 0xC177.

The Next MCI does not contain reference to service 0xC123 and contains reference to 0xC177.

The service label is signalled for service 0xC123 and for service 0xC177.

The FIG 0/20 is signalled as for the preparation phase.

C.6.3.3 Consolidation phase

The Current MCI does not contain reference to service 0xC123 and contains reference to 0xC177.

The service label is signalled for service 0xC177.

The FIG 0/20 contains a Service Component Information field with the following values.

Table C.8

Field	Value	Description
SId	0xC123	The Service Identifier of the service before the reconfiguration.
SCIdS	0	The whole service is affected: all service components receive the new SId.
Change flags	00b	The identity or source of the service is changing.
P-T flag	0b	A continuous service.
SC flag	0b	The SC description field is absent.
Date-time	Date: 01100b Hour: 11111b Minute: 111111b Second: 111111b	The SId has changed already.
SId flag	1b	The Transfer SId field is present.
EId flag	0b	The Transfer EId field is absent.
Transfer SId	0xC177	The Service Identifier of the service after the reconfiguration.

C.6.4 Receiver behaviour

Receivers recognize that an administrative change to the SId is scheduled, or has already taken place. Receivers that have acquired the SCI prior to the reconfiguration make a change to the SId for all the affected service elements in the service list and any presets at the time of the reconfiguration. If the receiver is playing the service at the time of the reconfiguration, the service presentation is not affected in any way. Receivers tuning to the ensemble for the first time after the reconfiguration (i.e. during the consolidation phase) make a change to the SId for all the affected service elements in the service list and any presets as soon as they decode the information.

C.7 Service moves to a new ensemble with SId change

C.7.1 Use case description

A service moves from one ensemble to another and also gets a new SId. The service remains in the service list with only the EId and SId changing. If the service is selected for playback it continues with minimum interruption. The broadcaster announces the change of SId and EId using FIG 0/20.

C.7.2 Concept

An administrative change is required but minimal impact should be experienced by users. FIG 0/20 is used to signal in advance in both ensembles that the EId and SId will change to allow receivers to know that they should continue to present the service after the reconfiguration without user intervention and should reassign any presets, etc., so that the user is unaware of the change. The signalling continues after the reconfigurations (one in each ensemble) to allow receivers that have not tuned to the ensembles for some time to still recognize that the "new" service is actually the same service with a new EId and SId.

At the moment the service leaves the original ensemble, a reconfiguration of that ensemble is required.

At the moment the service enters the new ensemble, a reconfiguration of that ensemble is required.

All the signalling can only be received while the receiver is tuned to that ensemble.

In this example, there is an overlap period where the service is present in both ensembles for a limited period - the service enters the new ensemble at 02:00 but does not leave the original ensemble until 23:59:59. The period of overlap is chosen by the broadcaster to suit his needs and may be zero (i.e. the service move is instantaneous).

C.7.3 Signalling

C.7.3.1 Preparation phase

In the original ensemble, EId = C111, preparation phase signalling stops at 23:59:59 on 2016-01-01.

The Current MCI contains reference to service 0xC249.

The service label is signalled for service 0xC249.

The FIG 0/20 contains a Service Component Information field with the following values.

Table C.9

Field	Value	Description
SId	0xC249	The Service Identifier of the service before the reconfiguration.
SCIdS	0	The whole service is affected: all service components receive the new SId.
Change flags	00b	The identity or source of the service is changing.
P-T flag	0b	A continuous service.
SC flag	0b	The SC description field is absent.
Date-time	Date: 01100b Hour: 10111b Minute: 111011b Second: 111011b	The service will leave on 2016-01-01 at 23:59:59 UTC. At the nearest second to this point of time the multiplex reconfiguration occurs.
SId flag	1b	The Transfer SId field is present.
EId flag	1b	The Transfer EId field is present.
Transfer SId	0xC262	The Service Identifier for the service after the reconfiguration.
Transfer EId	0xC222	The Ensemble Identifier for the ensemble that carries the service after the reconfiguration.

In the new ensemble, EId = C222, preparation phase signalling stops at 02:00:00 on 2016-01-01.

The Current MCI contains no reference to 0xC262.

The service label is signalled for service 0xC262.

The FIG 0/20 contains a Service Component Information field with the following values.

Table C.10

Field	Value	Description
SId	0xC249	The Service Identifier of the service before the reconfiguration.
SCIdS	0	The whole service is affected: all service components receive the new SId.
Change flags	00b	The identity or source of the service is changing
P-T flag	0b	A continuous service.
SC flag	0b	The SC description field is absent.
Date-time	Date: 01100b Hour: 00010b Minute: 000000b Second: 000000b	The service will enter on 2016-01-01 at 02:00:00 UTC. At the nearest second to this point of time the multiplex reconfiguration occurs.
SId flag	1b	The Transfer SId field is present.
EId flag	1b	The Transfer EId field is present.
Transfer SId	0xC262	The Service Identifier for the service after the reconfiguration.
Transfer EId	0xC222	The Ensemble Identifier for the ensemble that carries the service after the reconfiguration.

C.7.3.2 Reconfiguration phase

In the original ensemble, EId = C111, reconfiguration phase signalling occurs between approximately 23:59:53 and 23:59:59 on 2016-01-01:

- The Current MCI contains reference to 0xC249.
- The Next MCI does not contain reference to 0xC249.
- The service label is signalled for service 0xC249.

In the new ensemble, EId = C222, reconfiguration phase signalling occurs between approximately 01:59:54 and 02:00:00 on 2016-01-01:

- The Current MCI does not contain reference to 0xC262.
- The Next MCI contains reference to 0xC262.
- The service label is signalled for service 0xC262.

The FIG 0/20 is signalled as for the preparation phase in each ensemble.

C.7.3.3 Consolidation phase

In the original ensemble, EId = C111, consolidation phase signalling starts at 23:59:59 on 2016-01-01:

- The Current MCI does not contain reference to service 0xC249.
- The service label is not signalled for service 0xC249.

The FIG 0/20 contains a Service Component Information field with the following values.

Table C.11

Field	Value	Description
SId	0xC249	The Service Identifier of the service before the reconfiguration.
SCIdS	0	The whole service is affected: all service components receive the new SId.
Change flags	00b	The identity or source of the service is changing
P-T flag	0b	A continuous service.
SC flag	0b	The SC description field is absent.
Date-time	Date: 01100b Hour: 11111b Minute: 111111b Second: 111111b	The service has moved already.
SId flag	1b	The Transfer SId field is present.
EId flag	1b	The Transfer EId field is present.
Transfer SId	0xC262	The Service Identifier for the service after the reconfiguration.
Transfer EId	0xC222	The Ensemble Identifier for the ensemble that carries the service after the reconfiguration.

In the new ensemble, EId = C222, consolidation phase signalling starts at 02:00:00 on 2016-01-01:

- The Current MCI contains reference to service 0xC262.
- The service label is signalled for service 0xC262.

The FIG 0/20 contains a Service Component Information field with the following values.

Table C.12

Field	Value	Description
SId	0xC249	The Service Identifier of the service before the reconfiguration.
SCIdS	0	The whole service is affected: all service components receive the new SId.
Change flags	00b	The identity or source of the service is changing
P-T flag	0b	A continuous service.
SC flag	0b	The SC description field is absent.
Date-time	Date: 01100b Hour: 11111b Minute: 111111b Second: 111111b	The service has moved already.
SId flag	1b	The Transfer SId field is present.
EId flag	1b	The Transfer EId field is present.
Transfer SId	0xC262	The Service Identifier for the service after the reconfiguration.
Transfer EId	0xC222	The Ensemble Identifier for the ensemble that carries the service after the reconfiguration.

C.7.4 Receiver behaviour

Receivers tuning to either ensemble recognize that an administrative change to the EId and SId is scheduled, or has already taken place.

In this example there is an overlap period during which the service is available in both ensembles, but this may not always be the case. Receivers tuned to the original ensemble can determine from the SCI that the service will move to a new ensemble and know the date-time after which the service will no longer be available in this ensemble but will definitely be available on the new ensemble.

Similarly, receivers tuned to the new ensemble can determine from the SCI that the service will move to this ensemble and know the date-time after which the service will be available in this ensemble; however from this SCI alone they do not know when it will no longer be available on the original ensemble nor do they know the identity of the original ensemble.

Receivers that have gathered the SCI from both ensembles know the service origin, new identity and the date-times when the service enters the new ensemble and leaves the original ensemble. These receivers can therefore make the best decisions on when during the overlap period to change the information in their service list to minimize the impact on the user. When the overlap period is zero, the reconfigurations in the two ensembles will happen simultaneously.

If the receiver is tuned to the original ensemble and is playing the service at the time of the reconfiguration, the receiver retunes to the new ensemble, selects the service (now identified by the new SId) and resumes presentation - multi-tuner receivers are able to pre-tune and so minimize audio interruptions.

If the user selects the service during the overlap period, the receiver tunes to the new ensemble. Receivers tuning to either ensemble for the first time during the respective consolidation phase make a change to the EId and SId for all the affected service elements in the service list and any presets as soon as they decode the information.

Annex D (informative): Service lists - presentation

D.1 Entries in the service list

The service list is in fact a list of all the service components that can reasonably be expected to be available to the user of the device at the location and time that the user is looking. Different devices will have different implementations of what the designers considered to be "reasonable".

Different implementations may be expected for automotive, domestic, hand-held, etc., because these different types of devices have different usages and different capabilities. For example, mobile receivers (either automotive or hand-held) will pass through different ensemble coverage areas and may accumulate a great many service components over time, but only some will be tunable at a given location. In addition, some service components are only broadcast on a part-time basis and so are only tunable at certain times. New services are launched, and old services close down. To deal with these various possibilities, device manufacturers need to develop a strategy to present the service components that are available in an understandable and consistent way, so as to provide the best user experience.

Devices should only display to the user those service components that the device can present (for example DMB-tv services should not be selectable on an audio only receiver).

Devices should ensure that when a service component is available from more than one ensemble that only a single selection is presented to the user - all the alternatives should be kept hidden, but the best source should be chosen at the time and location that the user makes the selection. The definition of "best" may vary amongst implementations, but factors to consider are reception quality, audio bit-rate, etc. Devices need to ensure that they do not just pick the highest or lowest frequency that the service is available on!

D.2 Handling of part-time service components

The SCI provides a mechanism for maintaining part-time service components in the service list. During off-air periods devices may display a suitable message to the user. The **Date-time** field allows an indication to be made of when the service component will be available again, but the information may be outdated if the device has not been able to tune to the ensemble for some period.

If the part-time service component is a secondary service component, then if selected when off-air then the device will play the primary service component instead.

D.3 Automatic updates to the service list

Whenever the receiver is tuned to an ensemble, it should update the stored information about service components within the tuned ensemble. The receiver should update all information needed to build the service list, as well as all information needed to access a service element. This information will usually contain: EId, SId, labels, sub-channel addresses, SCId/SCIdS, ASCTy/DSCTy/UAId information. This may also include PTy, service linking information, announcement information, etc. All information the receiver stores about ensembles and their service components should be automatically kept up-to-date.

If the receiver detects a reconfiguration of the currently tuned ensemble, then this information is also updated.

If the receiver fails to tune to an ensemble, it may apply a service following process to continue the service presentation. This may include a change to other ensembles, linked services or to other bearers. However, the information stored about the unavailable ensemble should remain unchanged. This means that if an ensemble is no longer broadcast (or the receiver is outside its reception area), all its service elements will remain in the list.

After a "manual frequency scan" (see also clause 6.2.3) information about ensembles that can no longer be received will be removed from the service list.

If the receiver tunes to an ensemble and then finds another ensemble (e.g. if the receiver is moved to another location), the receiver should automatically store information describing this new ensemble.

NOTE 1: In this case the receiver will then have information about two ensembles on the same frequency. The combined service list may show service elements from both these ensembles even though only one ensemble can be tuned.

The list shown to the user is derived from all information collected on all ensembles.

NOTE 2: Receivers with a single tuner will not be able to detect changes outside the currently tuned ensemble. So it is possible that some services are listed although they are no longer broadcast. Only after tuning again to the ensemble that signalled that it carried this service will the receiver be able to determine whether the service is removed from this ensemble.

If the user selects a service and upon tuning the receiver detects that this service is no longer broadcast in any ensemble that it can currently receive, the receiver shows a suitable error message.

NOTE 3: A receiver with multiple tuners will be able to automatically and continuously update the information describing all available ensembles, so such a receiver can provide an up-to-date service list.

In order to minimize the risk of showing an outdated service list, a receiver might choose to automatically update the service list every now and then, e.g. before the receiver shuts down it might automatically attempt to tune to all known ensembles and/or tunable frequencies and retrieve the information about all currently receivable service elements.

D.4 Character sets in the service list

The part of a service list entry that is presented to the user, at a minimum, consists of the relevant label for the service element, i.e. the service label or the service component label. For a useful representation of a service element in the service list, the receiver will need to provide a glyph set suitable to fully support the DAB services in the target market.

Broadcasters should be aware that although the DAB system allows all ISO 10646 [i.1] characters to be transmitted, it is likely that only a sub-set of these characters will be presented on receivers. Care should be exercised to ensure that the labels used to identify service elements will be presented well by receivers in the market; the use of specified regional profiles (see ETSI TS 101 756 [2], annex E) is recommended.

D.5 Sorting of the service list

The sorting of the service list is implementation dependent. Various strategies may be applied for sorting. It is recommended to apply a lexicographical sorting of labels based on the character code used for display. It is essential that the sort order is independent of the character encoding used in label transmission.

The sort order of the service list should be static and consistent across ensembles. It should be predictable to the end user, so that a service can be found by its name. A receiver may provide an option to change the sorting order, e.g. from a lexicographical order top-down to bottom-up or v.v. Other than by user action, the sort order should not be changed.

D.6 Presentation of secondary service components

If a service uses secondary service components, then all secondary service components that the receiver is able to decode are shown in the service list and are directly accessible to the user in the same way as primary service components.

D.7 Updating an ensemble from one frequency to another

It may become necessary to move an ensemble from one frequency to a new frequency.

It is desirable for the receiver to automatically update the stored frequency without user intervention - i.e. the user should not be required to rescan the receiver.

To achieve this, the broadcaster should use FIG 0/21 to signal the new frequency as an alternative frequency for a period of up to 30 days before the switchover.

Capable receivers will store this alternative frequency alongside the current received frequency for the ensemble. After the switchover these receivers will attempt to select on the new frequency once the original frequency is no longer available. To a user this switchover will not be apparent.

D.8 Handling of unavailable service elements

After a valid service list entry has been created, the associated service element may become unavailable for a number of reasons. A service element is unavailable if it cannot be received on any of the ensembles it was previously received on and the receiver cannot re-establish reception of this or an alternative service element in another ensemble or at another frequency. Situations that result in a service element becoming unavailable include the following:

- A receiver is moved out of the coverage area of the ensemble transmission. Then all service elements of the ensemble become unavailable at the same time. The same ensemble may be available on another frequency.
- An ensemble stops being transmitted on its frequency and all its service elements become unavailable. The ensemble may have been moved to another frequency.
- A service element changes to an encoding scheme that is not supported by the receiver. It may be available with a supported encoding scheme on another ensemble.
- A service element changes its identification.
- A service element is moved from its ensemble to another ensemble.
- A service element stops transmission temporarily without signalling as part-time using SCI.

A service element is unavailable if one of the following cases applies:

- none of its ensembles can be received;
- no MCI can be found in its known ensembles referring to the service element;
- no SCI can be found in its known ensembles indicating a part-time service element in off-air status.

The service list entry of an unavailable service element should be retained and should remain selectable; it should be marked as unavailable. If the user attempts to select an unavailable service element, the receiver should make an attempt to re-establish reception, e.g. running the processes described in clauses 5 and 6. Once reception can be re-established, all unavailable marking should be removed.

D.9 Combined service list from multiple bearer systems

A combined FM-RDS and DAB receiver could list FM-RDS and DAB services in one combined service list. When creating a combined FM-RDS and DAB service list, the following points should be considered:

Implicitly linked services

- If FM-RDS and DAB are implicitly linked (i.e. no FIG 0/6 with FM-RDS information for the DAB service is broadcast), this means that if an FM-RDS service using the same PI-code as the DAB service identifier is broadcast, then FM-RDS service and DAB service can be considered the same service and only one service list entry might be shown. If the user selects this service list entry, the receiver would then determine the "best" source and play it.

Services linked by hard-links

- If a hard linkage set is signalled with one FM-RDS service and one DAB service within it, and the link is active, then the two services are carrying the same audio content at this time and a single service list entry could be shown. However, the link may become inactive at a later point, meaning that these services no longer carry the same content and will need separate service list entries.

- If the user selects this service list entry, the receiver would then determine the "best" source and play it. However, if the service linking information dynamically changes and these service are no longer hard linked, then both FM-RDS and DAB service would have to be shown individually in the service list.

- Linkage sets with more than one FM-RDS PI code and one DAB SId are unlikely to be reducible to a single service list entry without confusing the user.

- Another problem for the receiver is that it will only detect changes to the service linking information if it is tuned to the ensemble that carries the service linking information. Cached service linking information of other ensembles might be outdated and incorrect. Using service linking to combine service list entries is difficult and will lead to a dynamically changing service list. Therefore combining hard-linked DAB and FM-RDS services into one service list entry is generally discouraged.

Annex E (informative): Service lists - expected receiver behaviour

E.1 Domestic receiver implementations

E.1.1 Introduction

This annex addresses how a domestic receiver is able to present a service list that reflects what services the receiver can receive at a given time and location. Implementations in automotive or other sectors may differ. Depending on the receiver's architecture, different strategies can be used to achieve this goal.

Most domestic receivers will provide "presets" - buttons which allow quick access to a particular service element. In this context, two types of "preset" can be distinguished:

- Explicit preset: the user manually configures a preset. In addition to the SId/SCIdS, the receiver will usually store additional information such as service/service component label, frequency information and service linking information. If this preset is selected by the user, the receiver tunes to the configured service element.
- Implicit preset: if the receiver is powered on, the receiver will usually tune to the service element that was tuned at the time the receiver was powered off. Also in this case, the receiver might store additional information such as frequency information and service linking information.

Most domestic receivers have a single tuner. SCI information is particularly valuable for these types of receiver.

A single-tuner receiver can only decode the currently tuned ensemble. All information it has about service elements in other ensembles therefore tends to get out-of-date, and the longer the receiver has not tuned to an ensemble, the more out-of-date the information is likely to be.

Domestic receivers cache information about other ensembles so that they can present a full service list to the user. The receiver continues to play the currently selected service while the service list is shown, allowing the user to make a new selection whilst still listening to the current programme. This means that only the information shown about service elements in the currently tuned ensemble is sure to be accurate. Information about service elements in other ensembles might be outdated.

When the single-tuner receiver tunes to an ensemble, it automatically updates the cached information for the tuned ensemble. It adds all newly available service elements and removes all services and service elements that are no longer available, using the MCI and SCI. In addition, it updates the cache of SCI relating to future changes. By caching the SCI for this ensemble, the receiver is able to update its service list at the right time for service elements in this ensemble even if at that time the receiver is tuned to another ensemble.

The single-tuner receiver only knows for certain about the on-air status of service elements in the tuned ensemble. When the user selects a service element from the stored service list that is carried in another ensemble, the receiver tunes to the ensemble and inspects the MCI to determine if the service can be played. If the service element is off-air then a message will be provided to inform the user, and if the service element is part-time, then the expected on-air time may be displayed.

E.1.2 Addition of service elements

SCI with the change flags set to 01 provides information about the future addition of service elements to the tuned ensemble for up to 28 days in advance. The information is cached by the receiver to allow it to update its service list with an entry for the added service element at the date-time indicated when tuned to a different ensemble.

A service list entry created from SCI alone (before the respective MCI has been received) is termed a "tentative" service list entry. It should be marked as such to allow the user to understand it represents a service element that has not yet come into existence.

If a tentative service list entry is added to the service list, it should from then on be shown in the service list until the receiver can confirm that the new service element is broadcast. At that time the service list entry becomes permanent, any tentative marking is removed. This tentative service list entry should also be shown while the receiver is tuned to another ensemble than the ensemble that carried the SCI.

A tentative service list entry should be created at least at the date-time indicated in the SCI. If the receiver is not turned on at this date-time, then a tentative service list should be created after the receiver is turned on. This tentative service list entry should be created even and especially if the receiver is not tuned to this ensemble at this time.

NOTE: It is important that the receiver shows the tentative service list entry after the indicated date-time especially when not tuned to the ensemble because otherwise the listener would have no means to tune to this ensemble and detect and confirm the newly broadcast service element.

The tentative service element may also be created ahead of the indicated date-time. In this case, the receiver should also indicate when the service element will go on-air.

At the date-time indicated in the SCI, the new service element will be broadcast and a receiver tuned to this ensemble at or after the indicated date-time will receive the MCI information for the new service element and can then create the permanent service list entry.

If the receiver can successfully tune to the ensemble that should carry the service element, but the service element is not yet on air (i.e. SCI information was updated to a different date-time), then the receiver should update the date-time when this service element goes on-air.

If the service element is not broadcast and no SCI for this service element is broadcast or SCI indicates that the service element was removed, then the tentative service list entry should be removed.

If the user selects a tentative service list entry, then a message should be shown to the user if the selected service element is not yet broadcast or already removed again.

For part-time services, the information is used to update an existing service list entry with the time the service element will be back on-air, or, if there is no service list entry, to create one. If the **Date-time** field contains the special value, it indicates that the on-air time is not known.

E.1.3 Removal of service elements

SCI with the change flags set to 11 provides information about the removal of service elements from *all ensembles* for up to 28 days in advance. This is the "normal" case for removal.

For continuous service elements, the information is cached by the receiver to allow it to update its service list by deleting the entry for the removed service element at the date-time indicated if tuned to a different ensemble. When the primary service component is removed from the service list, all secondary service components are also removed because they cannot exist in isolation.

For part-time services during the on-air period, the information is used to update an existing service list entry with the time the service element will go off-air. If the **Date-time** field contains the special value, it indicates that the off-air time is not known.

SCI with the change flags set to 10 provides information about the future removal of service elements from *the tuned ensemble only* (local removal) for up to 28 days in advance. This is used by broadcasters when the service is removed from the tuned ensemble, but remains on-air on other ensembles. Those ensembles may or may not be receivable by the receiver at its current location.

For continuous services, the information is cached by the receiver to allow it to update its service list for the removed service element at the date-time indicated when tuned to a different ensemble. However, care needs to be used since the service element will continue to be available in other ensembles than the tuned ensemble and so the service list entry may remain if the receiver is able to tune to any of those other ensembles.

The signalling for local removal is not used for part-time service elements.

If the receiver is playing the affected service element at the time of the removal then it will try to present the listener with the best alternative. If the service element is only removed from the tuned ensemble (change flags = 10) then the service element may still be available on another ensemble. In order to continue service presentation, the receiver needs to run a process with several stages, see clause 6.4.3.1. If through that process no suitable alternative is available, the receiver will display a message.

E.1.4 Changed identity or ensemble

SCI with the change flags set to 00 provides information about changing the identity of services in the tuned ensemble for up to 28 days in advance. If the service is moving ensembles then this SCI appears in both ensembles.

For continuous services, the information is cached by the receiver to allow it to update its service list by updating the entry for the changed service identifiers (SId and/or EId) at the date-time indicated when tuned to a different ensemble.

For part-time services, the information is signalled during the off-air period prior to the new identity being used (i.e. in place of the usual 01 addition signalling). It allows an existing service list entry to be updated with the new identity. The service list entry is immediately updated with the changed service identifiers and the date-time indicates when the service will be back on-air. If no service list entry exists then one is created with the new identity. If the **Date-time** field contains the special value, it indicates that the on-air time is not known.

If the receiver is playing the affected service element at the time of the identity change and the change is within the same ensemble, then it will continue to play the service with no interruption.

If the receiver is playing the affected service element at the time of the identity change but the change is to another ensemble, then it will retune to the new ensemble and continue to play the service with minimal interruption. If the receiver cannot play the service it will try to find a hard-linked service instead.

E.1.5 Multi-tuner domestic receiver

A domestic receiver with more than one tuner is able to maintain its service list with greater accuracy, since it is able to decode the FIC from other ensembles whilst still playing the user's chosen service element.

The "background tuner" continuously scans the tuning range and compares the MCI and SCI received from each ensemble with the information stored in the service list. The on-air status of part-time services can be properly reflected and any updates to the **Date-time** field for identity changes can also be followed.

If the receiver has two complete tuning and demodulation chains then the playback of user selections can also be improved, for example, during and identity change from one ensemble to another, the handover can be made virtually seamless.

Annex F (informative): Example of coding of the FIC

An example of the FIC coding for an ensemble containing 20 DAB+ services, each with a SlideShow in the PAD, meeting the normal repetition rates is as follows:

MCI FIGs:

- FIG 0/0 and FIG 0/7 are required in each FIB0 every 96 ms
- FIG 0/1 is required per subchannel (20) every 96 ms
- FIG 0/2 is required per service (20) every 96 ms
- FIG 0/8 and FIG 0/13 are required per component (20) every 1 s

Labels:

- FIG 1/1 per service (20) every 1 s

SI FIGs:

- FIG 0/9 and FIG 0/10 are provided every 1 s
- FIG 0/5 is provided per subchannel (20) every 1 s
- FIG 0/17 is provided per service (20) every 1 s

In each 96 ms period, there are 12 FIBs (see ETSI EN 300 401 [1], clause 5.1, figure 5). Therefore in each 96 ms period, all the per 96 ms MCI and one tenth of the per second MCI and SI features need to be accommodated. Each FIB has 30 bytes to carry the FIGs, the figures in square brackets indicate the number of bytes used per FIG.

- FIB 0: FIG 0/0 [6]; FIG (0/7) [4]; FIG 0/1 [18 (2 + 4 × 4 subchannels)]; padding [2]
- FIB 1: FIG 0/1 [30 (2 + 4 × 7 subchannels)]
- FIB 2: FIG 0/1 [30 (2 + 4 × 7 subchannels)]
- FIB 3: FIG 0/1 [10 (2 + 4 × 2 subchannels)]; FIG 0/13 [16 (2 + 7 × 2 components)]; padding [4]
- FIB 4: FIG 1/1 [22]; FIG 0/8 [6]; padding [2]
- FIB 5: FIG 1/1 [22]; FIG 0/8 [6]; padding [2]
- FIB 6: FIG 0/2 [27 (2 + 5 × 5 services)]; padding [3]
- FIB 7: FIG 0/2 [27 (2 + 5 × 5 services)]; padding [3]
- FIB 8: FIG 0/2 [27 (2 + 5 × 5 services)]; padding [3]
- FIB 9: FIG 0/2 [27 (2 + 5 × 5 services)]; padding [3]
- FIB10: available for SI
- FIB11: available for SI

Each 960 ms FIB10 and FIB 11 provide 600 bytes for SI. Around 150 bytes are needed for FIG 0/9, FIG 0/10, FIG 0/5 and FIG 0/17 and so about 450 bytes per 960 ms are available for announcements, service following, etc.

Annex G (normative): Alarm announcements test mode

This annex provides a standardized way for public authorities to test alarm announcements without signalling an actual alarm: testing with an actual alarm announcement would cause unacceptable disruption and could lead to public unease. This method provides the necessary function to test the emergency procedures and emergency systems.

The method applies to tuned ensemble and other ensemble announcements.

Support of test alarm announcements is optional in consumer receivers and, if implemented, shall be explicitly enabled by the user and shall be automatically disabled after an alarm test announcement has ceased. Special receivers designed for use by public authorities may behave differently.

The Alarm flag in FIG 0/0 is set to 1 as for normal alarm announcements. Alarm test announcements shall be signalled using Cluster Id "1111 1110" (0xFE) in FIG 0/19 for the tuned ensemble; OE Cluster Id "1111 1110" (0xFE) in FIG 0/26 shall be used for OE announcements. Cluster Id "1111 1110" and OE Cluster Id "1111 1110" shall not be used for any regular announcement support (both FIG 0/18 and FIG 0/25) in ensembles that test alarm announcements using this method.

NOTE: A "normal" receiver will ignore these alarm test announcements because the cluster indicated in the announcement switching does not have corresponding announcement support, nor it is the alarm cluster ("1111 1111" (0xFF)).

Annex H (informative): Alarm announcement use cases

H.1 General

The alarm announcement feature targets all receivers within the coverage area of the ensemble. For this reason, DAB EWS [i.2] is now the preferred way to provide emergency warnings because it includes provision to target receivers within specific geographic areas and has explicitly defined behaviours for receiver wake-up from standby, amongst other features.

Usually the audio and multimedia data that informs the listeners about an emergency (the "alarm service") will be carried in the ensemble with the smallest coverage area that covers the area affected by the emergency. An emergency within a city would thus usually be covered by an ensemble that covers the city. Emergencies affecting larger areas might be covered by regional or even national ensembles.

To ensure that all receivers in the area affected by the emergency tune to the alarm service, OE alarm announcements are used. Other ensembles that can be received in the area affected by the emergency may signal such an OE alarm announcement that points to the ensemble that carries the alarm service.

For the alarm service, it is possible to use a service that is already broadcast (e.g. a news service) and that then carries emergency related content during the emergency. Alternatively the alarm service might be broadcast only during an emergency (i.e. a reconfiguration is needed to add this new service).

Since important information may also be conveyed by the multimedia data of the alarm service, it is important that a receiver also presents this multimedia data.

H.2 Signalling

The OE alarm announcement (FIG 0/26) indicates the alarm service, but it does not indicate the ensemble that carries this service. OE Service information (FIG 0/24) is provided so that the receiver knows which ensemble(s) carries the alarm service.

NOTE: FIG 0/24 will have the OE flag set to 1, since it describes a service not carried in the tuned ensemble.

Regarding the signalling of FIG 0/24, two use cases can be distinguished. These use cases affect whether this FIG 0/24 information is static or dynamic. Normally it can be assumed that for every ensemble there is only at most one alarm service and that this SId is known in order to generate the FIG 0/24 information.

1) For an ensemble with a small coverage area, there are only a small number of other ensembles covering the same area and so it is possible to provide static OE Service information. This information will signal the ensemble(s) for each possible alarm service for each of these ensembles.
   An example for this use case is an ensemble that covers a city and that might want to permit OE alarm announcement pointing to regional and national multiplexes receivable in this city. For this use case, in order to signal an OE alarm announcement, it is sufficient to signal the OE announcement switching (FIG 0/26) during the emergency. The static FIG 0/24 information will provide the ensemble(s) that carry the alarm service signalled with FIG 0/26.

2) For an ensemble with a large coverage area (e.g. a national ensemble), there may be many other ensembles with overlapping coverage areas that could provide alarm services. In this case it is likely that the OE Service information will only be provided slightly ahead of and during the OE alarm announcement.
   An example for this use case is a national ensemble that might want to permit OE alarm announcements pointing to all other national, all regional and all local multiplexes. For this use case, in order to signal an OE alarm announcement, it is necessary to signal both the required OE Service information (FIG 0/24) as well as the OE announcement switching (FIG 0/26) during the emergency.

No matter whether OE alarm announcement is accompanied by static or dynamic FIG 0/24 signalling, it is essential that at the time an OE alarm announcement is signalled, the relevant FIG 0/24 information is also signalled.

H.3 Receiver behaviour

A receiver should only react to an OE alarm announcement (i.e. tune to an alarm service on another ensemble) if the target ensemble is most likely available. For example, in use case 2 in clause H.2, an OE alarm announcement in a national multiplex points to a local multiplex covering a single city: only receivers within this city should try to tune to the alarm service, listeners of the national multiplex outside this city should not have their listening disrupted.

A multi-tuner receiver can verify successful reception of the alarm service before switching the audio. If the alarm service cannot be received, no switching and so no interruption to the audio occurs.

A single tuner receiver should only try tuning to the target ensemble (as indicated by FIG 0/24) if the ensemble is already in pre-tuning memory and thought likely to be receivable: otherwise the disruption to users from a failed tuning exercise causing a mute of several seconds across a whole country for an emergency occurring only in a small city would be disproportionate to the benefit to those in the affected city.

Annex I:
(Void)

History

Version	Date	Status
V1.1.1	August 2012	Publication
V1.1.2	July 2013	Publication
V1.2.1	May 2016	Publication
V2.1.1	August 2017	Publication
V2.2.1	October 2018	Publication
V2.3.1	November 2019	Publication
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